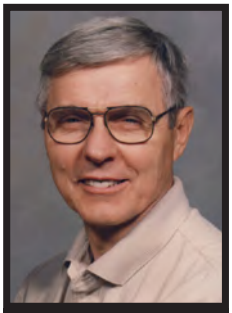
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FLUID POWER BASICS

A compendium
of tutorial articles from
Hydraulics & Pneumatics®

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From the Editor


Trinkel

Fluid Power Basics is a comprehensive collection of more than 200 pages of useful information on hydraulics and pneumatics technologies. It was written by Edgar (Bud) Trinkel, a consultant and educator in fluid power technology. He developed this material to use in the classes he taught and gave exclusive rights to *Hydraulics & Pneumatics* to edit and republish this valuable information.

Since we first published this e-book on our website more than ten years ago, it has consistently been extremely popular with readers, who have repeatedly asked for a version in print. We proudly present this PDF version, which allows you to print individual chapters for your own use.

Unfortunately, Bud passed away in 2009, but his legacy lives on with this compendium of educational information for those seeking a greater understanding of fluid power components, circuits, and systems.
-Alan Hitchcox
Editor, Hydraulics & Pneumatics

Hydraulics & Pneumatics

This initial version of the Fluid Power Basics eBook contains only the first 4 chapters. Versions with subsequent chapters will be released periodically over the several few months until the entire publication is available as a PDF file.

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CHAPTER 1

Fluid power in **industrial applications**

WHAT IS FLUID POWER?

Any media (liquid or gas) that flows naturally or can be forced to flow could be used to transmit energy in a fluid power system. The earliest fluid used was water — hence the name hydraulics was applied to systems using liquids. In modern terminology, hydraulics implies a circuit using mineral oil. Figure 1-1 shows a basic power unit for a hydraulic system. (Note that water is making something of a comeback in the late 1990s; and some fluid power systems today even operate on seawater.)

The other common fluid in fluid power circuits is compressed air. As indicated in Figure 1-2, atmospheric air — compressed seven to10 times — is readily available and flows easily through pipes, tubes, or hoses to transmit energy to do work. Other

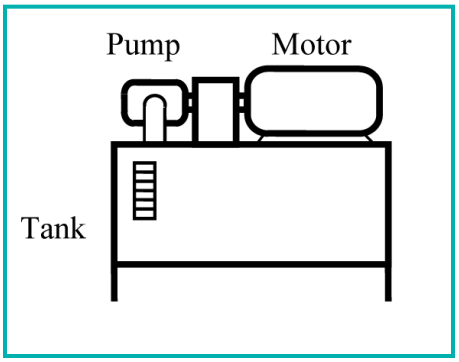


Fig. 1-1: Basic pneumatic power arrangement.

gasses, such as nitrogen or argon, could be used but they are expensive to produce and process.

WHO USES AND IS RESPONSIBLE FOR FLUID POWER?

Of the three main methods of transmitting energy — mechanical, electrical, and fluid — fluid power is least understood by industry in general. In most plants there are few people with direct responsibility for fluid power circuit design or maintenance. Often, general mechanics maintain fluid power circuits that originally were designed by a salesperson from a fluid-power distributor.

In most facilities, the responsibility for fluid power systems is part of the mechanical engineers' job description. The problem is that mechanical engineers normally receive little if any fluid power training at college, so they are ill equipped to carry out this duty. With a modest amount of fluid power training and more than enough work to handle, the engineer often depends on a fluid power distributor's expertise. To get an order, the distributor salesperson is happy to design the circuit — and often assists in installation and

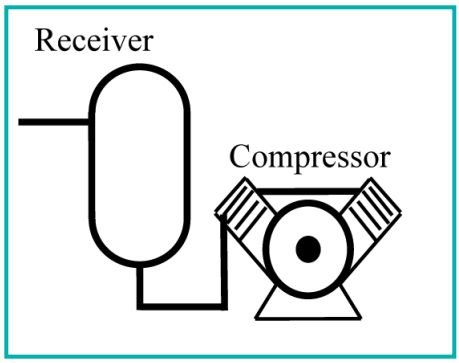


Fig. 1-2: Basic pneumatic power arrangement.

startup. This arrangement works reasonably well, but as other technologies advance, fluid power is being turned down on many machine functions. There is always a tendency to use the equipment most understood by those involved.

ADVANTAGES AND DISADVANTAGES OF FLUID POWER

Fluid power cylinders and motors are compact and have high energy potential. They fit in small spaces and do not clutter the machine. These devices can be stalled for extended time periods, are instantly reversible, have infinitely variable speed, and often replace mechanical linkages at a much lower cost. With good circuit design, the power source, valves, and actuators will run with little maintenance for extended times.

The main disadvantages are lack of understanding of the equipment and poor circuit design, which can result in overheating and leaks. Overheating occurs when the machine uses less energy than the power unit provides. (Overheating usually is easy to design out of a circuit.) Controlling leaks is a matter of using straight-thread O-ring fittings to make tubing connections or hose and SAE flange fittings with larger pipe sizes. Designing the circuit for minimal shock and cool operation also reduces leaks.

PNEUMATICS OR HYDRAULICS?

A general rule to use in choosing between hydraulics or pneumatics for cylinders is: if the specified force requires an air cylinder bore of 4 or 5 in. or larger, choose hydraulics. Most pneumatic circuits are under 3 hp because the efficiency of air compression is low. A system that requires 10 hp for hydraulics would use approximately 30 to 50 air-compressor horsepower. Air circuits are less expensive to build because a separate prime mover is not required, but operating costs are much higher and can quickly offset low component expenses. Situations where a 20-in. bore air cylinder could be economical would be if it cycled only a few times a day or was used to hold tension and never cycled. Both air and hydraulic circuits are capable of operating in hazardous areas when used with air logic controls or explosion-proof electric controls. With certain precautions, cylinders and motors of both types can operate in high-humidity atmospheres . . . or even under water.

When using fluid power around food or medical supplies, it is best to pipe the air exhausts outside the clean area and to use a vegetable-based fluid for hydraulic circuits.

Some applications need the rigidity of

liquids so it might seem necessary to use hydraulics in these cases even with low power needs. For these systems, use a combination of air for the power source and oil as the working fluid to cut cost and still have lunge-free control — with options for accurate stopping and holding as well. Air-oil tank systems, tandem cylinder systems, cylinders with integral controls, and intensifiers are a few of the available components.

PASCAL'S LAW

The reason fluids can transmit energy when contained is best stated by a man from the 17th century named Blaise Pascal. Pascal's Law is one of the basic laws of fluid power. This law says: "Pressure in a confined body of fluid acts equally in all directions and at right angles to the containing surfaces." In other words, if you poke a hole in a pressurized container or line, fluid will squirt out. Figure 1-3 shows how this law works in a cylinder application.

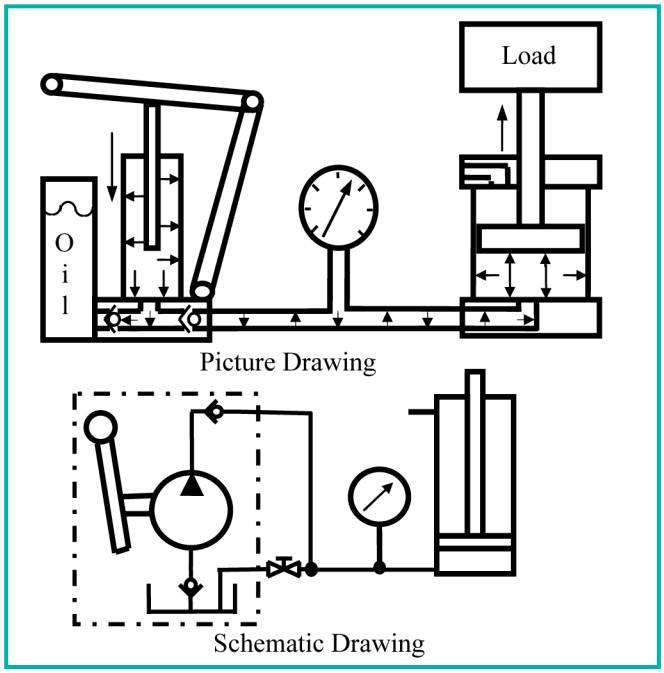


Fig:1-3: How Pascals Law affects a cylinder

Oil from a pump flows into a cylinder that is lifting a load. The resistance of the load causes pressure to build inside the cylinder until the load starts moving. While the load is in motion, pressure in the entire circuit stays nearly constant. The pressurized oil is trying to get out of the pump, pipe, and cylinder, but these mechanisms are strong enough to contain the fluid. When pressure against the piston area becomes high enough to overcome the load resistance, the oil forces the load to move upward. Understanding Pascal's Law makes it easy to see how all hydraulic and pneumatic circuits function.

Notice two important things in this example. First, the pump did not make pressure; it only produced flow. Pumps never make pressure. They only give flow. Resistance to pump flow causes pressure. This is one of the basic principles of fluid power that is of prime importance to troubleshooting hydraulic circuits. Suppose a machine with the pump running shows almost 0 psi on its pressure gauge. Does this mean the pump is bad? Without a flow meter at the pump outlet, mechanics might change the pump, because many of them think pumps make pressure. The problem with this circuit could simply be an open valve that allows all pump flow to go directly to tank. Because the pump outlet flow sees no resistance, a pressure gauge shows little or no pressure. With a flow meter installed, it would be obvious that the pump was all right and other causes — such as an open path to tank — must be found and corrected.

Another area that shows the effect of Pascal's law is a comparison of hydraulic and mechanical leverage. Figure 1-4 shows how both of these systems work. In either case, a large force is offset by a much smaller force due to the difference in lever-arm length or piston area.

Notice that hydraulic leverage is not restricted to a certain distance, height, or physical location like mechanical leverage is. This is a decided advantage for many

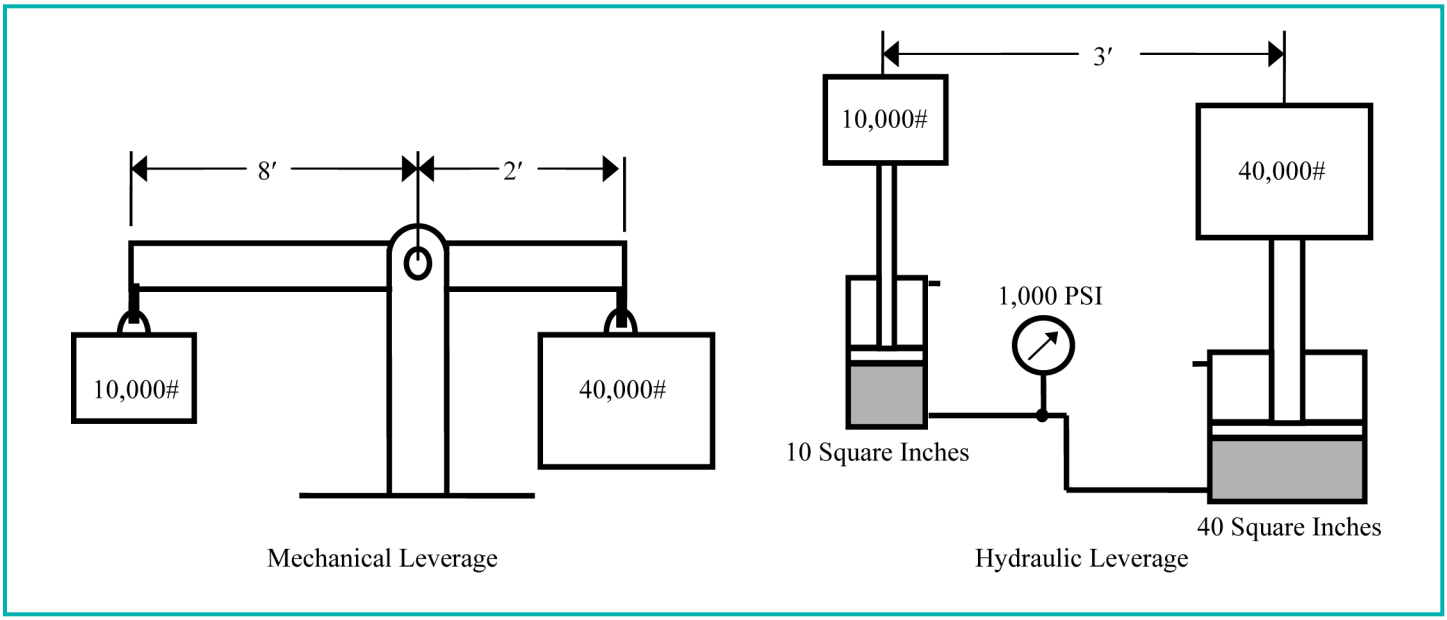


Fig. 1-4: Comparison of mechanical and hydraulic leverage

mechanisms because most designs using fluid power take less space and are not restricted by position considerations. A cylinder, rotary actuator, or fluid motor with almost limitless force or torque can directly push or rotate the machine member. These actions only require flow lines to and from the actuator and feedback devices to indicate position. The main advantage of linkage actuation is precision positioning and the ability to control without feedback

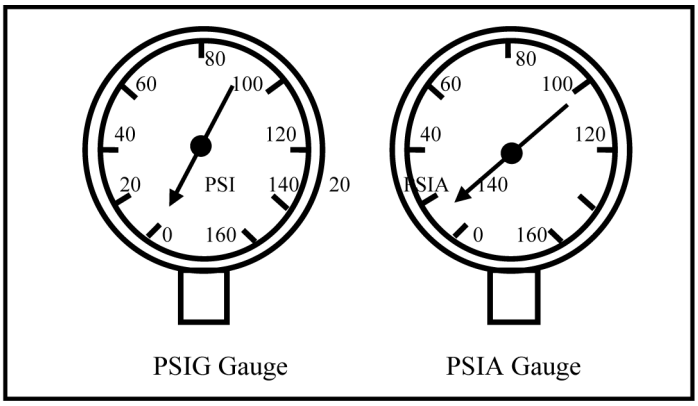


Fig: 1-5: Measurement of gauge and absolute pressure

At first look, it may appear that mechanical or hydraulic leverage is capable of saving energy. For example: 40,000 lb is held in place by 10,000 lb in Figure 1-4. However, notice that the ratio of the lever arms and the piston areas is 4:1. This means by adding extra force say to the 10,000-lb side, it lowers and the 40,000-lb side rises. When the 10,000-lb weight moves down a distance of 10 in., the 40,000-lb weight only moves up 2.5 in.

WORK AND POWER

Work is the measure of a force traversing through a distance. (Work = Force × Distance.). Work usually is expressed in lb-ft and, as the formula states, it is the product of force in pounds times distance in feet. When a cylinder lifts a 20,000-lb load a distance of 10 ft, the cylinder performs 200,000 ft-lb of work. This action could happen in three seconds, three minutes, or three hours without changing the amount of work.

When work is done with in a certain time, it is called *power*. {Power = (Force × Distance) ÷ Time.} A common measure of power is horsepower — a term taken from early days when most persons could relate to a horse's strength. This allowed the average person to evaluate to new means of power, such as the steam engine. Power is the rate of doing work. One horsepower is defined as the weight in pounds (force) a horse could lift one foot (distance) in one second (time). For the average horse this turned out to be 550 lb. one foot in one second. Changing the time to 60 seconds (one minute), it is normally stated as 33,000 ft-lb per minute.

COMPRESSIBILITY OF FLUIDS

No consideration for compressibility is necessary in most hydraulic circuits because oil can only be compressed a very small amount. Normally, liquids are considered to be incompressible, but almost all hydraulic systems have some air trapped in them. The air bubbles are so small even people with good eyesight cannot see them, but these bubbles allow for compressibility of approximately 0.5% per 1,000 psi. Applications where this small amount of compressibility does have an adverse effect include: single-stroke air-oil intensifiers; systems that operate at very high cycle rates; servo systems that maintain close-tolerance positioning or pressures; and circuits that contain large volumes of fluid. In this book, when presenting circuits where compressibility is a factor, it will be pointed out — along with ways to reduce or allow for it.

Another situation that makes it appear there is more compressibility than stated previously is if pipes, hoses, and cylinder tubes expand when pressurized. This requires more fluid volume to build pressure and perform the desired work. In addition, when cylinders push against a load, the machine members resisting this force may stretch, again making it necessary for more fluid to enter the cylinder before the cycle can finish.

As anyone knows, gasses are very compressible. Some applications use this feature. In most fluid power circuits, compressibility is not advantageous; in many, it is a disadvantage. This means it is best to eliminate any trapped air in a hydraulic circuit to allow faster cycle times and to make the system more rigid.

BOYLE'S LAW

Boyle's Law for gasses states that for relatively low pressures, the absolute pressure of an ideal gas kept at constant temperature varies inversely with the volume of the gas. This means if a ten cubic foot volume of atmospheric air is squeezed into a one cubic foot container, pressure increases ten times. (10 × 14.7 psia = 147 psia.) Notice that pressure is stated as psia.

Normally, pressure gauges read in psi (with no additional letter). Commonly called *gauge pressure*, psi disregards the earth's atmospheric pressure of 14.7 psia, because it has no effect — either negative or positive — on a fluid power circuit. The a on the end of psia stands for absolute, and would be shown on a gauge with a pointer that never goes to zero unless it is measuring vacuum. Another type of gauge that shows both negative and positive pressures would have a pointer with an inches-of-mercury (in. Hg) scale below zero and a psig scale above zero. Both of these gauges could read pressure or vacuum. (They are always found in a refrigeration repairperson's tool kit. Refrigeration units have both vacuum and pressure in different sections of the system at the same time.) Figure 1-5 illustrates a typical psig gauge and one type of psia gauge.

In the example above, when ten cubic feet of air was squeezed into a one cubic-foot space, both pressures were given in psia. To see what gauge pressure (psig) would be, subtract one atmosphere from the 147-psia reading. (147 psia – 14.7 psia = 132.3 psig.) To calculate the amount of compression of air in a system, always use absolute pressure, or psia, not psig.

For example: the cylinder in Figure 1-6 contains eight cubic feet of air at 70 psig. To what will pressure increase when an external force pushes the piston back until the space behind the piston is two cubic foot? It is obvious the pressure will rise four times. At first it might look easy to take 70 psig × 4 = 280 psig, but this answer is wrong. For the correct answer, gauge pressure must be changed to absolute

pressure. In this case, we must add one atmosphere to the 70-psig reading. (70 psig + 14.7 psia = 84.7 psia.) Now multiply the 84.7-psia pressure by 4 to see what the absolute pressure is when the cylinder stops at one cubic foot volume. (84.7 × 4 = 338.8 psia.) Finally, to return to gauge pressure, subtract one atmosphere from the absolute pressure. (338.8 psia – 14.7 psia = 324.1 psig.) Notice that the correct pressure is 44.1 psig higher than when gauge pressure is the multiplier.

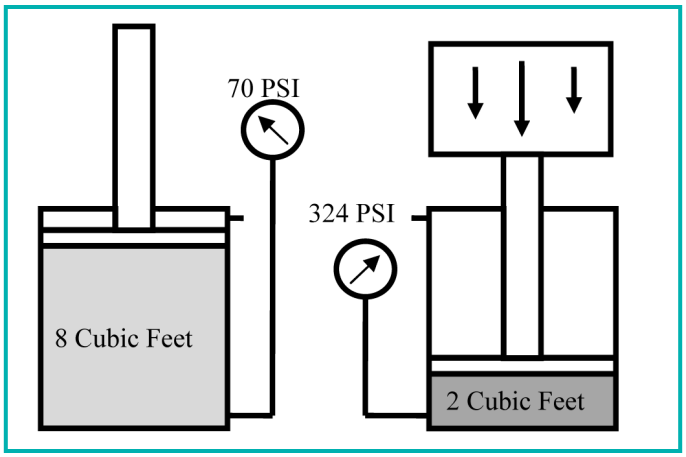


Fig: 1-6: Pressure change as air is compressed

Temperature was not considered in either of the preceding cases, but notice that the law says "kept at constant temperature." Compressing a gas increases its temperature because the heat in the larger volume is now packed into a smaller space. The next law says that increasing temperature increases pressure if the gas cannot expand. This means the pressures given are measured after the gas temperature returns to what it was originally.

Gauges today read in psi and bar. Bar is a metric, or SI, unit for pressure and is equal to approximately the barometer reading or one atmosphere. One atmosphere is actually 14.696 psi, but the SI unit for bar is 14.5 psi.

CHARLES' LAW

Heating a gas or liquid causes it to expand. Continuing to heat a liquid will result in it changing to the gaseous state and perhaps spontaneous combustion. If the gas or liquid cannot expand because it is confined pressure in the contained area increases. This is stated in Charles' Law: "The volume of a fixed mass of gas varies directly with absolute temperature, provided the pressure remains constant." Because fluid power systems have some areas in which fluid is trapped, it is possible that heating this confined fluid could result in part damage or an explosion. If a circuit must operate in a hot atmosphere, provide over pressure protection such as a relief valve or a heat- or pressure-sensitive rupture device. Never heat or weld on any fluid-power components without proper preparation of the unit.

STATIC HEAD PRESSURE

The weight of a fluid in a container exerts pressure on the containing vessel's sides and bottom. This is called *static head pressure*. It is caused by earth's gravitational pull. A good example of head pressure is a community water system. Figure 1-7 shows a water tower with a topmost water level of 80 ft. A cubic inch of water weighs 0.0361 lb. Therefore, a one square-inch column of water will exert a force of 0.0361 psi for every inch of elevation. This works out to 0.433 psi per foot of elevation.

For the water tower in Figure 1-7, the pressure at the base would be: 80 ft × 0.433 psi/ft = 34.6 psi. This pressure is always available, even when no pump is running. Of course, if the water level drops, static head pressure also will drop.

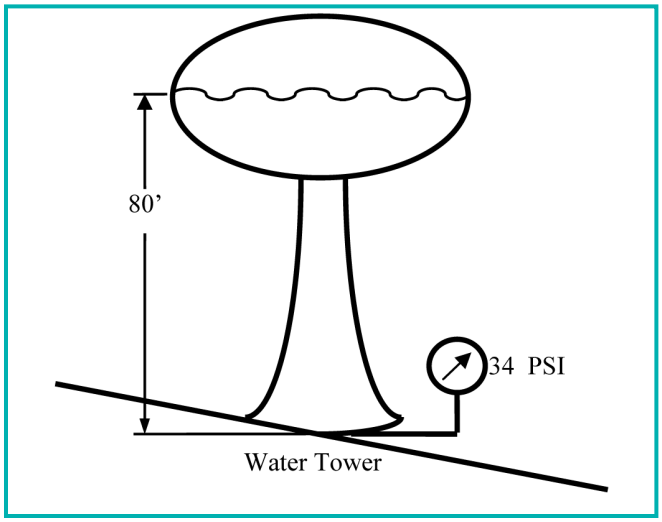


Fig: 1-7:Pressure measurement for water tower

SEE FIG 1-7

The specific gravity of hydraulic oil is approximately 0.9, so multiplying water's 0.433 psi/ft by 0.9 shows oil exerts 0.39 psi/ft of elevation. This fraction is usually rounded to 0.4 for simplicity. If the water tower were filled to 80 ft with oil, it would exert a pressure of 32 psi at ground level. Other fluids would develop a higher or lower static pressure according to their specific gravity.

This pressure is only realized at ground level at the tower. Outlets at other levels would be higher or lower according to their distance below the fluid surface.

Tanks seen on most water towers simply store volume. Pressure does not drop rapidly or require frequent pump starts to maintain the fluid level. The size or shape of the tank does not affect pressure at the base. Pressure at the base of a straight 80-ft pipe would be the same, but useful volume before pressure drop would change drastically. Always remember: it is not the physical size of a body of fluid that determines pressure but how deep it is.

Head pressure can have an adverse effect on a hydraulic system because many pumps are installed above the fluid level. This means the pump must first create enough vacuum to raise the fluid and then create even higher vacuum to accelerate and move it. Therefore there is a limit to how far a pump can be located above the oil level. Most pumps specify a maximum suction pressure of 3 psi. At 4- to 5-psi suction pressure, pumps start to cavitate . . . causing internal damage. At 6- to 7-psi vacuum, cavitation damage is severe and noise increases noticeably. (The effects of cavitation are covered fully in Chapter 8, Fluid Power Pumps and Accessories.) Axial- or in-line-piston pumps are especially vulnerable to high inlet vacuum damage and should be positioned below the fluid level to produce a positive head pressure.

Many modern hydraulic systems place the pump next to the reservoir so the fluid level is

always above the pump inlet. With this type of installation the pump always has oil at startup and has a positive head pressure at its inlet. A better arrangement puts the tank above the pump to take advantage of even greater head pressure. Everything possible should be done to keep pressure drop low in the pump inlet line because the highest possible pressure drop allowable is one atmosphere (14.7 psi at sea level).

VACUUM

The earth's atmosphere exerts a force of 14.7 psi at sea level on an average day. This pressure covers the whole earth's surface, but at elevations higher than sea level, it is reduced by approximately 0.5 psi per 1,000 ft. This pressure of earth's atmosphere is the source of the power of vacuum. The highest possible vacuum reading at any location is the weight of the air above it at that time.

A reading of maximum vacuum available is given during the local weather forecast as the barometer reading. Divide the barometer reading by two to get the approximate atmospheric pressure in psi. This force could be directly measured if it were possible to isolate a one square-inch column of air one atmosphere tall at a sea level location. Because this is not possible, the method used to measure vacuum is demonstrated in Figure 1-8.

Submerge a clear tube with one closed end in a container of mercury and allow it to fill completely. (The tube must be more than 30-in. long for this example to work when mercury is the liquid.) After the mercury displaces all the air in the tube, carefully raise the tube's closed end, keeping the open end submerged so the mercury can't run out and be replaced by air. When the tube is positioned vertically, the liquid mercury level will lower to give the atmospheric pressure reading in inches of mercury (29.92-in. Hg at sea level). The mercury level will fluctuate from this point as high and low-pressure weather systems move past. If the tube had been 100-in. tall, the mercury level would still have dropped to whatever the atmospheric pressure was at its location. The reason the mercury does not all flow out is that atmospheric pressure holds it in.

This barometer could have been built using another liquid but the tube would have been longer because most other liquids have a much lower specific gravity than mercury's 13.546. Water, with a specific gravity of 1.0, would require a closed-end tube at least 33.8 ft long, while oil, with a specific gravity of approximately 0.9, would have to be even longer.

Vacuum pumps can be similar in design to air compressors. There are reciprocating-piston, diaphragm, rotary-screw, and lobed-rotor designs. (See air compressor types in Chapter 8, Fluid power pumps and accessory items.) Imagine hooking the inlet of an air compressor to a receiver tank and leaving the outlet open to atmosphere. As the pump runs, it evacuates air from the receiver and causes a negative pressure in it.

Vacuum pumps are an added expense and normally are only found in facilities that use a constant supply of negative pressure to operate machines or make products.

Vacuum generators that use plant compressed air as a power source are also available. These components have no moving parts but use plant air flowing through a venturi to produce a small supply of negative pressure. Figure 1-9 shows a simplified cutaway view of a venturi-type vacuum generator. The device consists of body A with compressed-air inlet B that passes air flow through venturi nozzle C. The air exhausts at a higher velocity to atmosphere through orifice D. As air at increasing velocity flows past opening E near the venturi nozzle, it creates a negative pressure and draws in atmospheric air through port F. Port F can connect to any external device that needs a vacuum source.

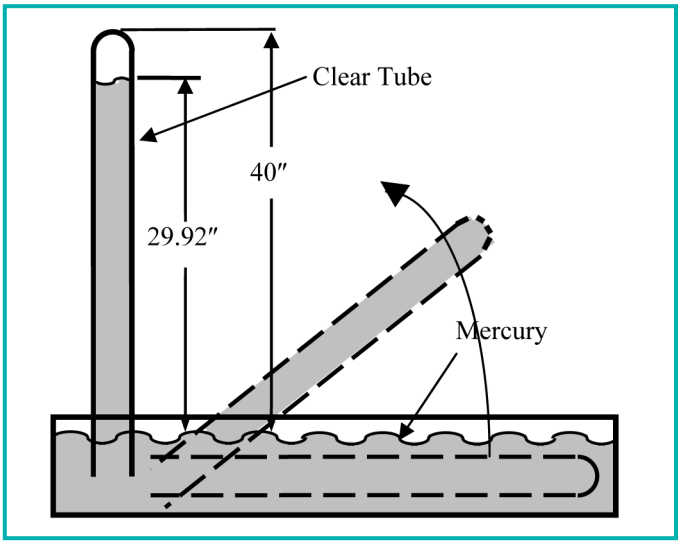


Fig: 1-8: Vacuum measurement with mercury

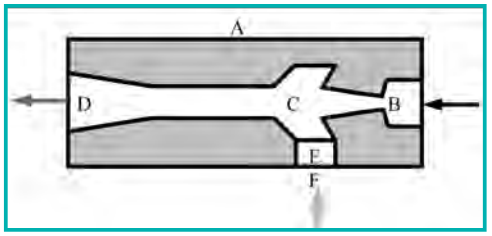


Fig: 1-9: Cross-sectional view of venutri vacccum generatior

A vacuum gauge at port F shows negative pressure when compressed air is supplied to port B.

Vacuum generators are inexpensive, but can be costly to operate. For every 4 cfm of air supply required to power them, they use approximately one compressor horsepower. For this reason, venturi-type vacuum generators usually are installed with a control valve to turn them on only when needed.

Vacuum is limited to one atmosphere maximum at any location, and standard vacuum pumps only reach about 85% (approximately 12 psi) of this on average. As a result, vacuum is not powerful enough to do much work unless it acts on a large area.

Many industrial vacuum applications have to do with handling parts. Large-area suction cups can lift a large heavy part with ease, as illustrated in Figure 1-10. When the lift rises, negative pressure (vacuum) inside the suction cups causes atmospheric pressure on the opposite side of the part to push it up.

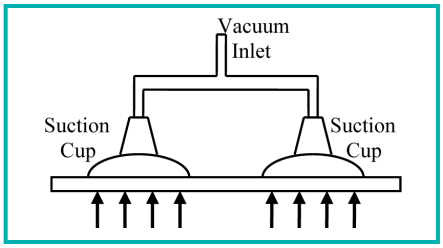


Fig 1-10: Simplified representation of lifting with vacuum

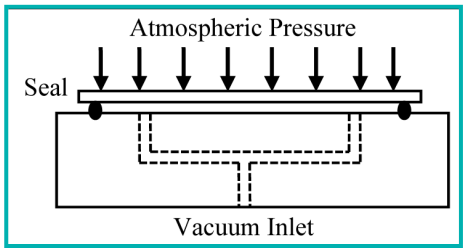


Fig 1-11: Simplified representation of work holding with vacuum

Industries such as glass and wood manufacturing use vacuum to hold work pieces during machining or other operations, as shown in Figure 1-11. The pieces are held firmly in place as the negative pressure under them causes atmospheric pressure to push against them. A resilient seal laid in a groove in the fixture keeps atmospheric air from entering the cavity beneath the part. This groove can be cut to match the contour of the part. In machining operations, the seals can isolate interior cutouts, allowing them to be removed while firmly holding the rest of the piece.

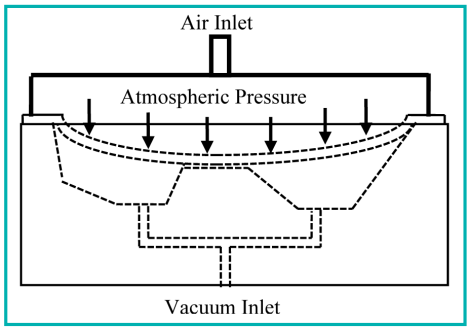


Fig 1-12: Simplified representation of plastic-sheet forming with vacuum

Heated plastic sheet can be vacuum-formed to make some products at a much lower cost than other types of plastic forming, as suggested in Figure 1-12. Forming heated plastic sheet in a cavity or over a shape is quick and positive. When atmospheric pressure tries to fill the negative-pressure area under the softened sheet, the sheet is forced into the desired shape. Large parts such as pickup-truck bed liners are formed by this method. ■

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QUIZ ON CHAPTER 1

1. Most hydraulic circuits are designed by:
A. mechanical engineers.
B. fluid power engineers.
C. fluid power distributor salesmen.

2. Atmospheric pressure at sea level is equal to approximately:
A. 14.7 Hg
B. 14.7 psi
C. 30 psi

3. Gauges that read pounds per square inch absolute (psia):
A. show atmospheric pressure on their scales.
B. do not show atmospheric pressure on their scales.
C. have nothing to do with atmospheric pressure.

4. Static head pressure of hydraulic oil is equal to approximately:
A. .1 psi per foot of elevation.
B. .2 psi per foot of elevation.
C. .4 psi per foot of elevation.

5. The power of vacuum comes from:
A. the power of a pump.
B. earth's atmosphere.
C. the size of a pump.

6. Vacuum generators use:
A. compressed air to create a vacuum.
B. an electric motor to create vacuum.
C. an air motor to create vacuum.

7. Pascal's Law states that pressure is produced by the pump and motor of a hydraulic system.
A. True
B. False

8. Pressure in a fluid power system comes from:
A. resistance to flow.
B. motor horsepower.
C. pump volume.

9. A pressure gauge reading of 100 bar would be equivalent to:
A. 2940 psi
B. 1070 psi
C. 1450 psi

10. When a gas or liquid is heated it will:
A. contract.
B. stay the same.
C. expand.

CHAPTER 2:

Hydraulic **Fluids**

For long service life, safety reasons, and reliable operation of hydraulic circuits, it is very important to use the correct fluid for the application. The most common fluid is based on mineral oil, but some systems require fire resistance because of their proximity to a heat source or other fire hazard. (Water is also making its return to some hydraulic systems because it is inexpensive, fireproof, and does not harm the environment.

TRANSMIT ENERGY

The main purpose of the fluid in any system is to transmit energy. Electric, internal combustion, steam powered, or other prime movers drive a pump that sends oil through lines to valves that control actuators. The fluid in these lines must transmit the prime movers energy to the actuator so it can perform work. The fluid must flow easily to reduce power losses and make the circuit respond quickly.

LUBRICATE

Hydraulic systems require fluid with good lubrication qualities. Pumps, motors, and cylinders need ample lubrication to make them efficient and extend their service life. Mineral oils with anti-wear additives work well and are available from most suppliers. Some fluids may need special considerations in component design to overcome their lack of lubricity.

SEAL

Fluid thickness can be important also because one of its requirements is for sealing. Almost all pumps and many valves have metal to metal sealing fits that have minimal clearance but can leak at elevated pressures. Thin watery fluid can flow through these clearances, reducing efficiency and eroding the mating surfaces. Thicker fluids keep leakage to a minimum and efficiency high.

There are several areas that apply to specifying fluids for a hydraulic circuit. *Viscosity* is the measure of a fluid's resistance to flow, sometimes thought of as thickness. Viscosity is specified by a SUS or SSU designation, similar to the SAE designation used for automotive fluids. SUS stands for Saybolt Universal Seconds (or as some put it, Saybolt Seconds Universal). It is a measuring system set up by a man named Saybolt. Simply stated, the system takes a sample of fluid, heats it to 100° F, and them measures how much fluid passes through a specific orifice in a certain number of seconds.

Viscosity is most important as it applies to pumps. Most manufacturers specify viscosity limits for their pumps and it is best to stay within the limits they suggest. The prime reason for specifying a maximum viscosity is that pressure drop in the pump suction line typically is low and if the oil is too thick, the pump will be damaged due to cavitation. A pump can move fluid of any viscosity if high enough pressure exists at its inlet. On the other hand, if fluids are too thin, pump bypass wastes energy and generates extra heat. All other components in the circuit could operate on any viscosity fluid because they only use what is fed to them. However, thicker fluids waste energy because they resist movement. Thin fluids waste energy because they allow too much bypass.

Viscosity index (VI) is a measure of a fluid's reistance to change in viscosity with temperature. It is common knowledge that heating any oil makes it thinner. A normal industrial hydraulic circuit runs at temperatures between 100° and 130° F. Cold starts could be as low as 40° to 50° F. Using an oil with a low VI might start well but wind up with excessive leakage and wear or cause cavitation damage at startup and run well at temperature. Most industrial hydraulic oils run in the 90- to 105-VI range and are satisfactory for most applications.

Pour point is the lowest temperature at which a fluid still flows. It should be at least lower than the lowest temperature to which the system will be exposed so the pump can always have some lubrication. Consider installing a reservoir heater and a circulation loop on circuits that start or operate below 60° F.

Refined mineral oil does not have enough lubricating qualities to meet the needs of modern day hydraulic systems. Several lubricity additives to enhance that property are added to mineral oil as a specific manufacturers package. These additives are formulated to work together and should not be mixed with others additives because some components may be incompatible.

Refined mineral oil also is very much affected by temperature change. In its raw state it not only has low lubricity but also would thin out noticeably with only a small increase in temperature. Viscosity modifiers enhance the oil's ability to remain at a workable viscosity through a broad temperature range.

OXIDATION

Any of several conditions can cause of oxidation of hydraulic oil. These include contamination, air, and heat. The interaction of these outside influences cause sludge and acids to form. *Oxidation inhibitors* slow or stop the fluid's degradation and allow it to perform as intended.

Wear inhibitors are additives that bond with metal parts inside a hydraulic system and leave a thin film that reduces metal-to-metal contact. When these additives are working, they extend part life by reducing wear.

In most hydraulic systems, fast and turbulent fluid flow can lead to the formation of small bubbles known as foam. *Anti-foaming* agents make the fluid less likely to form bubbles and allow those

that do form to dissipate more rapidly.

Moisture in the air can condense in a hydraulic reservoir and mix with the fluid. *Rust inhibitors* negate the effect of this unwanted water and protect the surfaces of the systems metal components. All of these additives are necessary to extend system life and improve reliability.

Overheating the fluid can counteract the additives and decrease system efficiency. Overheating also thins the oil and reduces efficiency because of internal bypassing. Clearances in pump and valve spools let fluid pass as pressure increases, causing more heating until the fluid breaks down. External leaks through fittings and seals also increase as fluid temperatures rise.

Another problem caused by overheating is a breakdown of some seal materials. Most rubber compounds are cured by controlled heat over a specific period of time. Continued heating inside the hydraulic system over long periods keeps the curing process going until the seals lose their resiliency and their ability to seal. It is best if hydraulic oil never exceeds 130° F for any extended period. Installing heat exchangers is the most common cure for overheating but designing heat out of a circuit is the better way.

Cold oil is not a problem as far as the oil is concerned but cooling does increase viscosity. When viscosity gets too high, it can cause a pump to cavitate and damage itself internally. Thermostatically controlled reservoir heaters easily eliminate this problem in most cases.

FIRE-RESISTANT FLUIDS

Certain applications must operate near a heat source with elevated temperatures or even open flames or electrical heating units. Mineral oil is very flammable. It not only catches fire easily but will continue to burn even after removing the heat source. This fire hazard situation can be eliminated by several different choices of fluids. These fluids are not fireproof, only fire-resistant, which means they will burn if heated past a certain temperature but they will not continue to burn after removing them from the heat source.

Generally, the fire-resistant fluids do not have the same specifications as mineral oil-based fluids. Pumps often must be down rated because the fluids lubricity or specific gravity is different and would shorten the pumps service life drastically at elevated pressures or high rotary speeds. Some fire-resistant fluids are not compatible with standard seal materials so seals must be changed. Always check with the pump manufacturer and fluid supplier before using or changing to a fire-resistant fluid.

WATER

Originally, hydraulic circuits used water to transmit energy (hence the word hydraulics). The main problem with water hydraulics is low viscosity. Therefore, water-based hydraulic systems generally operate a reduced pressure (500 to 600 psi) or require very expensive pumps and valves to prevent excessive internal leakage. However, water's low viscosity means it flows with less resistance than oil does, so potentially, it can operate at higher efficiency.

ISeveral companies have developed reliable pumps and valves for water that operate at pressure to 2,000 psi. There are still limitations (such as freezing) to using water, but in certain applications it has many benefits. One big advantage is that there are fewer environmental problems during operation or in disposing of the fluid. Price also is a factor because water costs so little and is readily available almost anywhere.

HIGH WATER-CONTENT FLUIDS

Some types of manufacturing still use water as a base and add some soluble oil for lubrication. This type of fluid is known as high water-content fluid (HWCF). The common mixture is 95% water and 5% soluble oil. This mixture takes care of most of the lubricity problems but does not address low viscosity concerns. Therefore, systems using HWCF still need expensive pumps and valves to make them efficient and extend their life.

Rolling mills and other applications with molten metals are one area where HWCF is prevalent. Often the soluble oil is the same compound used for coolant in the metal-rolling process. This eliminates concerns about cross-contamination of fluids and the problems it can cause.

WATER-IN-OIL EMULSIONS

Some systems use around 40% water for fire resistance and 60% oil for lubrication and viscosity considerations. Again, these are not common fluids because they require special oil and continuous maintenance to keep them mixed well and their ratio within limits. Most manufacturers do not want the problems associated with water-in-oil emulsions so their use is very limited.

WATER GLYCOL

Water glycol is a common fire-resistant fluid. This fluid uses water for fire resistance and a product like ethylene glycol (permanent anti-freeze) for lubricity, along with thickeners to enhance viscosity. Ethylene glycol will burn, but the energy it takes to vaporize the water present quickly quells the fire once it leaves the heat source. This means a fire would not spread to other parts of the plant. Always remember fire-resistant not fire-proof.

Water glycol fluids are heavier than mineral oil and do not have its lubricating qualities, so most pump manufacturers specify reduced rpm and lower operating pressures for water glycol. In addition, the water in this fluid can evaporate, especially at elevated temperatures, so it must be tested regularly for the correct mixture.

Cost is also a consideration. Water glycol is more expensive than oil and requires most of the same considerations when disposing of it.

Always check with the pump manufacturer before specifying water glycol fluid to see what changes are necessary to run the pump with this fluid. Seal compatibility is usually not a problem, but always check each manufacturers specifications before implementing this fluid. In addition, it is imperative to completely flush a system of any other fluids before refilling with water glycol.

SYNTHETICS

The other main fire-resistant fluids are synthetic types. They are made from mineral oil, but have been processed and contain additives to obtain a much higher flash point. It takes more heat to start them burning but there is not enough volatile materials in them to sustain burning. These fluids may catch fire from a pot of hot metal but quickly self-extinguish after leaving the heat source.

Synthetic fluids retain most of the

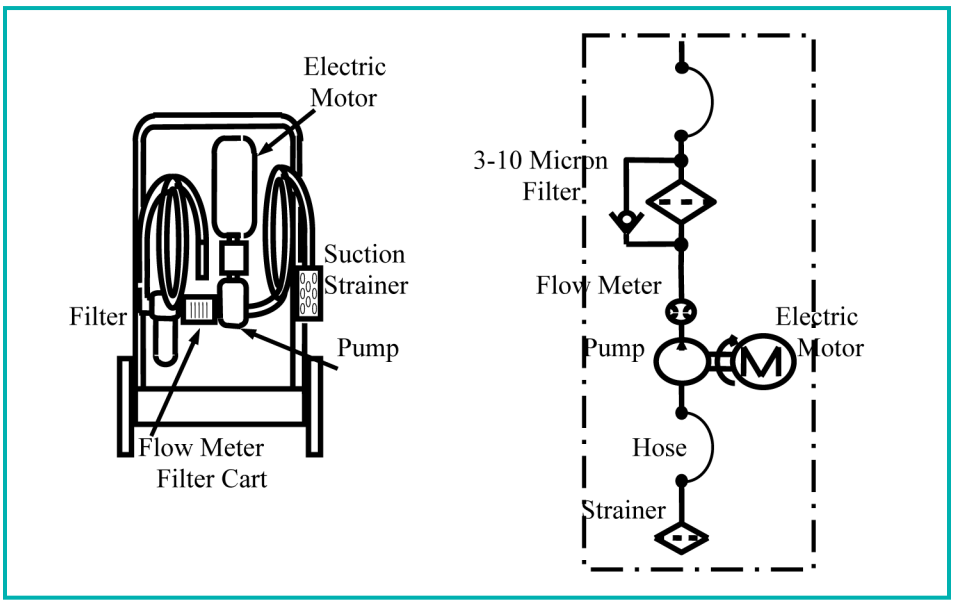


Fig 2-1: Filter cart (used to transfer hydraulic fluids) and its circuit schematic diagram.

qualities of the mineral oil from which they are derived, so most hydraulic components specify no operating restrictions. However, most of these fluids are not compatible with common seal materials so seal specification changes are usually necessary. Special consideration must be given to handling of synthetics because they can cause skin irritation and other health hazards. Also most synthetic fluids require protective epoxy paint for all components in contact with them.

Of all the fluids discussed, synthetics are the most expensive. They can cost up to five times more than mineral oil. But no matter which fluid is chosen, design the circuit to work in a reasonable temperature range; install good filters and maintain them; and check the fluids regularly to see if they are within specification limits.

A good operating temperature range is between 70° and 130° F with the optimum being around 110° F. A rule of thumb would be: warm enough to feel hot to the touch but cool enough to hold tightly for an extended period. Overheating hydraulic fluids is second only to contamination when it comes to reasons for fluid failure.

Continuous filtration of any hydraulic system is necessary for long component life. Fluids seldom wear out but they can become so contaminated that the parts they drive can fail. (The filter section of this book offers some good recommendations on keeping fluid clean.)

Even with the best of care, any hydraulic fluid should be checked at least twice a year. Systems located in dirty atmospheres may need to be checked more often to see if a pattern exists that requires special consideration. Pay close attention to the sampling process and packaging procedures recommended by the test facility that will process the sample. Expect a report on the level of contamination plus an analysis of the additive contents, water content, ferrous and non-ferrous material amounts, and any other problem areas the test facility finds. Use this information to know when to change fluids and to check for abnormal part wear problems.

New oil or other fluids from the supplier are not necessarily clean. The fluids are shipped in drums or by bulk, and there is no way of knowing how clean these containers are. Some suppliers offer filtered oil with a guaranteed contamination level at added cost. Otherwise, about the lowest level of contamination from most manufacturers is 25 microns.

Anytime a system needs new fluid, it is best to use a transfer unit, Figure 2-1, with a 10-micron or finer filter in the loop. Another way of filtering new or refill fluid is with a filter permanently attached to the reservoir, Figure 2-2. In this arrangement, the breather or other possible fill points should be made inaccessible.

The filter cart shown in Figure 2-1 can also be used to filter any hydraulic unit in the plant. Instead of this filter unit sitting idle except when filling systems, set it up at a machines power unit for a timed run. Place the suction hose in one end of the reservoir and the return hose in the opposite end. This adds a continual filtration loop to any machine even when the machines main pump is shut off. Run the cart until the fluid is clean and then move is to another power unit. Repeating this process on a regular schedule can assist the hydraulic units filters and add extra life to the fluid and the hydraulic components. This process may also show a pattern on machines that have a contamination problem.

Hydraulic fluids should be stored in a clean dry atmosphere. Keep all containers closed tightly and reinstall covers on any partially used drums.

Never mix fluids in any hydraulic system. Make sure all containers are clearly marked and segregated so fluids will not be mixed with one another. Mixing fluids can result in damage to components and some combinations are very difficult to clean up. Be especially careful when mineral oils and synthetic or water-glycol fluids are used in different parts of the same plant.

Fluids are the lifeblood of any hydraulic system and should be selected using the utmost care. ■

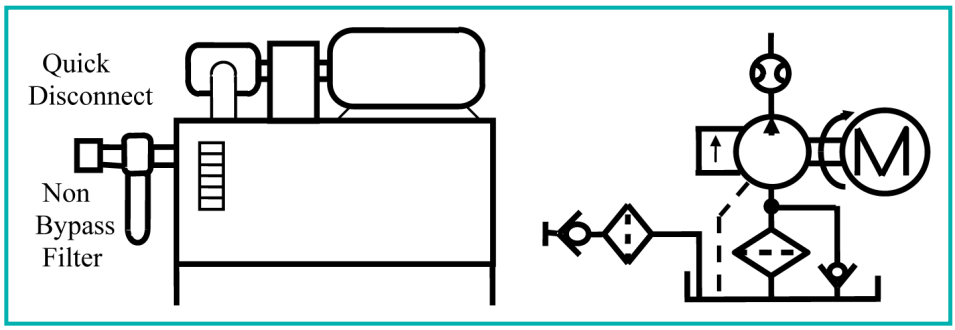


Fig 2-2: Hydraulic power unit and circuit diagram of its filter arrangement

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QUIZ ON CHAPTER 2: HYDRAULIC FLUIDS

1. Hydraulic oil is compressible:
A. not at all
B. by a great amount
C. by a small amount

2. Viscosity is a measure of the hydraulic fluids:
A. color
B. temperature
C. thickness

3. Synthetic fluids:
A. cost more than mineral based fluids
B. will not burn
C. are compatible with most seal materials

4. The main purpose for hydraulic fluids is:

A. to fill the tank
B. to leak from every opening
C. transmit energy, lubricate and seal

5. Over heating hydraulic fluid:
A. makes it last longer
B. deteriorates the additives
C. makes it smell better

6. New hydraulic oil is clean to:
A. 5 µm
B. 10 µm
C. 25 µm

7. Viscosity Index is a measurement of the change of flow characteristics of a fluid from one temperature another.
A. True
B. False

8. Hydraulic fluid should be stored:
A. in a damp moist atmosphere
B. in a closed and sealed container
C. in an open container

9. A high viscosity number gives:
A. better sealing characteristics
B. poorer sealing characteristics
C. has no effect on sealing characteristics

10. Refined mineral oil has:
A. excellent lubricating qualities
B. good lubricating qualities
C. poor lubricating qualities

Notes:

CHAPTER 3:

System and **machine plumbing**

Poor plumbing practices can permanently cripple a fluid power circuit even if it was designed with the best engineering practices and assembled with the most up-to-date components. Undersized lines, elbows instead of bends, incorrect component placement, and long piping runs are a few of the items that strangle fluid flow.

Other problems, such as using tapered pipe threads or lines with thin walls, can make a circuit a maintenance nightmare that requires daily attention. Fortunately, there are numerous publications that assist in specifying correct line size and conductor thickness to give low pressure drop and safe working-pressure limits.

SIZING PNEUMATIC LINES

Because pneumatic circuits generally are less complicated and operate at lower pressures, they are not as vulnerable to plumbing problems. One important aspect often overlooked is the length and size of lines between the valves and actuators. Piping between the valve and actuator should be as short as possible and of the minimum diameter to carry the required flow. The reason for this is that all the air in the pipes between the actuator and valve is wasted every cycle. These runs must be filled to make the device move but the air it takes to fill them does no work. During each cycle, air in the actuator lines exhausts to atmosphere without helping cycle time or force. For this reason, always mount the valve close to the actuator ports.

Another aspect of plumbing a pneumatic system is the in-plant pipe installation procedure. To get the required amount of compressed air to the point of usage requires some planning — or the site may be starved at times.

Pipe materials and size: Air systems are normally plumbed with Schedule 40 black iron pipe. (Galvanized pipe is not recommended because some galvanizing material may flake off and get into moving parts.) Several other available plumbing materials could be used for air piping because pressure is relatively low. Some mechanics recommend plastic pipe, but be aware a few synthetic compressor lubricants attack plastic and cause it to lose strength. This type of damage weakens the plastic until it can burst, sending shards of plastic flying everywhere in the plant. Never use any piping material not specifically designated by code.

To help select pipe size, the chart in Figure 3-1 shows flow (in cfm) down the left-hand side, length of run (in feet) across the top, and minimum Schedule 40 pipe size in the body at the intersection of these two.

This chart is based on a 1-psi pressure drop for the run lengths given. The right-hand column shows approximate compressor horsepower for the flow figures on the left. Using larger than specified pipe is of little help in reducing pressure drop, but provides more storage volume to handle short brief-flow needs. This chart does not consider fittings and valves, but they also must be considered when figuring the length of a run. Add 5 to 7 ft of pipe length for each fitting or valve — to be on the safe side.

Not having enough air to run the equipment is expensive, so never try to save a few cents at installation by skimping on pipe size. One or two pipe sizes over minimum add little to cost up front, but can make a big difference later. It is less expensive to run oversize pipe initially than to have to add a line later.

There are three basic compressed-air

	Length Of Pipe Run In Feet													
CFM	50’	100’	150’	200’	250’	300’	350’	400’	450’	500’	550’	600’	650’	700’	HP
10	¾”	¾”	1”	1”	1”	1”	1”	1”	1”	1”	1”	1”	1 ¼”	1 ¼”	2 ½
25	1”	. 1”	1 ¼”	1 ¼”	1 ¼”	1 ¼’	1 ¼”	1 ½’	1 ½”	1 ½”	1 ½”	1 ½’	1 ½”	1 ½”	6
50	1 ¼”	1 ½”	1 ½”	1 ½”	2”	2”	2”	2”	2”	2”	2”	2’	2”	2”	12
100	1 ½”	2”	2”	2”	2 ½”	2 ½”	2 ½”	2 ½”	2 ½”	2 ½”	3”	3”	3”	3”	25
200	2”	2 ½’	2 ½”	3”	3”	3”	3”	3”	3 ½”	3 ½”	3 ½”	3 ½”	3 ½”	3 ½”	50
300	2 ½”	3”	3”	3 ½”	3 ½”	3 ½”	3 ½”	3 ½”	4”	4”	4”	4”	4”	4”	75
400	3’	3”	3 ½”	3 ½”	4”	4”	4”	4”	4”	5”	5”	5”	5”	5”	100
600	3 ½”	3 ½”	4”	4”	5”	5”	5”	5”	5”	5”	5”	5”	6”	6”	150
800	3 ½”	4”	5”	5”	5”	5”	6”	6”	6”	6”	6”	6”	6”	6”	200

Fig. 3-1. Pipe size selection chart (in feet) for plant-air systems

piping layouts that meet the requirements of most industrial plants. Some facilities may have two or more of these systems to handle special needs. In general, smaller plants use a modified grid system, especially when the facility is growing. A unit distribution system offers flexibility, but can be expensive up front. A loop system is best suited to new construction; it provides extra storage capacity and dual supply for short bursts of high flow.

Figure 3-2 shows a typical grid-system layout using a centrally located air compressor. All air from the receiver goes to a large header pipe that runs down the center of the plant or department. Branch lines from the header go to separate areas where working drops come down to specific machines. With preplanning for future working drops, this arrangement is very flexible.

Figure 3-3 shows a typical loop piping system for compressed air. Again, the compressor and receiver are at a central location. The oversized loop around the periphery of the plant — or department — adds storage and allows flow with low pressure drop. It also allows for short bursts of high-volume flow to any section because flow in the loop is bi-directional. (Another way to get short high-volume flows with any of these piping systems is to install extra receiver tanks at or near areas that need such flow.)

Figure 3-4 illustrates a unit distribution layout that works well in plants that run departments on different days or shifts — or plants that started out small and added compressors as business grew. It is the most expensive configuration of the three for a new installation, so is not often used there. One advantage of the multiple compressors is that there is always backup air available for critical operations should a single compressor fail. The disadvantage—besides higher price—is that some compressors might be neglected by maintenance personnel because they are spread throughout a facility.

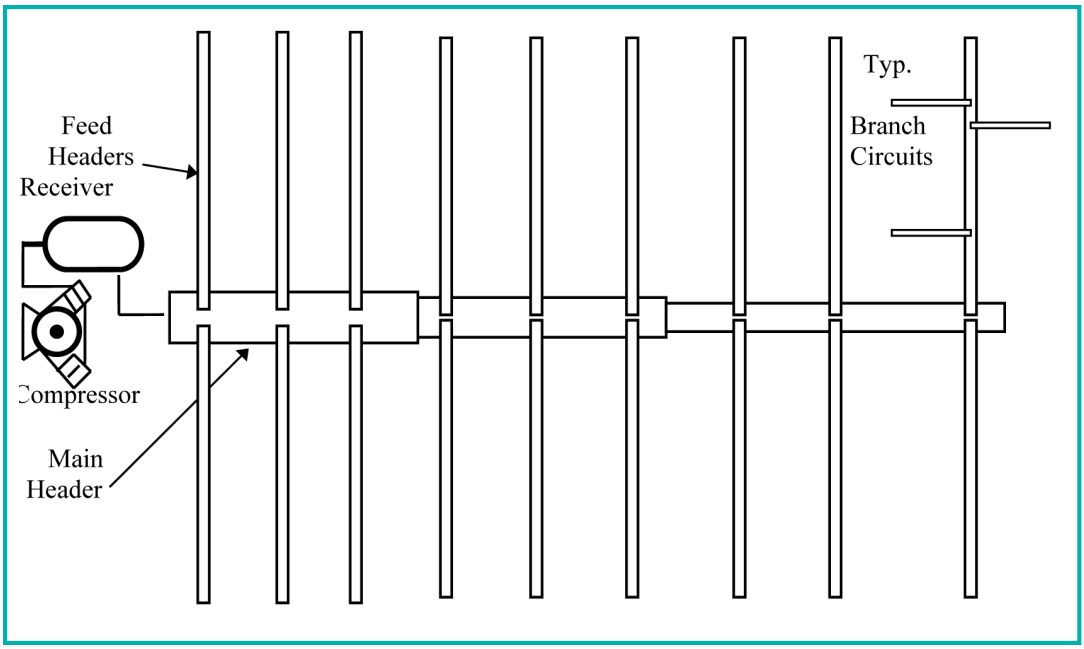


Fig. 3-2. Typical grid system layout for plant air

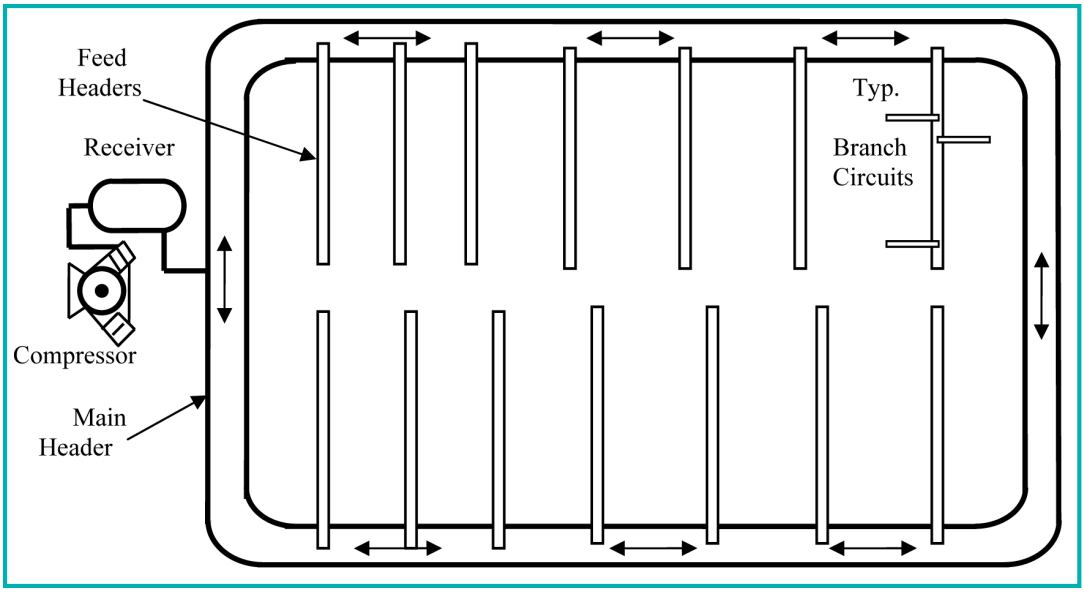


Fig. 3-3. Typical loop-piping system layout for plant air

Figure 3-5 shows a typical pipe run layout for optimum performance from a compressed air system. Strict attention to the details shown here ensures a smooth-operating and trouble-free air system.

Individual machines can be plumbed with any of the materials recommended for plant piping. However, because piping at the machine is usually much smaller, polyethylene, nylon, or vinyl tubing with push-to-connect fittings will work very well. Such tubing and fittings come in a variety of sizes (and colors) and require only a few tools to install. The push-to-connect fittings also release easily for troubleshooting checks or rework.

PIPE MATERIALS AND SIZES

Though many hydraulic circuits are plumbed using black-iron pipe with tapered

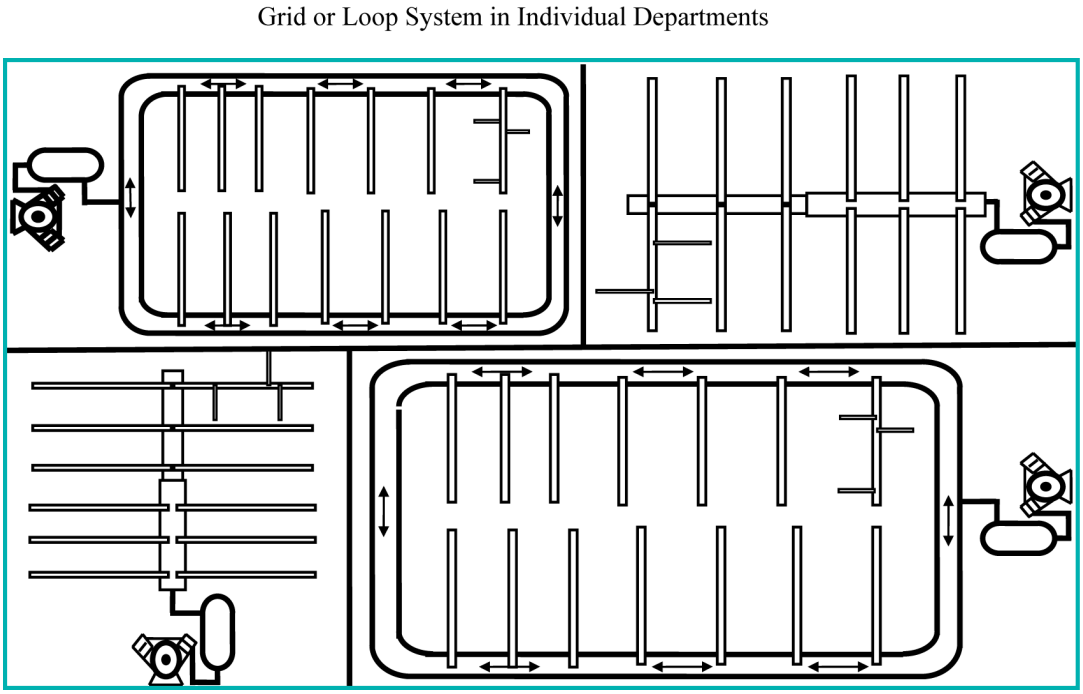


Fig. 3-4. Typical unit distribution layout for plant air

pipe threads, this is not the recommended way. It is nearly impossible to maintain leak-free operation of a 1,000- to 3,000-psi hydraulic system for any period of time with tapered pipe threads. Even if pipe-connection compound is used, expansion and shock soon loosen the taper interference and fluid weeps through the resulting openings. Another problem with tapered pipe threads occurs on circuits that must be routinely dismantled. Every time a tapered pipe thread is unscrewed, it must be tightened past where it was originally to get a good seal at reassembly. This can only happen so often before the pipe and/or valve must be replaced.

The recommended plumbing material is steel tubing with straight-thread O-ring fittings up to 2-in. outside diameter (OD), Figure 3-6. In sizes larger than 2 in., use steel pipe with welded SAE O-ring-sealed flange fittings on each end, Figure 3-7. For flexible connections, reinforced rubber hose is most common; however, some prefer sealed steel swivel joints.

A good reason for using steel tubing is that it is easily formed to allow for direction changes. Instead of installing fittings that can cause turbulence, use a tube bender to make sweeping turns that eliminate most of the pressure drop associated with elbows. This produces less pressure drop and less heat. Tubing is designated by its OD. As wall thickness increases, inside diameter (ID) decreases. (Black-iron pipe is measured by its nominal ID, but also has a decrease in ID with increase in wall thickness.)

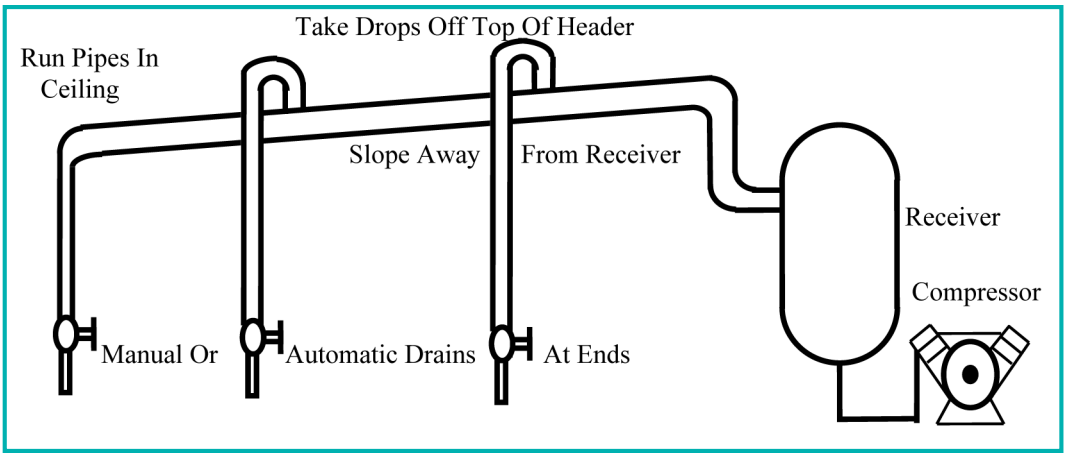


Fig. 3-5. Side view of typical compressed-air header and drop arrangement

HYDRAULIC HOSE

There are places on many machines where rigid pipe or tubing cannot be used because of their inflexibility. Rigid lines can cause problems at cylinders with pivot mountings, pumps on noise-isolation mounts, or connections between separate units. Hose avoids these problems.

However, wholesale use of hose in place of rigid lines it is not generally recommended. Hose is expensive, must be replaced on a regular basis, and flexes and stretches under pressure surges. This flexing produces extra volume and adds to cycle time. It is never recommended to use hose in a servo circuit (although there are times it can't be avoided). Servo circuits are for actuators that need precise control and flexing of hose lines can cause these valves to respond slowly and then go into high frequency oscillations.

Hose is specified by its ID and, unlike pipe and tube, this dimension does not change. Thicker walls for higher pressures make the OD of hose greater. Pressure is specified in working and burst values (similar to pipe). Working pressure should always be equal to or higher than maximum system operating pressure. Flow rates of hose are slightly higher than pipe and about the same as steel tubing due to hose's smooth bore. However, many of the end connectors for hose are restrictive because they always go inside the inner liner. These fittings are only short restrictions, but can raise pressure drop noticeably in some cases.

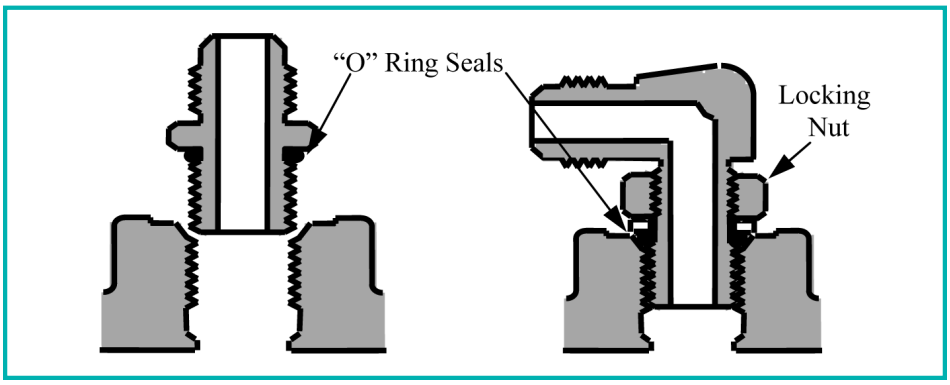


Fig. 3-6. O-ring sealed straight-thread fittings

Several factors influence hose service life and each one is controllable by some up-front fact finding and planning. First: never go under the manufacturers recommendation for minimum bend radius. Bending hose always causes stress but flexibility is the main reason for using it. Standard hose construction entails wire- or fiber-braided material laid down when the product is straight. Bending these braided materials puts extra stress on the outside of the bend and bunches up those on the inside. Add the constant expansion and contraction from pressure fluctuations during operation and it is easy to see the adverse effect.

Second: don't use hose above its rated working pressure. While maximum pressure might be set at or below the hose rating, higher shock pressures could be damaging during every cycle. Make sure the pressure rating of the hose on all machines is at or above operating pressure — and design out system shock to protect the hose and other hydraulic components.

Third: avoid operating at temperatures above the rating of the hose. Most hose manufacturers make hoses in different temperature ratings. Of course, the higher the temperature rating, the more expensive the hose is, but it is false economy to use the wrong hose to save a few pennies.

Fourth: don't install hose where it must twist during each cycle or make it operate in a twist because of poor tightening procedures. Always hold the hose straight while tightening a connection. Either case stresses the hose and causes premature failure and its accompanying extra expense.

Hose distributors know of these pitfalls and can help with installation suggestions, as well as troubleshooting hose problems. The causes of hose problems are usually quite evident to someone who works regularly with hose, even when all he or she sees is the damaged part.

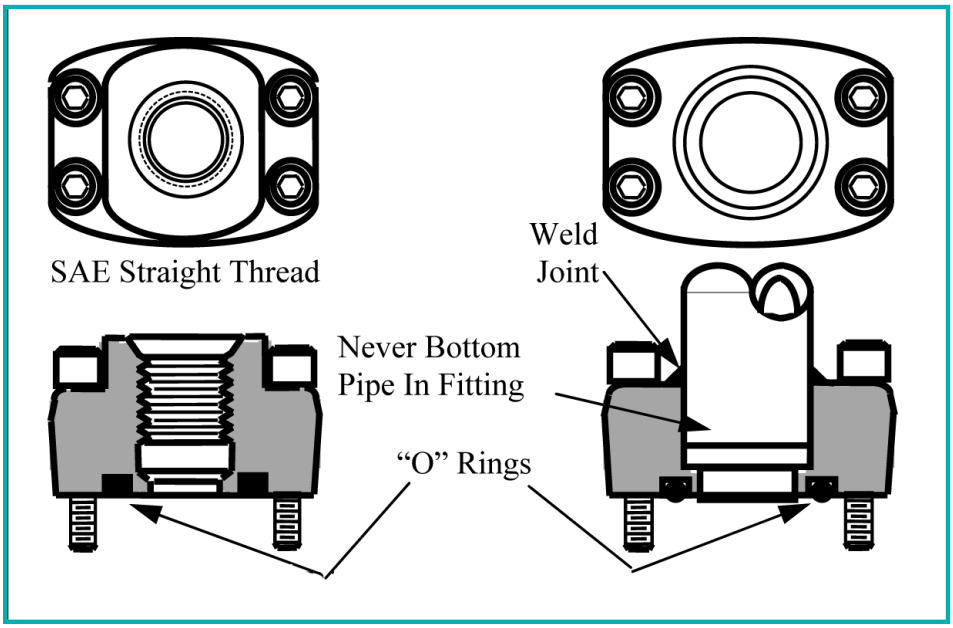


Fig. 3-7. Two types of O-ring sealed flange fittings

SIZING HYDRAULIC LINES

Fluid flow is measured in feet per second (fps), so the type of conduit is irrelevant. Many books have charts that relate gpm to fps for all standard piping systems. Use these charts to pick out the correct size fluid carrier for the required flow.

PUMP INLET LINE (SUCTION LINE)

Fluid velocity should not exceed 2 to 4 fps. The reason for this recommendation is that the highest possible pressure drop in the pump inlet line is one atmosphere. Actually, no type of hydraulic pump can even come close to this, so most inlet lines never see much more than 3- to 4-psi vacuum. Using velocity higher than 2 to 4 fps dramatically increases pressure — causing cavitation and pump damage. It is best to use a suction line equal to or larger than the size of the pump inlet being plumbed. There are circumstances when a smaller suction line is satisfactory, but only do this when absolutely necessary and with the supplier's approval.

The suction line should be full size; as straight as possible; have no or the minimum number of fittings; never include a standard pipe union; and be completely sealed. Using hose in place of pipe or tube can overcome many possible suction-line problems. Hose is a viable alternative and is quite satisfactory if certain precautions are addressed. Always use hose designed and specified for suction (vacuum) use.

Hose normally used for pressure may be rated at 3,000 psi but is not suited for suction lines. The reason for this is pressure hose uses an inner lining like a tube in a tube-type tire. The outer layers are strong but they are porous and would leak high-pressure fluid except for the inner tube. High-pressure hose as a suction line sees constant negative pressure trying to collapse the inner tube. After some time, it is possible for the inner liner to be drawn in, restricting flow and causing pump cavitation. This phenomenon may not happen immediately, but usually does cause problems in time.

RETURN LINES

Fluid velocity in return lines should be limited to between10 to 15 fps. The pump can push oil returning to tank, but any backpressure in these lines must be overcome by extra pressure at the pump outlet. To maintain a high-efficiency circuit, it is important to keep pressure drop in all lines as low as possible. All energy used to push oil through the lines is wasted and shows up as excessive heat.

WORKING-PRESSURE LINES

Medium pressure lines (up to 2,000 psi) should not exceed 15 to 20 fps. Flow in systems that operate above 2,000 psi can go as high as 30 fps. Unlike air systems, there is usually excess pressure capacity in hydraulic circuits when actuators are in motion. Typically, high pressure only comes into play when the actuators near the end of stroke. In an effort to keep line and valve sizes small, it is common practice to use these higher velocities — but keep in mind this practice wastes energy.

Several fluid power handbooks are available with excellent charts showing tubing and pipe in all different wall thickness, along with flow in gallons per minute (gpm) for all standard sizes. Remember each fitting or valve in the circuit has its own pressure-drop adders and they must be taken into consideration as part of the overall pressure-loss picture. ■

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QUIZ ON CHAPTER 3: PIPE, TUBE, AND HOSE

1. Which type of threads is recommended for plumbing hydraulic circuits?
A. SAE straight threads
B. NPT tapered pipe threads
C. BSP tapered pipe threads

2. What is the minimum size pipe to handle 400-cfm air flow for 500 feet?
A. 4 in.
B. 5 in.
C. 6 in.

3. What is the recommended fluid velocity for a suction line?
A. 2 to 4 fps
B. 10 to 15 fps
C. 30 fps

4. A grid system for plant air piping requires:
A. several small compressors
B. several large compressors
C. one adequately sized compressor

5. For hydraulic lines up to and including 2 in., use:
A. steel tubing
B. black-iron pipe
C. plastic pipe

6. Always take drop lines in an air piping circuit from the:
A. bottom of the header
B. side of the header
C. top of the header

7. Elbows are better than smooth tubing bends.
A. True
B. False

8. The pipes in an air plumbing system should be:
A. sloped away from the receiver tank
B. sloped toward the receiver tank
C. level

9. Hose size is designated by its:
A. ID
B. OD
C. ND

10. Velocity in hydraulic circuit return lines should be:
A. 2 to 4 fps.
B. 10 to 15 fps.
C. 15 to 20 fps.

QUIZ ANSWERS FOR CHAPTER 1

1. Most hydraulic circuits are designed by:
C. fluid power distributor salespeople.

2. Atmospheric pressure at sea level is equal to approximately:
B. 14.7 psi.

3. Gauges that read pounds per square inch absolute (psia):
A. show atmospheric pressure on their scale.

4. Static head pressure of hydraulic oil is equal to approximately:
C. 0.4 psi/ft elevation.

5. The power of vacuum comes from:
B. Earth's atmosphere.

6. Vacuum generators use:
A. compressed air to create a vacuum.

7. Pascal's Law states that pressure is produced by the pump and motor of a hydraulic system.
B. False

8. Pressure in a fluid power system comes from:
A. resistance to flow.

9. A pressure gauge reading of 100 bar would be:
C. 1450 psi.

10. When a gas or liquid is heated it will:
C. expand.

QUIZ ANSWERS FOR CHAPTER 2

1. Hydraulic oil is compressible:
C. by a small amount.

2. Viscosity is a measure of the hydraulic fluids:
C. thickness.

3. Synthetic fluids:
A. cost more than mineral based fluids.

4. The main purpose for hydraulic fluids is:
C. transmit energy, lubricate and seal.

5. Overheating hydraulic fluid:
B. deteriorates its additives.

6. New hydraulic oil is clean to:
C. 25 µm.

7. Viscosity Index is a measurement of the change of flow characteristics of a fluid from one temperature another.
A. True

8. Hydraulic fluid should be stored:
B. in a closed and sealed container.

9. A high viscosity number gives:
A. better sealing characteristics.

10. Refined mineral oil has:
C. poor lubricating qualities.

QUIZ ANSWERS FOR CHAPTER 3

1. Which type of threads are recommended for plumbing hydraulic circuits?
A. SAE straight threads.

2. What is the minimum size pipe to handle 400 cfm air flow 500 ft?
B. 5 in.

3. What is the recommended feet per second (ft/sec) velocity for a suction line?
A. 2 to 4 ft/sec

4. A grid system for plant air piping requires:
C. one adequately sized compressor.

5. For hydraulic lines up to 2-in. diameter, use:
A. steel tubing.

6. Always take drop lines in an air piping circuit from the:
C. top of the header.

7. Elbows are better than smooth tubing bends.
B. False

8. The pipes in an air plumbing system should be:
A. sloped away from the receiver tank.

9. Hose size is designated by its:
A. I.D.

10. Velocity in hydraulic circuits return lines should be:
B. 10 to 15 ft/sec.

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CHAPTER 4

ISO Symbols and Glossary

A family of graphic symbols has been developed to represent fluid power components and systems on schematic drawings. In the United States, the American National Standards Institute (ANSI) is responsible for symbol information. The Institute controls the make-up of symbols and makes changes or additions as required. The International Standards Organization (ISO) has the same control over symbols used internationally. Both systems have almost the same format (especially since ANSI changed its symbols in 1966 to eliminate all written information).

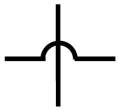
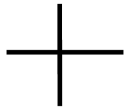
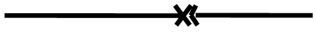
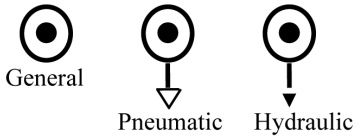
Standard symbols allow fluid power schematic diagrams to be read and understood by persons in many different countries, even when they don't speak the same language. Either symbol set (ANSI or ISO) may be -- and is -- used in the United States. However, many companies today use the ISO symbols as their standard for work with foreign suppliers and customers.

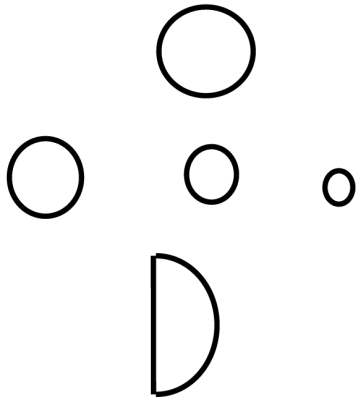
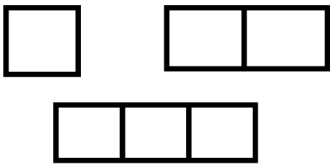
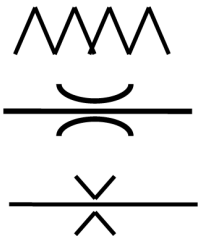
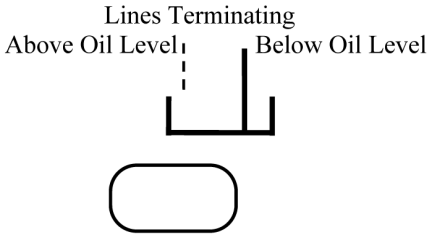
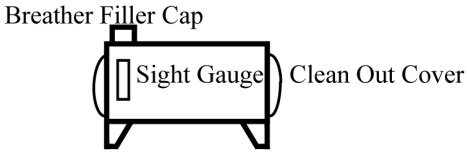
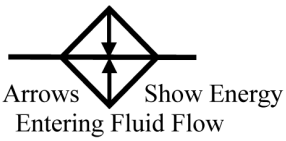
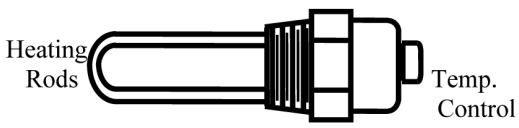
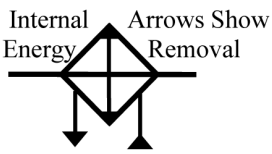
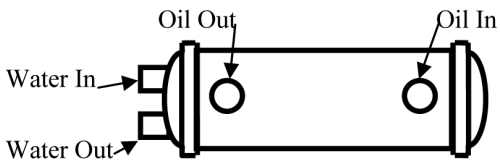
The following pages go through all standard ISO symbol information as it applies to hydraulic and pneumatic schematics. There are still many plants that modify the standards to suit some individual's taste. This widespread practice may be confusing to novices. Symbols have been developed to represent most

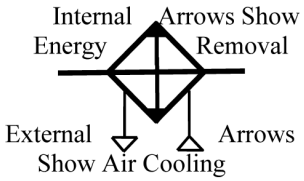
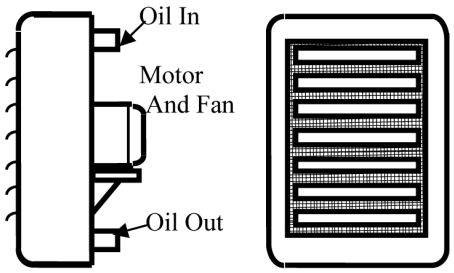
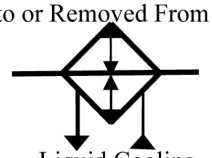
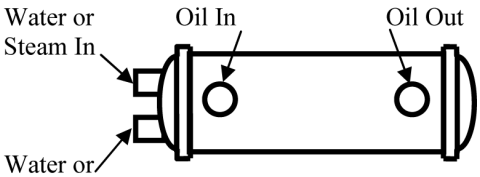
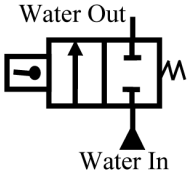
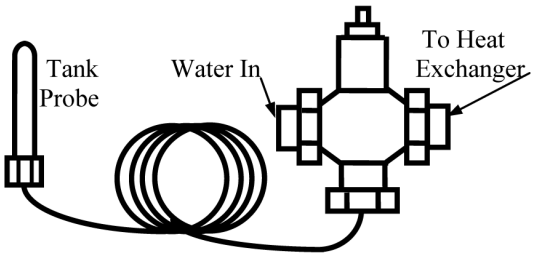
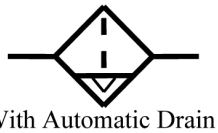
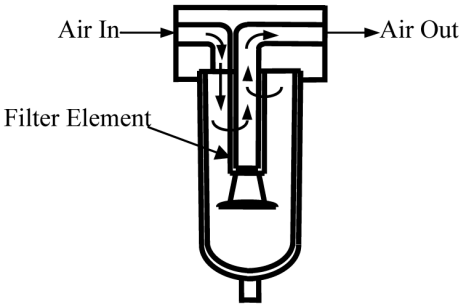
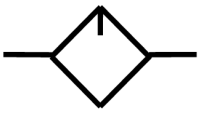
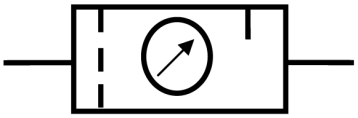
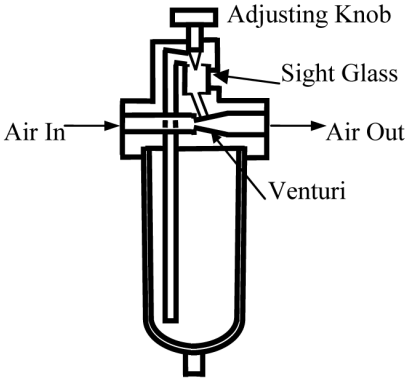
ISO Designation	Symbol	Picture Representation
Basic Information		
Lines		
Continuous		
Non-Flowing	Black	Represents a working fluid line. This fluid comes from a prime mover and goes to the actuator to perform work. May be a 1/8" plastic air line or any size pipe or tube in a hydraulic system.
Pump Flow	Red	
Tank Flow	Blue	
Suction Flow	Green	
Metered Flow	Yellow	
Reduced Pressure	Orange	COLOR CODING FOR OVERHEADS
Intensified Fluid	Purple	
Long Dashes	Orange	Represents pilot lines that supply a small amount fluid to another valve or device making it operate. The length of these dashes should be at least ten times their thickness .
Short Dashes	Green	Represents drain lines for hydraulic circuits. Many hydraulic valves have internal leakage that can get trapped and cause a malfunction .A drain line is a small line giving trapped fluid a free flow path to tank. The length of these dashes is five times their thickness.
Double Lines		Represents a mechanical connection Between components. A pump motor shaft, feedback connections between valves and actuators, etc. The outside dimension of these lines should be at least five times the line thickness.
Center Line		Represents an enclosure outline that indicates the parts inside it are a unit. This unit may be a casting with the parts machined in it or it might be a several components assembly.
Electric Line		Denotes a line carrying electrical power or signal.

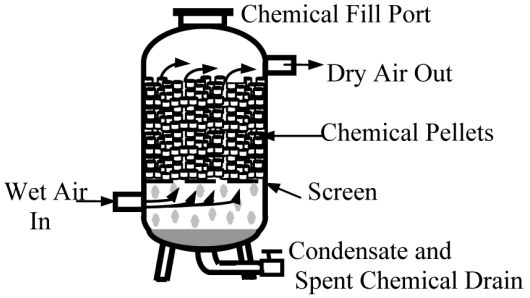
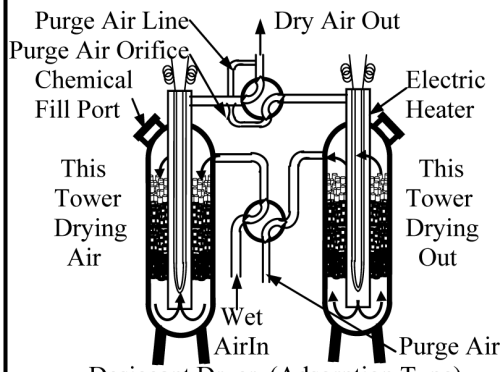
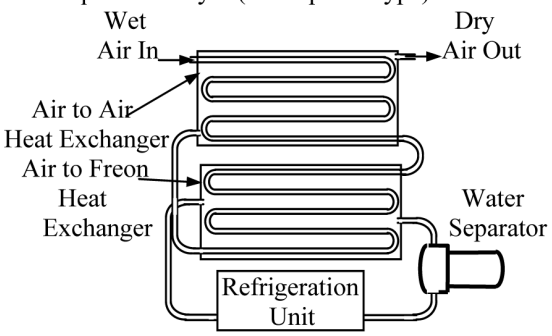
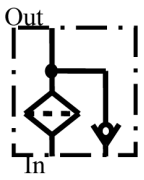
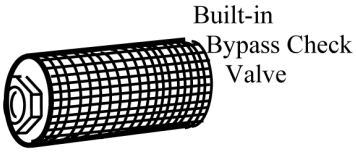
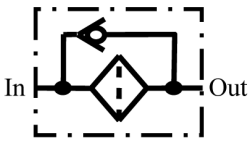
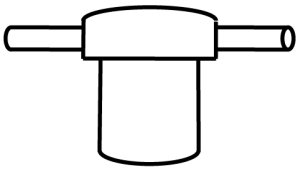
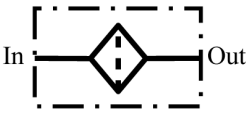
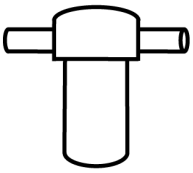
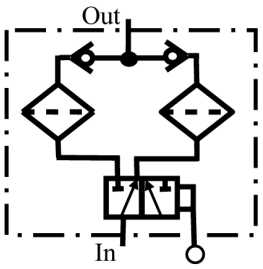
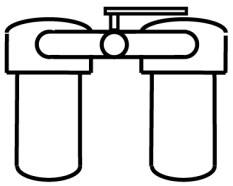
of the available fluid power components. However, some parts must be made up of combinations of different symbols to show how they function. Other times there is no standard symbol and one must be made up. In such cases, look first in the supplier's catalog for the symbol they show. If the supplier did not make a symbol, the only other option is design one for the new part. Try to design the new symbol using standard practices shown here.

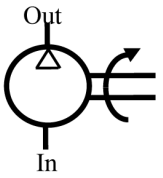
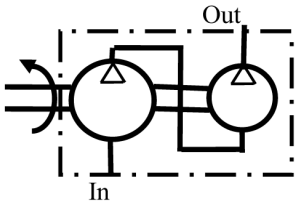
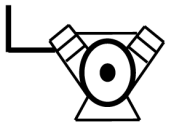
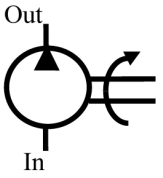
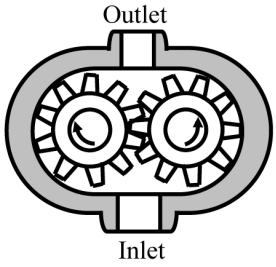
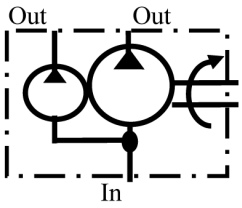
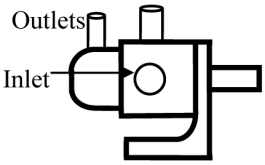
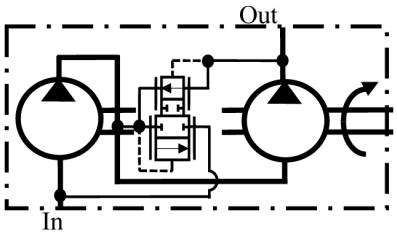
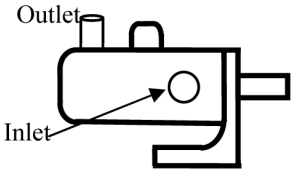
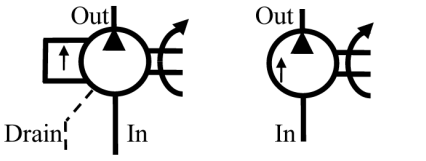
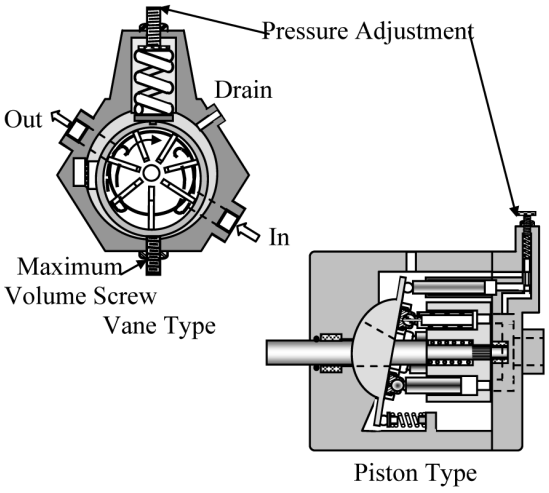
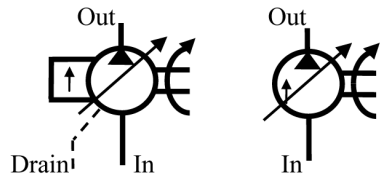
As the phrase fluid power implies, these symbols cover both hydraulic and pneumatic components. Any exceptions are noted. ■

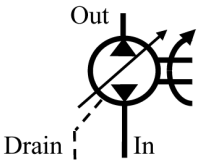
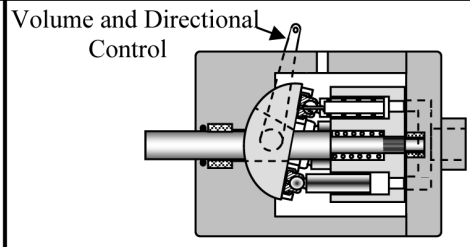
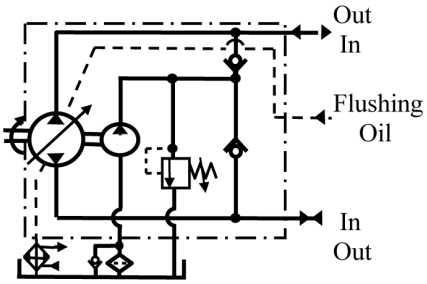
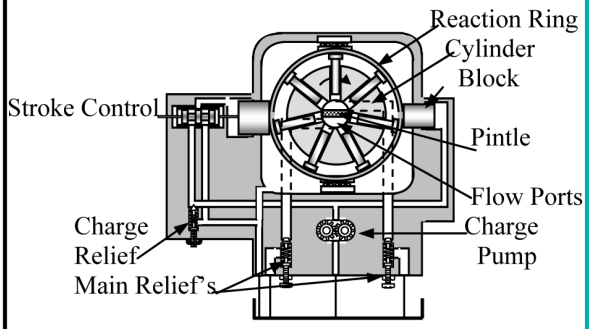
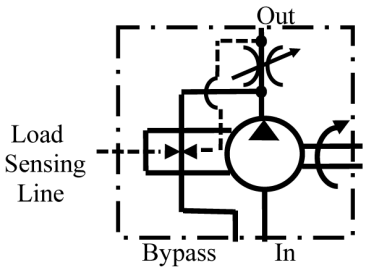
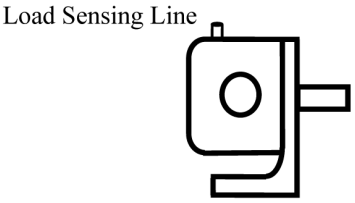
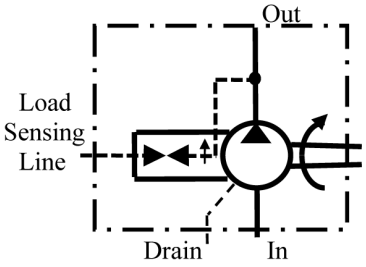
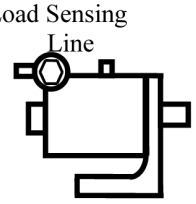
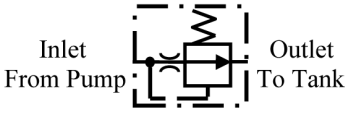
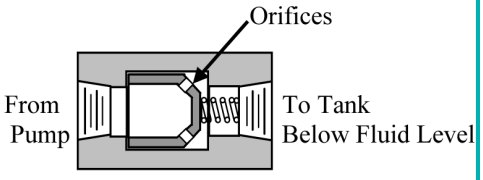
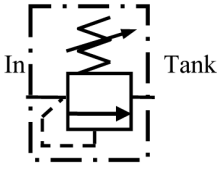
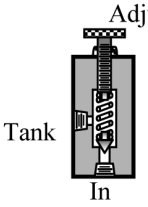
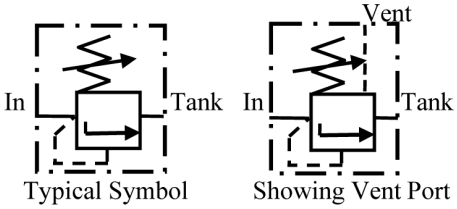
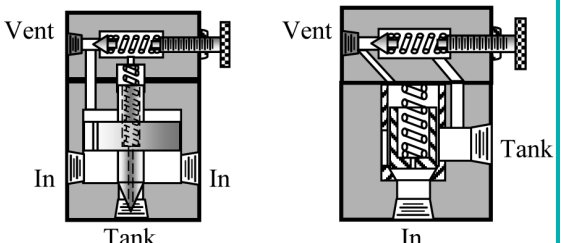
ISO Designation	Symbol	Picture Representation
Flexible Line		Shows hose or other flexible line. Could be whole length or only a portion of the conduit.
Pipe Junction		Denotes a tee or cross where lines connect to one another. The connecting Dot is always used with the Jumper below. Preferred way.
Pipe Junction		Denotes pipes connecting at a tee or cross Optional way.
		Denotes pipes crossing. Preferred way
Pipes Crossing		Denotes pipes crossing. Optional way
Plug		
Air Bleed		Shows a connection for bleeding trapped air
Pressure Takeoff		A connection for taking power from a line or for pressure testing. Shown plugged.
		With takeoff line connected .
Energy Triangle		Pneumatic / Hydraulic — Shows direction of flow and type of fluid.
Power Source		Denotes a power source from another part of the schematic or a another source
Arrows		Indicates direction of rotation of a pump or motor shaft, valve actuator or other actuator.
		Indicates direction of movement of a component.
		Arrows used for flow direction in valves. Arrows with a perpendicular line opposite the arrow head Arrows with a perpendicular line at the head end indicates the path stays connected to its outlet when it moves
Sloping Arrow		A sloping arrow through a pump, valve, spring, cushion plunger, solenoid or other device indicates it is adjustable or variable.

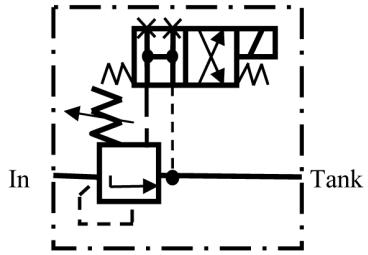
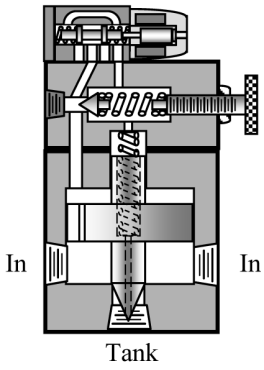
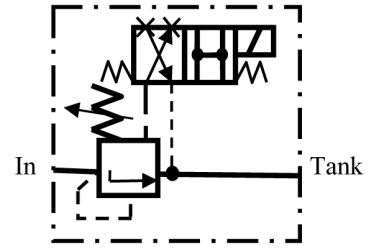
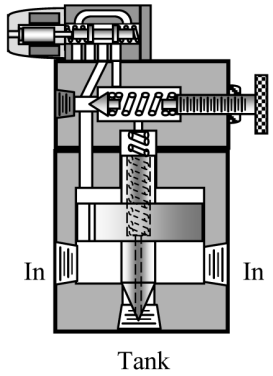
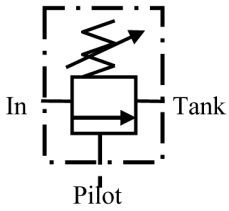
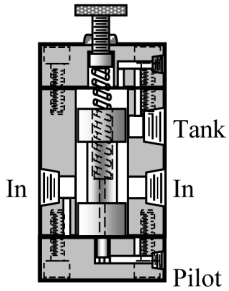
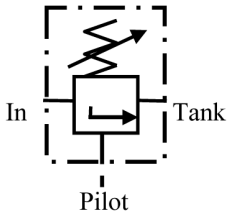
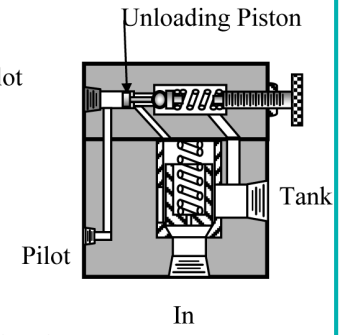
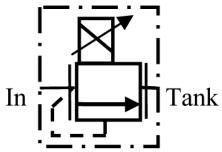
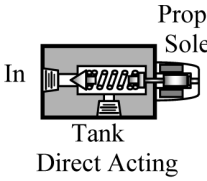
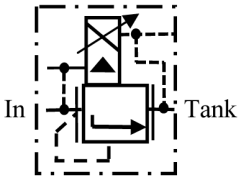
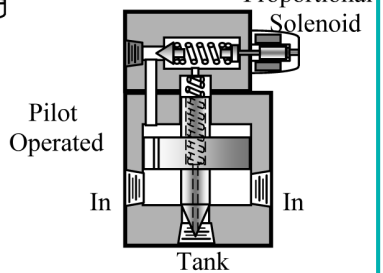
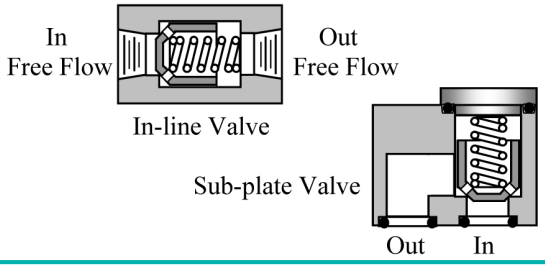
ISO Designation	Symbol	Picture Representation
Circles and Semi-circles		Represents a rotary device such as a pump or motor. Denotes the device is capable of continuous rotation in one or both directions. Circles of different sizes are the basis of unit of measurement instruments such as gauges. Also represents mechanical rollers on cam operated valves. Semi-circles denote rotary output devices that are not capable of continuous rotation. These actuators only oscillate through some arc.
Squares and Rectangular Boxes (Envelopes)		Square or rectangular boxes or envelopes are the basic unit of pressure and directional control valves. Single boxes denote pressure controls while multiple boxes show directional controls. The valve operator box is also a rectangle
Diamond Shaped Boxes		Diamond shaped boxes indicate a fluid conditioning device like a filter, lubricator, or heat exchanger.
Miscellaneous Symbols		A sawtooth line represents a spring. Back to back semi-circles on a line show a standard orifice that is affected by viscosity. Back to back "V's" on a line denotes an orifice That is not affected by viscosity.
Reservoirs Chapter 6 Reservoir Open To Atmosphere Pressurized Reservoir		 For systems that operate in contaminated locations or when using pressure to force fluid flow.
Heaters Chapter 6 Tank Heater		
Heat Exchangers (Coolers) Chapter 6 Water Cooled Heat exchanger		

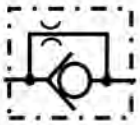
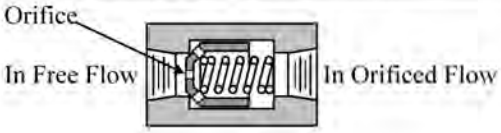
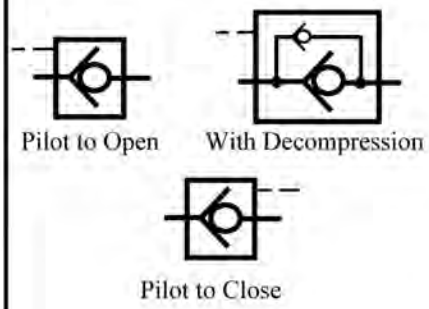
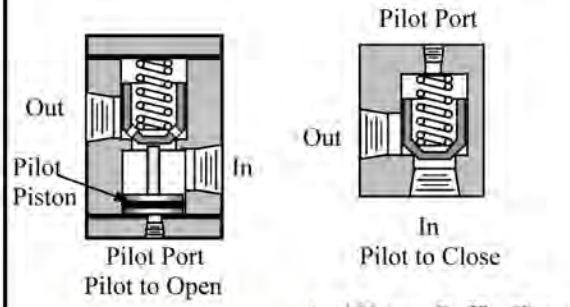
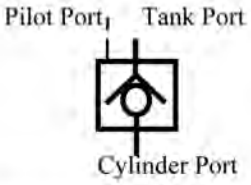
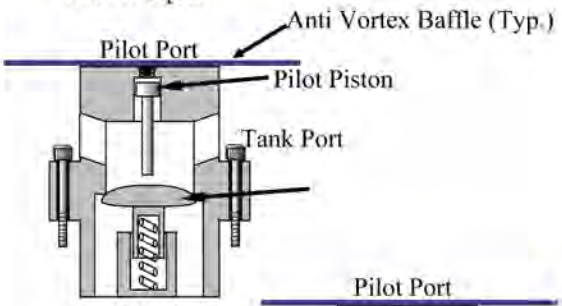
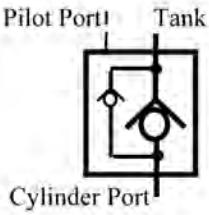
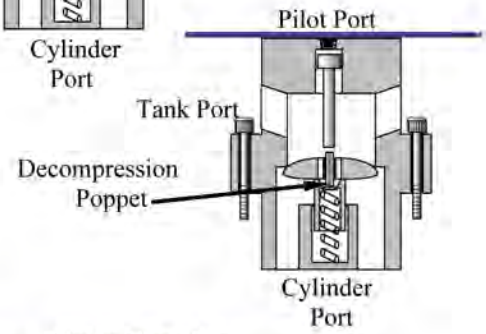
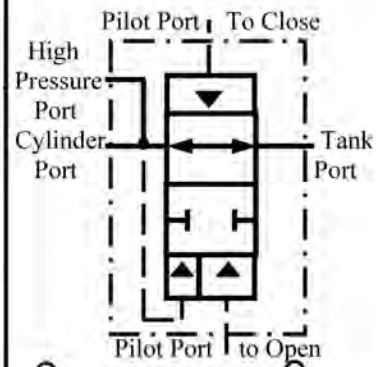
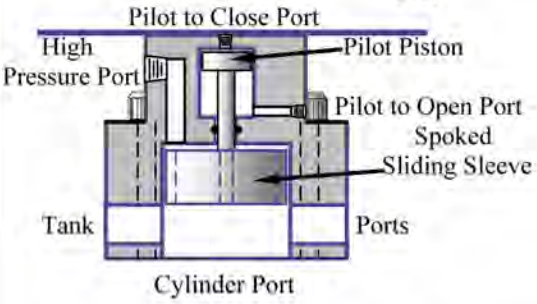
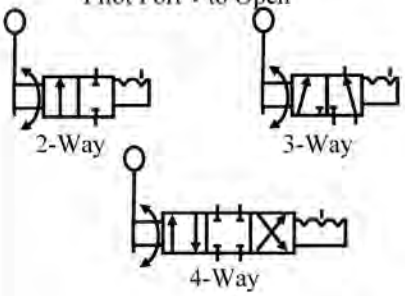
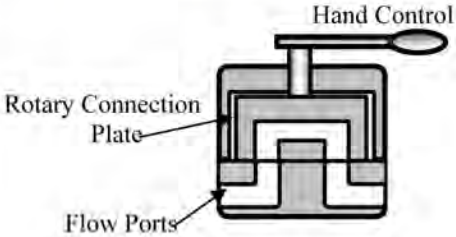
ISO Designation	Symbol	Picture Representation
Heat Exchangers (Coolers) Chapter 6 Air Cooled Heat Exchanger		 Fan Cooled Radiator Type Heat Exchanger
Temperature Controller Chapter 6	Internal Arrows Show Energy Added to or Removed From Fluid Flow  Liquid Cooling	 Shell and Tube Temperature Controller
Temperature Controlled Water Valve Chapter 6		 Temperature Controlled Water Valve
Air Line Filter Chapter 7	 With Manual Drain  With Automatic Drain	
Air Line Lubricator Chapter 7 **Air Line Filter, , Regulator and Lubricator Chapter 7**	  Simplified Symbol	

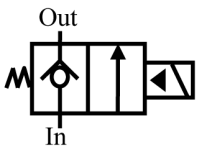
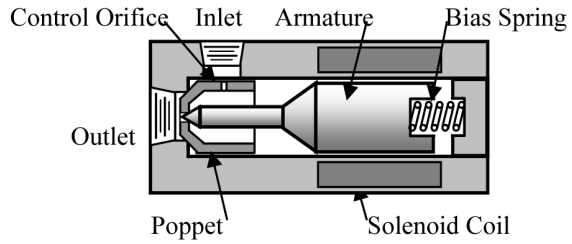
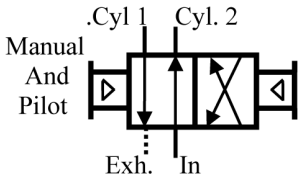
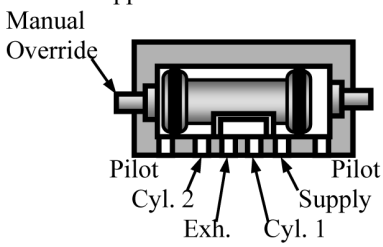
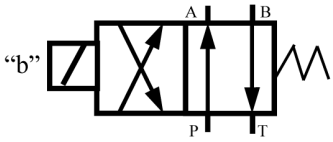
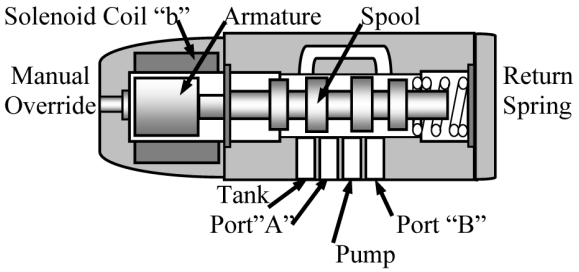
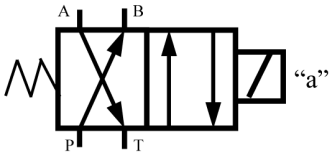
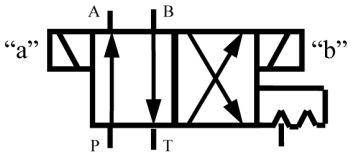
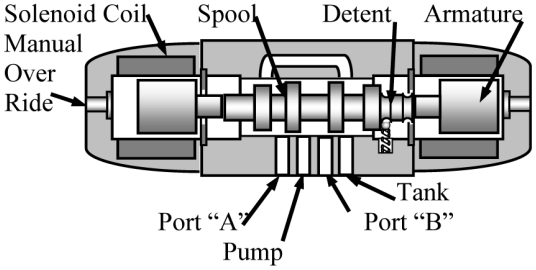
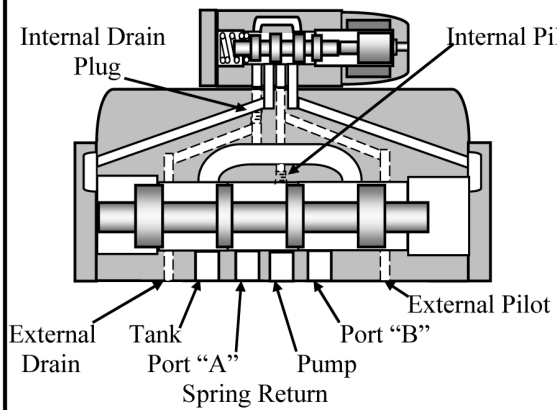
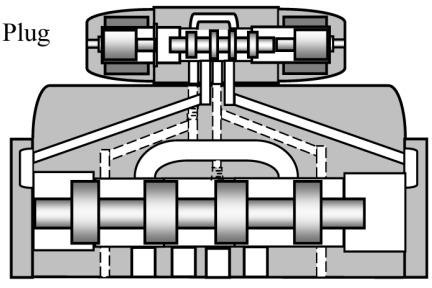
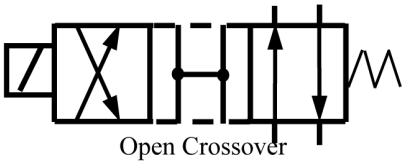
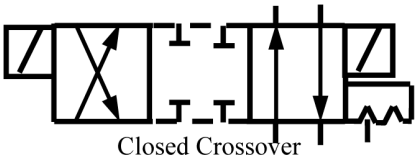
ISO Designation	Symbol	Picture Representation
Air Dryer **Chapter 7**		 Deliquescent Dryer (Absorption Type)
	 Desiccant Dryer (Adsorption Type)	 Refrigeration Dryer (Condensing Type)
Hydraulic Filters **Chapter 7** In Tank Hydraulic Suction Strainer With Bypass		 In Tank Type Suction Strainer
In-Line Return Filter With Bypass		
In-Line Pressure Filter Without Bypass		
Dual Pressure Filter For Changing Elements While Circuit is Running		

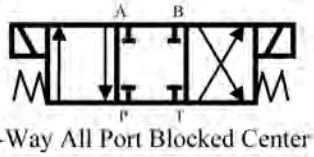
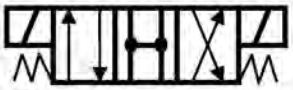
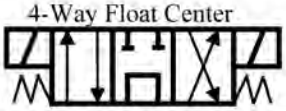
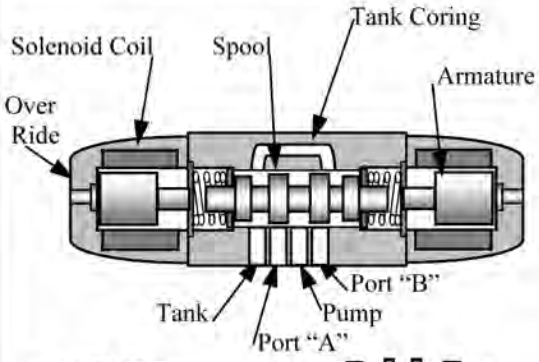
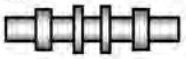
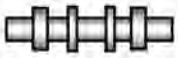
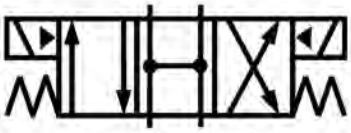
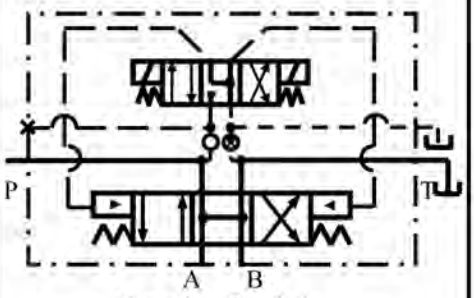
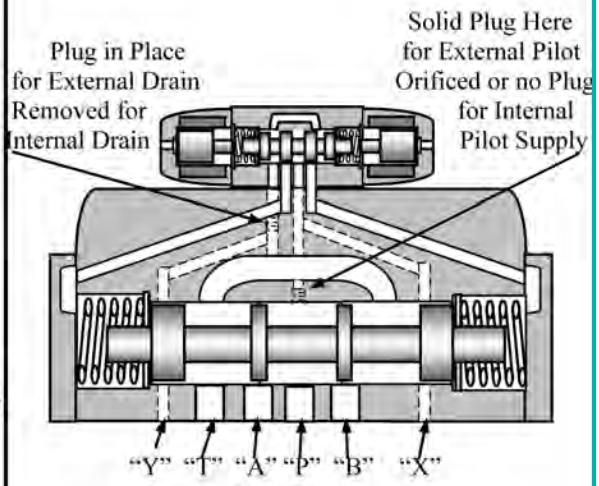
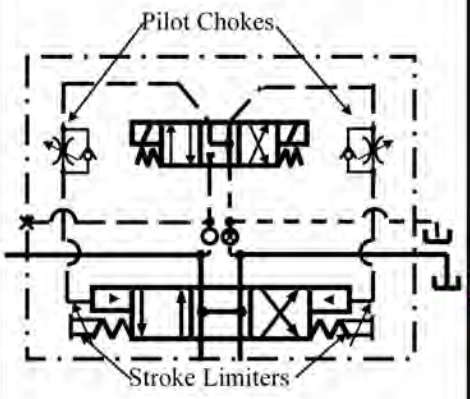
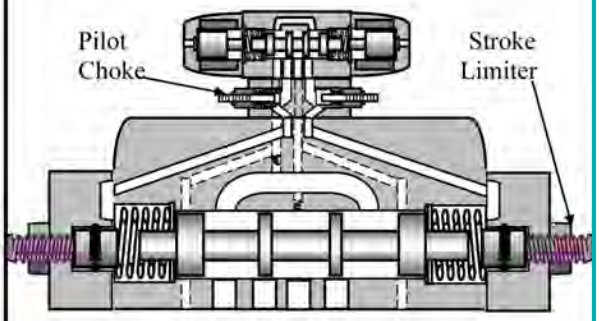
ISO Designation	Symbol	Picture Representation
Pumps **Chapter 8** Air Pump Single Stage (Compressor)		
Air Pump Two Stage (Compressor)		
Hydraulic Pump Fixed Volume Single Flow Single Stage		
Hydraulic Pump Fixed Volume Double Flow Single Stage		
Hydraulic Pump Fixed Volume Single Flow Two Stage		
Hydraulic Pump Pressure Compensated Single Flow Single Stage	 Complete Symbol Simplified Symbol	
Hydraulic Pump Pressure Compensated Variable Volume Single Flow Single Stage		

ISO Designation	Symbol	Picture Representation
Pumps (Cont'd.) Hydraulic Pump Variable Volume Bi-directional Flow Single Stage		
Hydraulic Pump Variable Volume Bi-directional Single Stage With Charge Pump For Closed Loop Circuits		
Hydraulic Pump Fixed Volume Single Flow Single Stage With Load Sensing		
Hydraulic Pump Pressure Compensated Single Flow Single Stage With Load Sensing		
Air Bleed Valve for closed center circuits to let trapped air escape from pump outlet		
Relief and Unloading Valves Chapter 9 Relief Valve Direct Acting Adjustable		
Relief Valve Pilot Operated Adjustable With Vent Port		

ISO Designation	Symbol	Picture Representation
Relief and Unloading Valves (Cont'd.)		
Solenoid Operated Relief Valve Normally Open		
Solenoid Operated Relief Valve Normally Closed		
Unloading Valve External Pilot Operated Direct Acting		
Unloading Valve External Pilot Operated Pilot Operated Also Used as an Accumulator Circuit Pump Unloading Valve Through Unloading Spool		
Proportional Relief Valve Direct Acting		
Proportional Relief Valve Pilot Operated		
Directional Control Valves Chapter 10		
Check Valves Inline, Sub-plate Mounted, Screw In Cartridge	  As a Back Pressure Valve	

ISO Designation	Symbol	Picture Representation
Directional Control Valves (Cont'd.)		
Orificed Check Valve		
Pilot Operated Check Valves		
Pre-fill Valves Without Decompression Feature		
With Decompression Feature		
Sliding Sleeve Pre-fill Valve		
Rotary Shear Valve		

ISO Designation	Symbol	Picture Representation
Directional Control Valves (Cont'd.)		
Poppet Type Valve		
Sliding Plate Valve		
Spool Type Valve		
4-Way 2 Position Valve Spring Return Direct Acting Solenoid		
		
4-Way 2 Position Valve Detented Direct acting Solenoid		
4-Way 2 Position Valve Spring Return and Detented Solenoid Pilot Operated		
2 Position Valve Typical Transition or Crossover Condition		

ISO Designation	Symbol	Picture Representation
Directional Control Valves (Cont'd.) Spool Type Valves Direct Solenoid Operated, Three Position, Spring Centered Directional Control Valves	 Manual 4-Way All Port Blocked Center  4-Way All Ports Open Center  4-Way Float Center  4-Way Tandem Center **Four Standard Center Conditions**	  All Ports Blocked Spool  All Ports Open Spool  Float Center Spool  Tandem Center Spool
Solenoid Pilot Operated Three Position, Spring Centered Directional Control Valve	 Simplified Symbol  Complete Symbol	
Solenoid Pilot Operated, Three Position, Spring Centered Directional Control Valves With Pilot Chokes and Main Spool Stroke Limiting Screws		

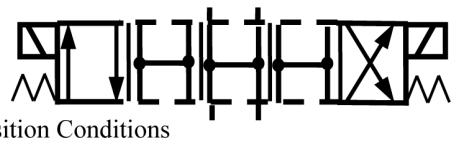
ISO Designation	Symbol	Picture Representation
Directional Control Valves (Cont'd.) Solenoid Pilot Operated, Three Position, Hydraulic Centered, Directional Control Valve	External Pilot Supply External Drain P T A B Complete Symbol Internal Pilot Supply, External Drain	This Solenoid Energized Differential Area Sleeve With Centering Washer Shifting Area Valve Shown in Shifted Condition



Other Available Center Conditions From Most Manufacturers

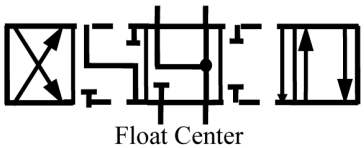


4-Way 2 Position Single Solenoid
All Ports Blocked Center



4-Way 3 Position Double Solenoid Spring Centered
All Ports Open Center

Typical Crossover or Transition Conditions



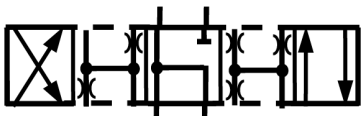
Float Center



Tandem Center



Pump "A" and "B" Connected
Tank Blocked



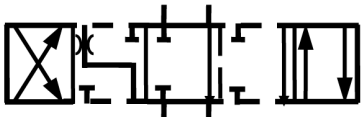
Pump "A" and Tank Connected
"B" Blocked



Pump "B" and Tank Connected
"A" Blocked



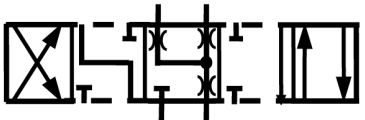
Pump and "B" Blocked
A" to Tank



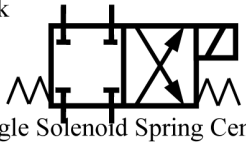
Pump and "A" Blocked
"B" to Tank



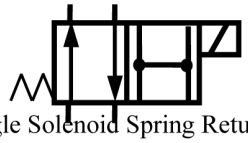
Open Center Orificed Flow



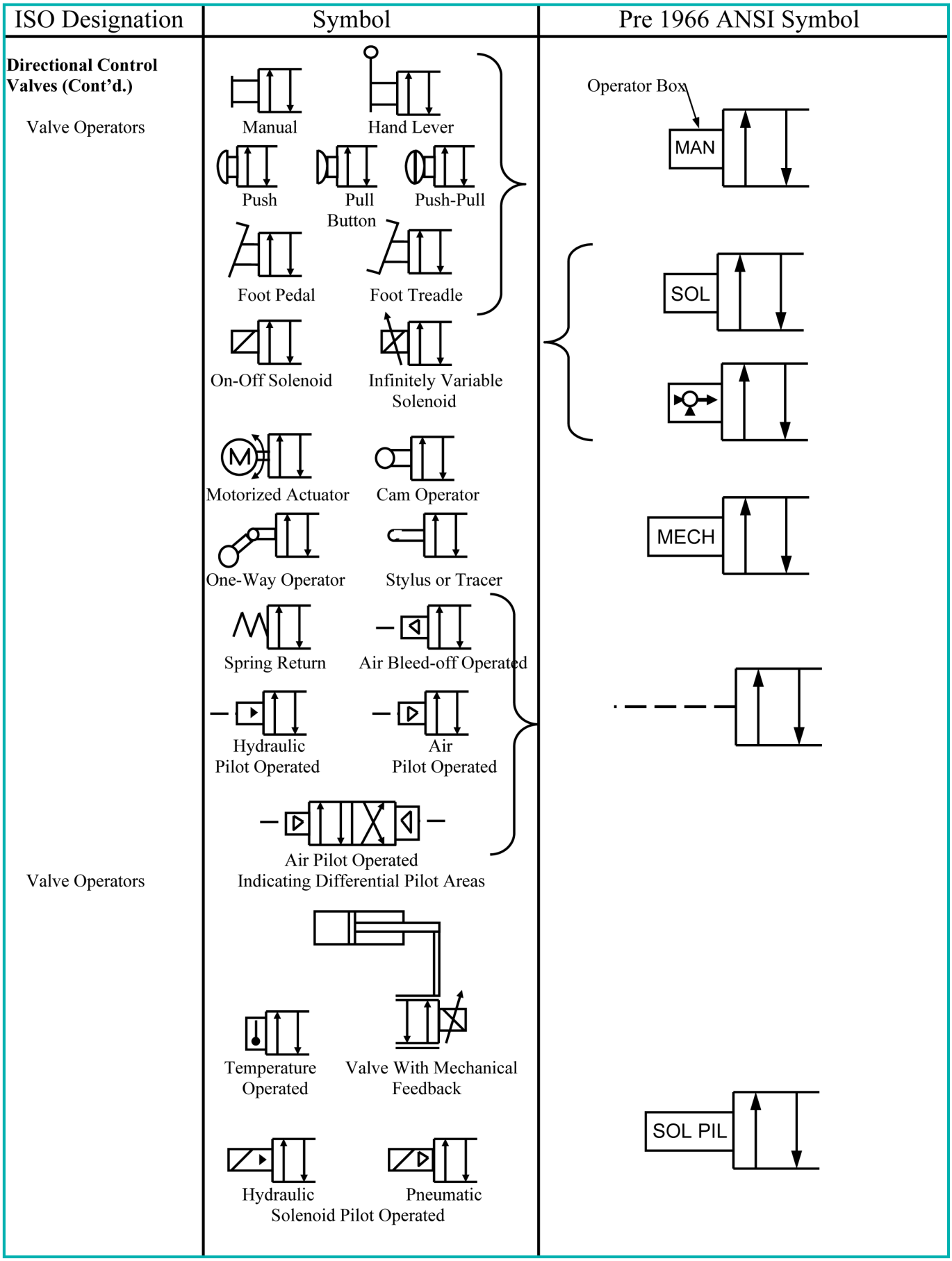
Float Center Orificed Flow



Single Solenoid Spring Centered
At Rest in Center Condition Shift to Flow Path



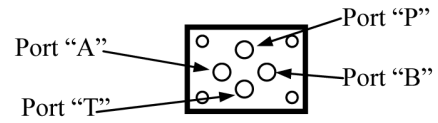
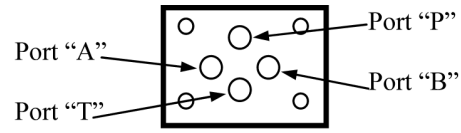
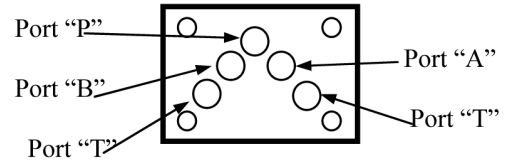
Single Solenoid Spring Return
At Rest in Flow Path, Shift to a Center Condition


ISO Designation
Symbol
Pre 1966 ANSI Symbol
Directional Control Valves (Cont'd.)
Valve Operators
Manual
Hand Lever
Push
Pull
Push-Pull
Button
Foot Pedal
Foot Treadle
On-Off Solenoid
Infinitely Variable Solenoid
Motorized Actuator
Cam Operator
One-Way Operator
Stylus or Tracer
Spring Return
Air Bleed-off Operated
Hydraulic Pilot Operated
Air Pilot Operated
Air Pilot Operated Indicating Differential Pilot Areas
Valve Operators
Temperature Operated
Valve With Mechanical Feedback
Hydraulic
Pneumatic
Solenoid Pilot Operated
Operator Box
MAN
SOL
MECH
SOL PIL

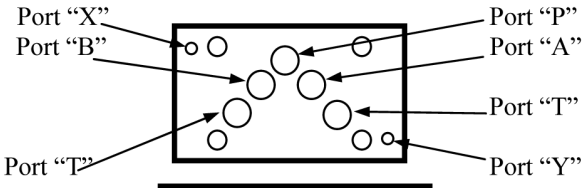
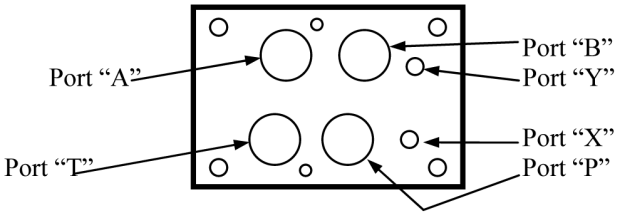
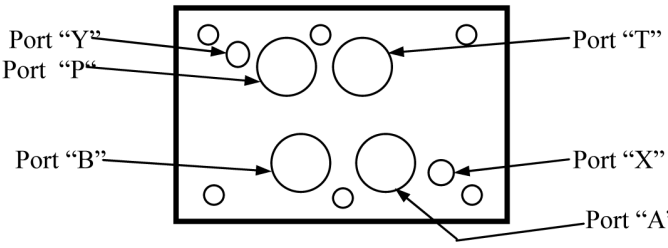
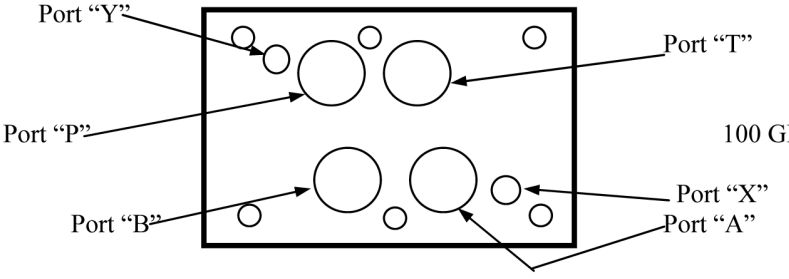
ISO Designation	Symbol	Picture Representation

Directional Control Valves (Cont'd.)

NFPA AND ISO STANDARD INTERFACE LAYOUTS FOR HYDRAULIC DIRECTIONAL CONTROL VALVES

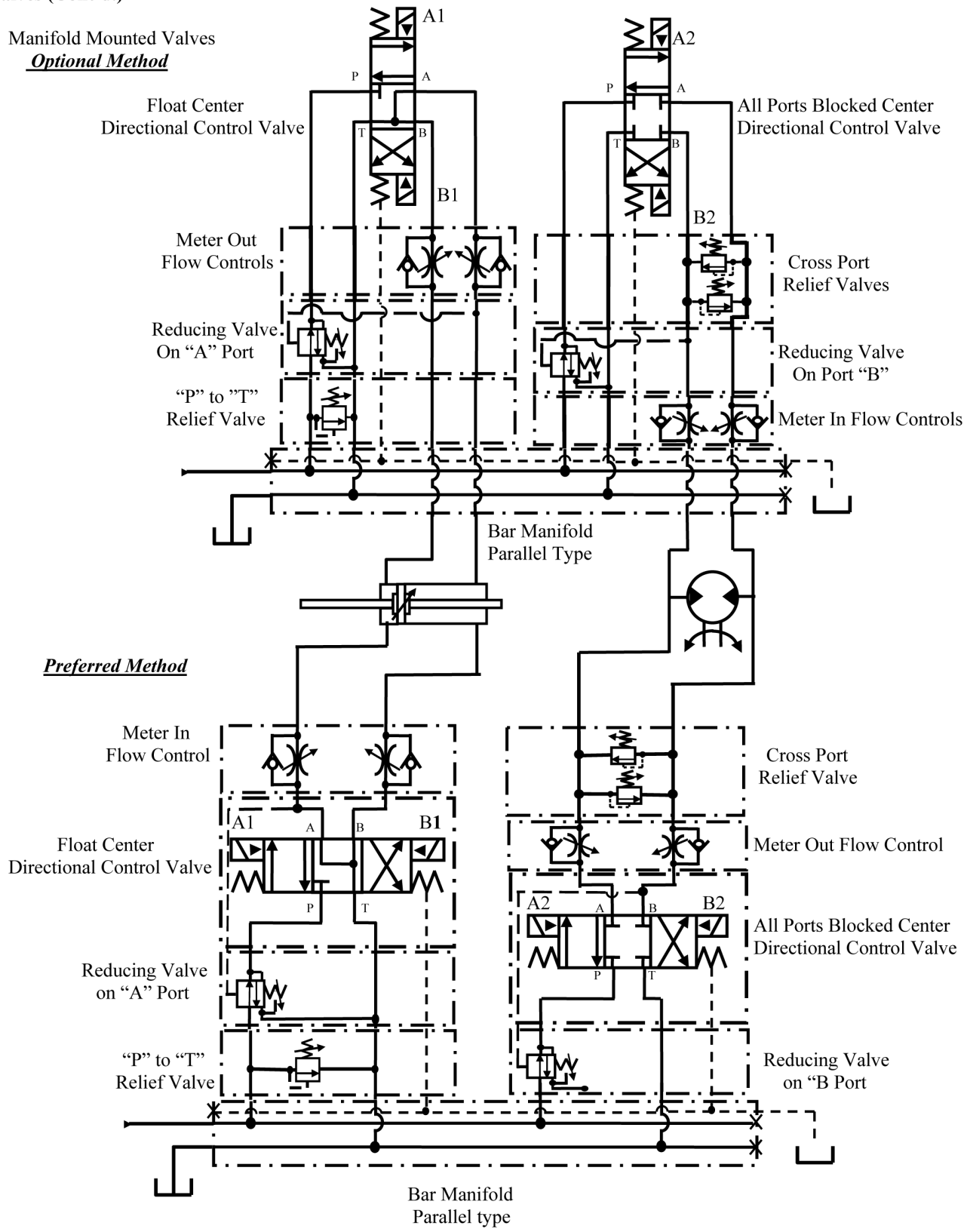
NFPA.	ISO	*CETOP*	NG	PORT DIA	PORT CONFIGURATION	NOMINAL FLOW
DO2	02	*2*	4	.177		5 GPM
D03	03	*3*	6	.295		10 GPM
D05	05	5	10	.440		20 GPM

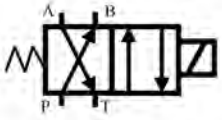
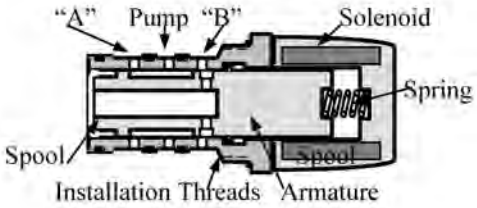
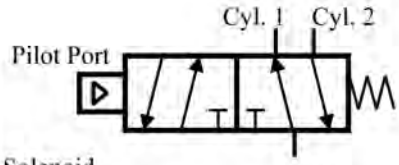
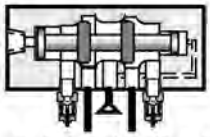
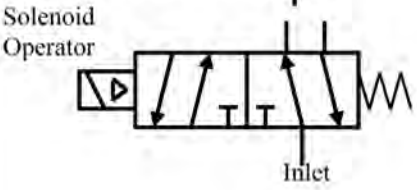
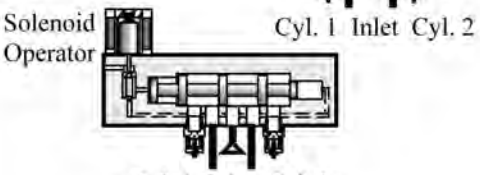
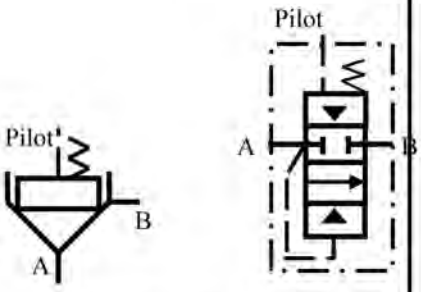
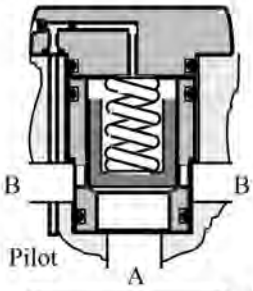
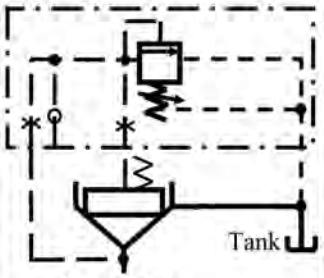
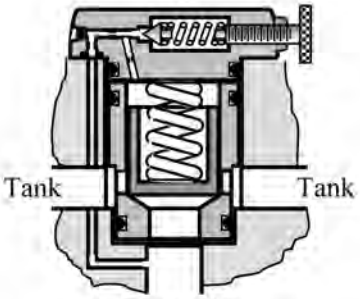
All Views Shown Looking Into Ports On The Valve

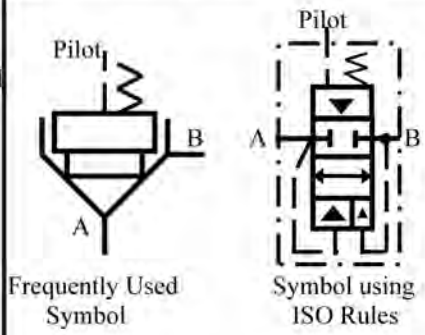
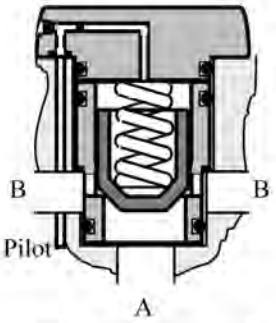
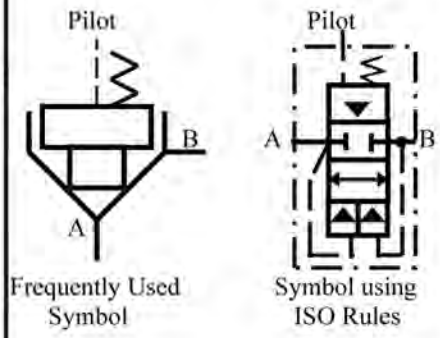
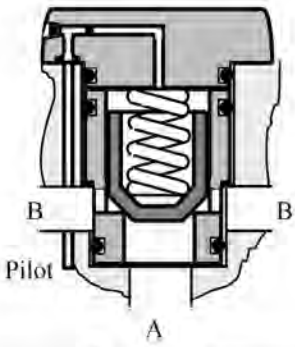
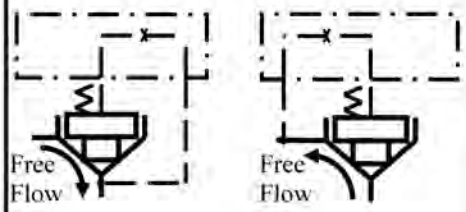
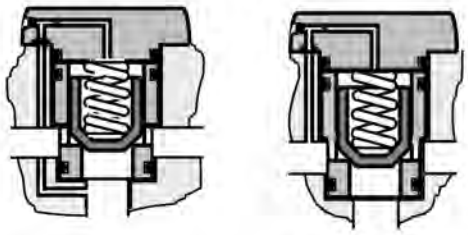
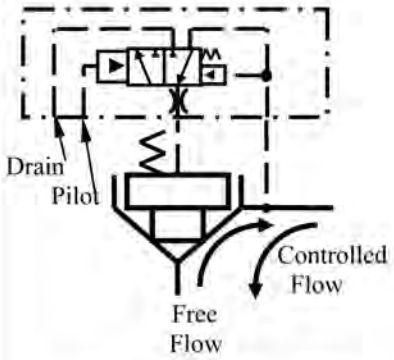
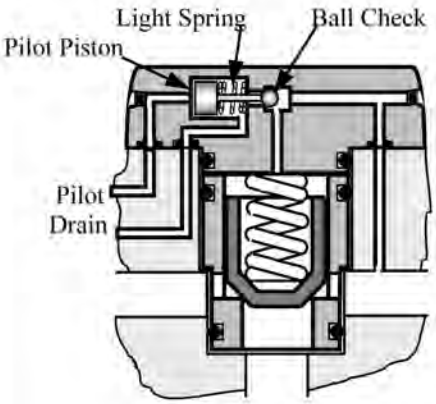
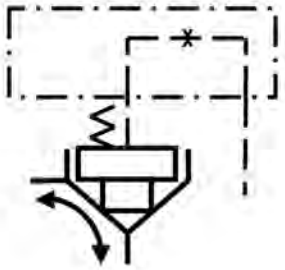
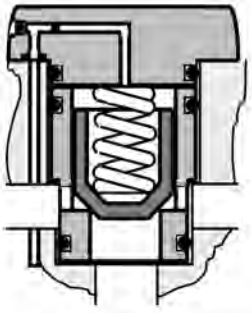
D05H	05			.440		25 GPM
D07	07	7	16	.690		30 GPM
D08	08	*8*	25	.984		60 GPM
D10	10	*10*	32	1.250		100 GPM

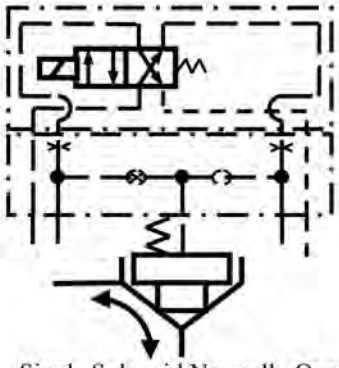
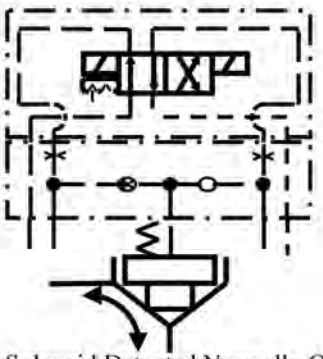
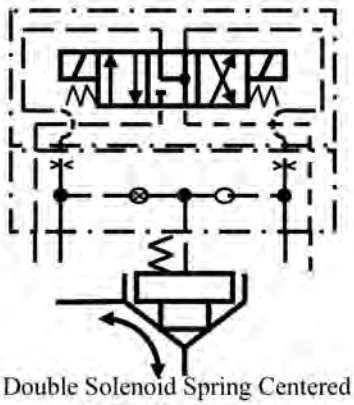
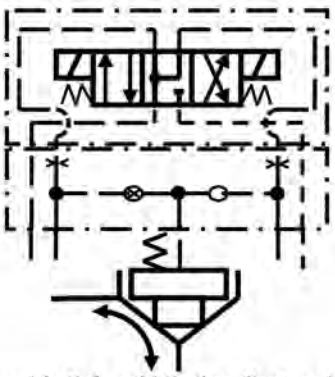
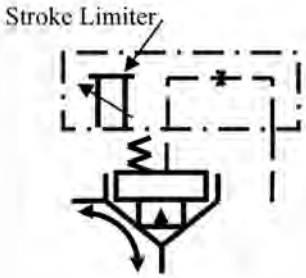
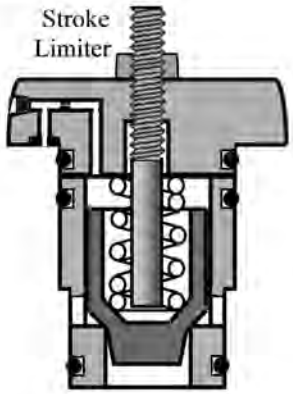
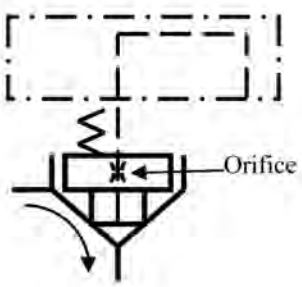
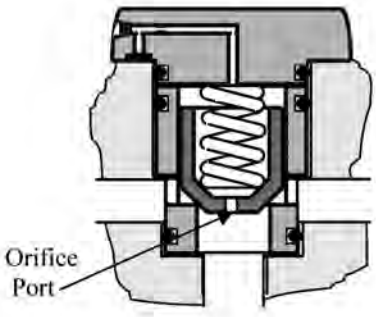
ISO Designation	Symbol	Picture Representation

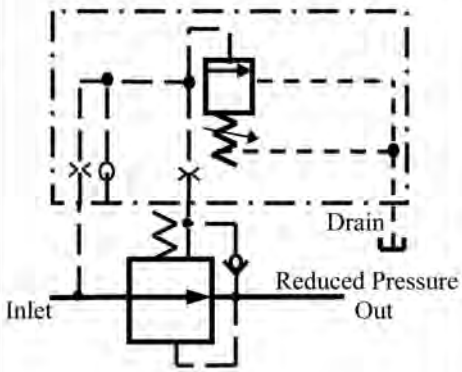
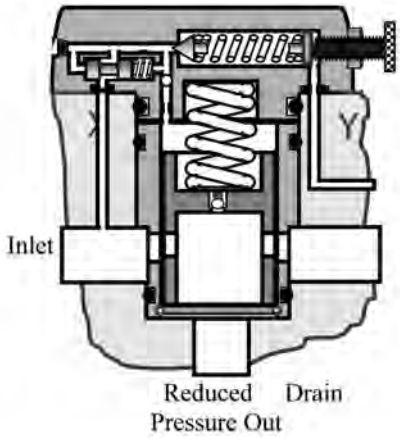
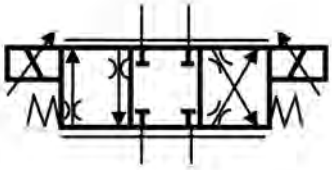
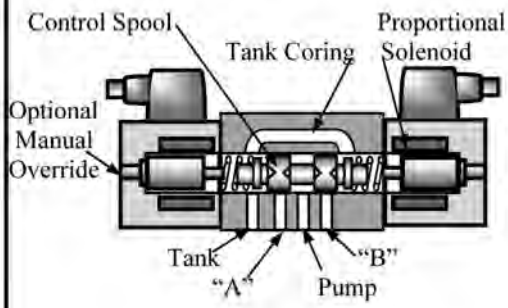
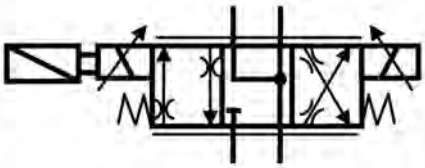
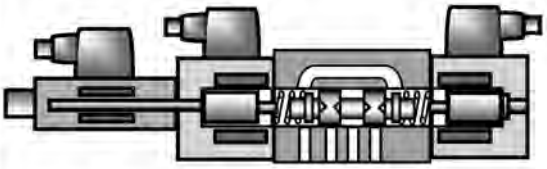
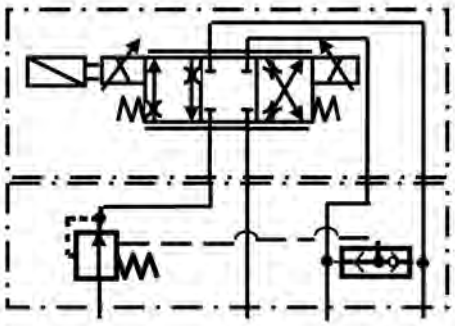
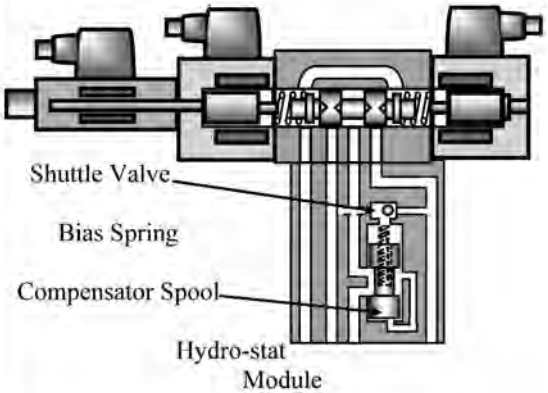
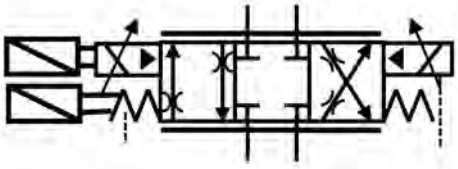
Directional Control Valves (Cont'd.)

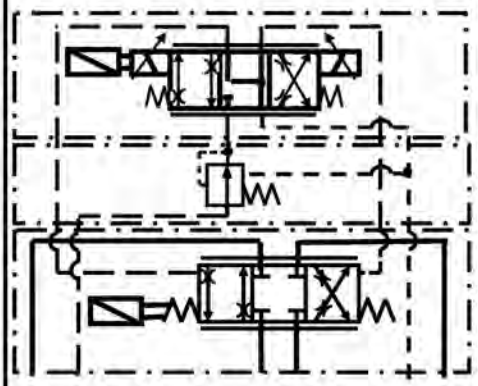
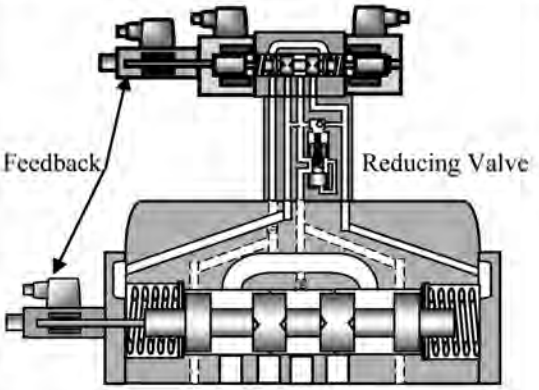
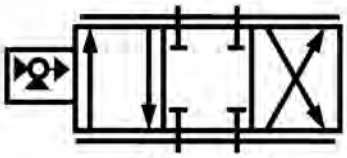
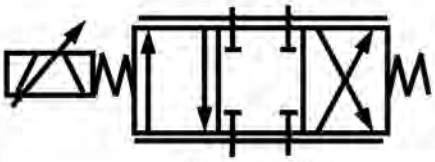
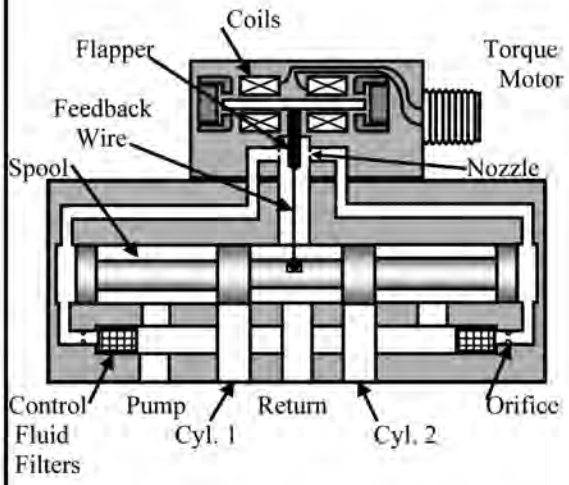
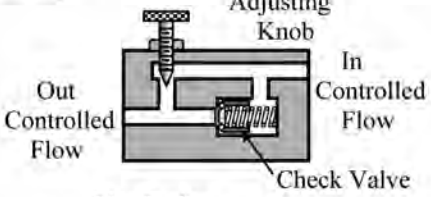
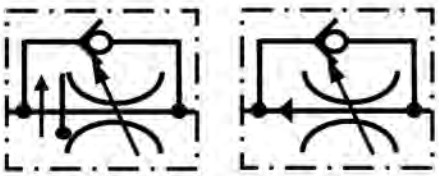
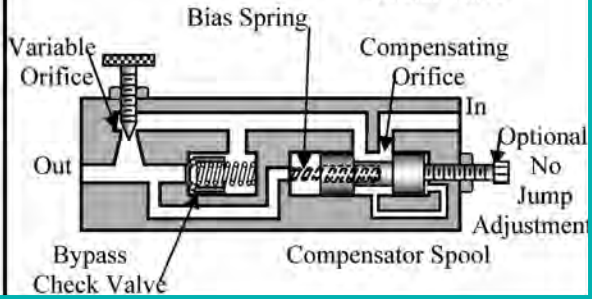


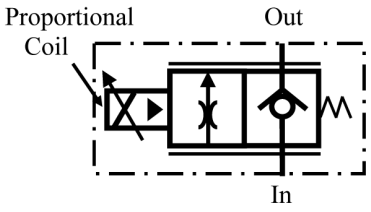
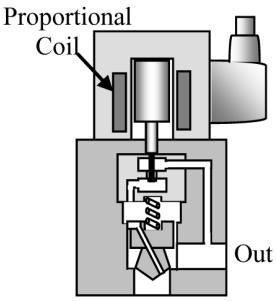
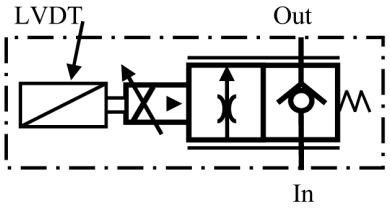
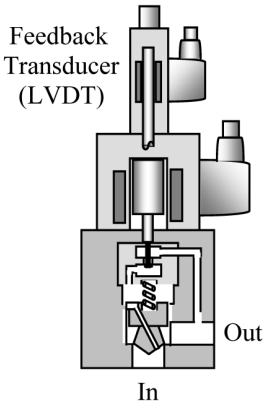
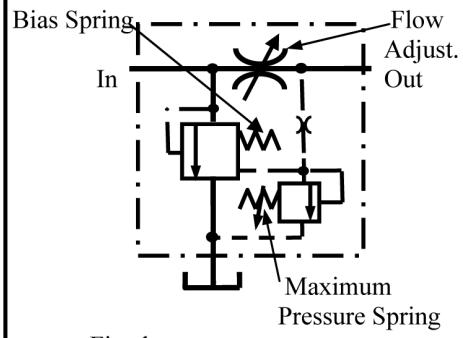
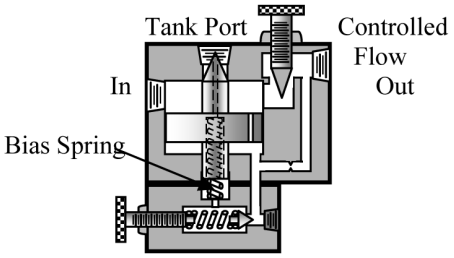
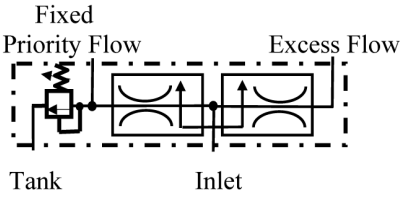
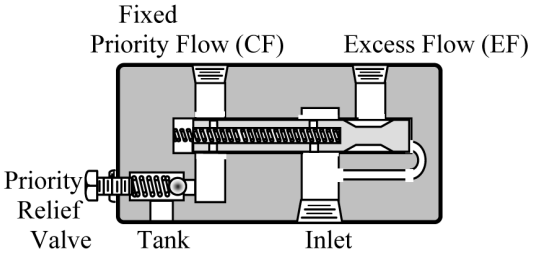
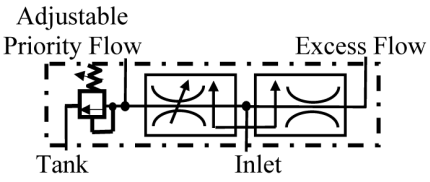
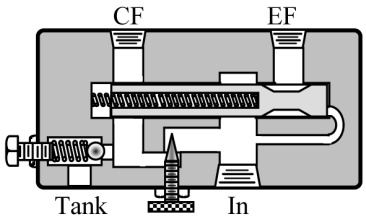
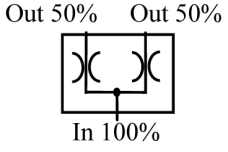
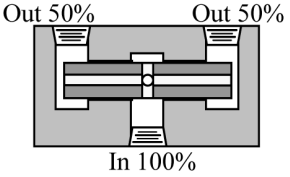
ISO Designation	Symbol	Picture Representation
Directional Control Valves (Cont'd.) Screw-in Cartridge Valves, All the same functions as In-line and Sub-Plate mounted Valves.		
5-Way Air Pilot Operated, Spring Return Air Valve, Poppet Type		
5-Way Solenoid Air Pilot Operated, Spring Return Air Valve, Spool Type		
1:1 Poppet Ratio used for Pressure Control		
1:1 Poppet Ratio as a Relief Valve		
Dual-Pressure and Pump Unload and Relief Valve		
Normally open Solenoid Operated Relief Valve		
Proportional Relief Valve		

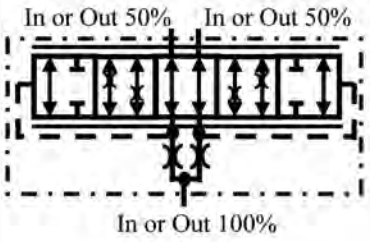
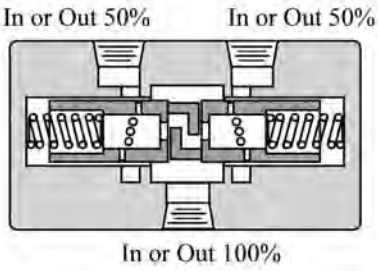
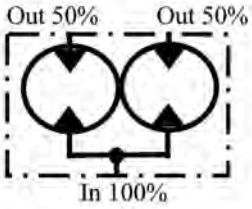
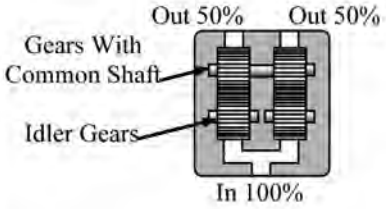
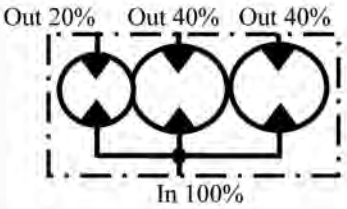
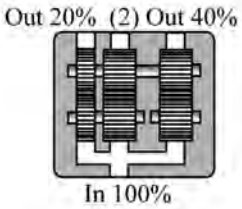
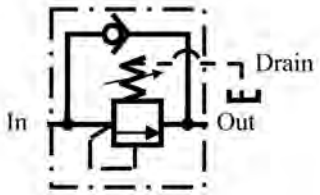
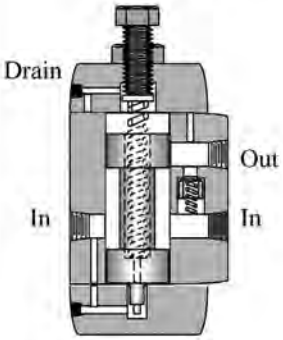
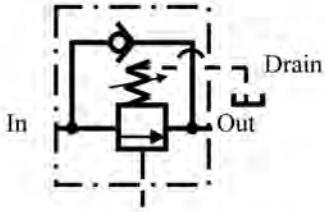
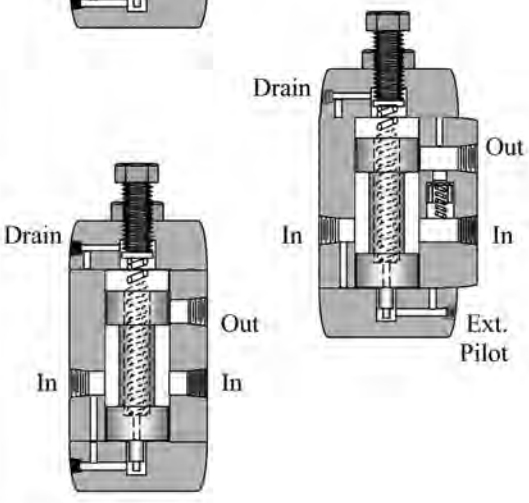
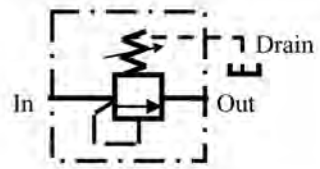
ISO Designation	Symbol	Picture Representation
Slip-in Cartridge Valves (Cont'd.) 1:1.1 Poppet Ratio Used for Pressure Control or Directional Control		
1:2 Poppet Ratio Used For Directional Control		
1:2 Poppet Ratio as a Check Valve With Plain Cover and Internal Connection		
1:2 Pilot Ratio as a Pilot Operated Check Valve With Pilot Operated Check Valve Cover		
1:2 Pilot Ratio as a Directional Control Valve With Plain Cover and External Connection		

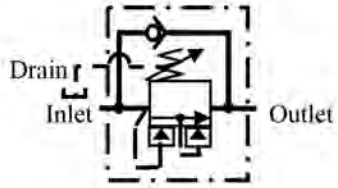
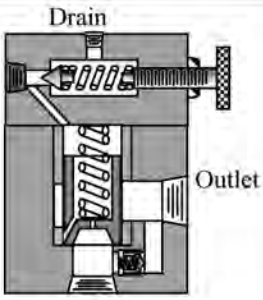
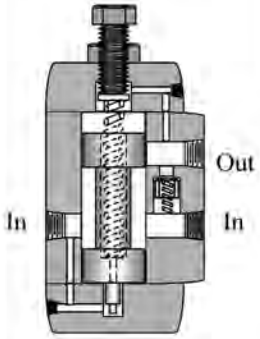
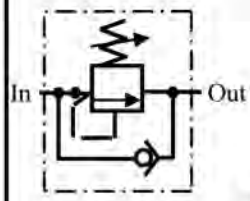
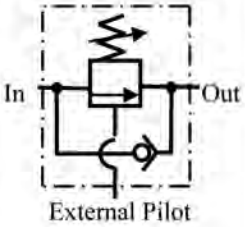
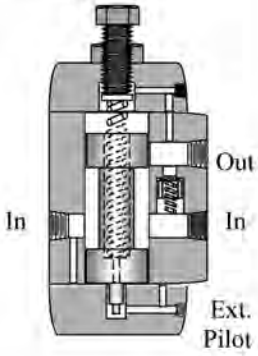
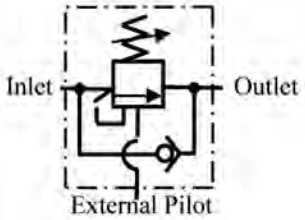
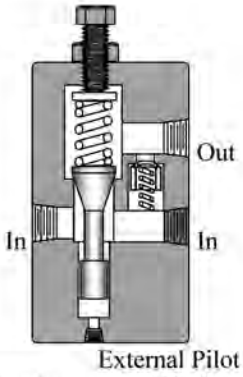
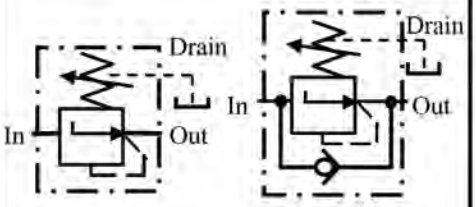
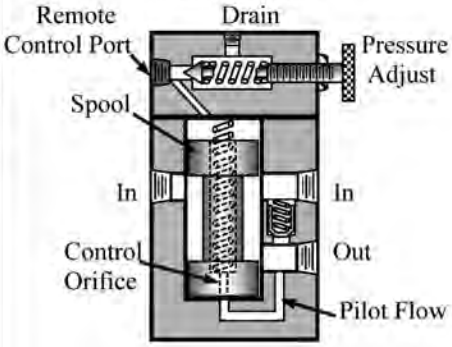
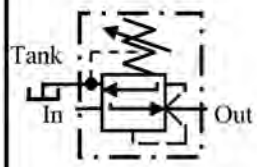
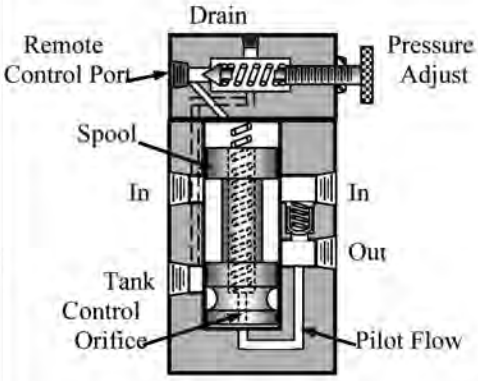
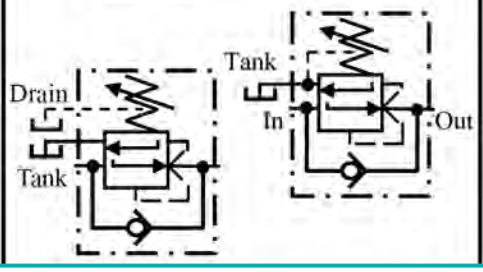
ISO Designation	Symbol	Picture Representation
Slip-in Cartridge Valves (Cont'd.) 1:2 Pilot Ratio as:	 Single Solenoid Normally Open	 Double Solenoid Detented Normally Open
	 Double Solenoid Spring Centered Normally Open	 Double Solenoid Spring Centered Normally Closed
Slip-in Cartridge Valve With Stroke Limiter and Deceleration Skirts		
Slip-in Cartridge valve With Orifice port As a Check Valve		

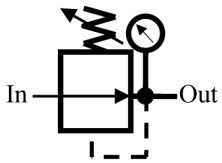
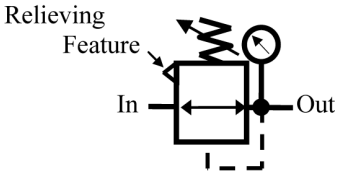
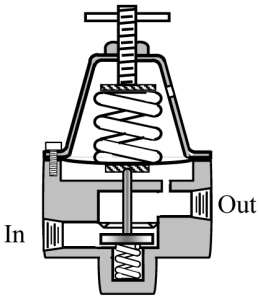
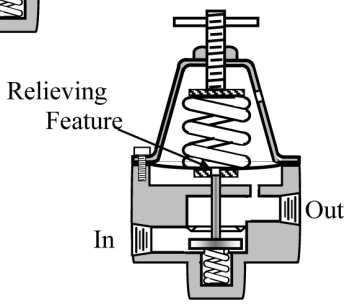
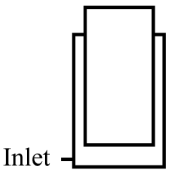
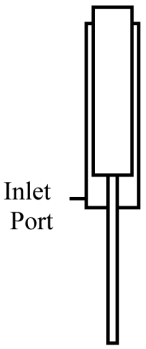
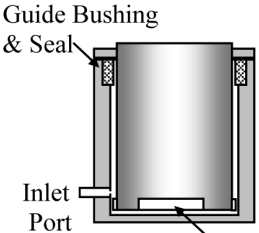
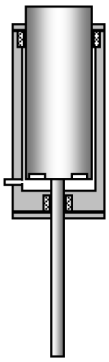
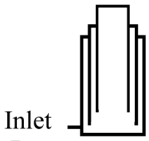
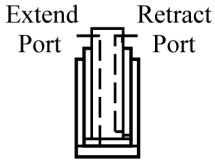
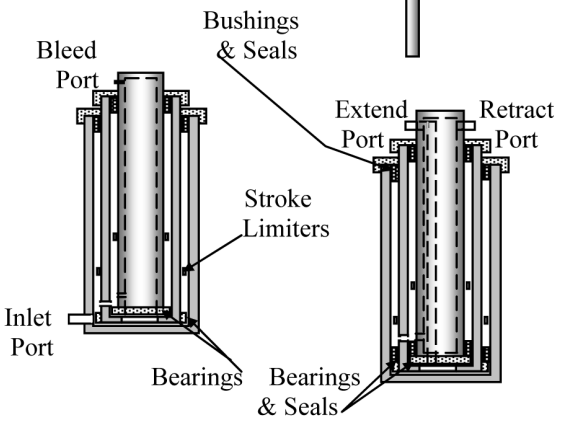
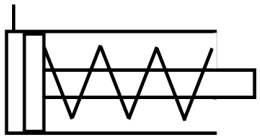
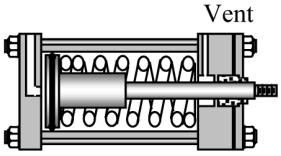
ISO Designation	Symbol	Picture Representation
Slip-in Cartridge Valves (Cont'd.)		
Slip-in Cartridge Valve as a Reducing Valve		
Infinitely Variable Directional Valves Chapter 12		
Proportional Controls Direct Solenoid Operated Without Spool Position Feedback		
Direct Solenoid Operated With Spool Position Feedback		
Direct Solenoid Operated With Spool Position Feedback and Pressure Compensating Module (Hydro-Stat)		
Simplified Symbol for a Solenoid Pilot Operated Valve With Spool Position Feedback		See next page for proportional valve complete symbol

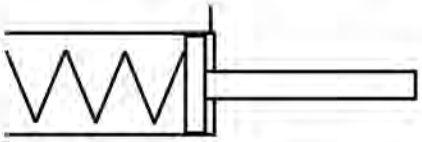
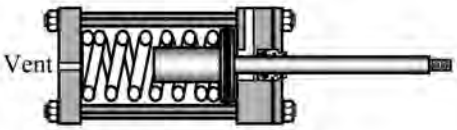
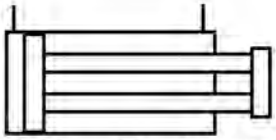
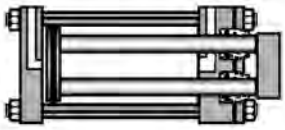
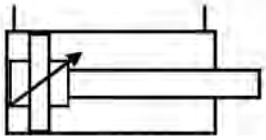
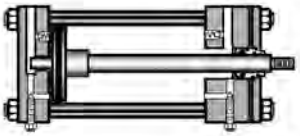
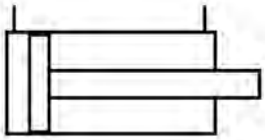
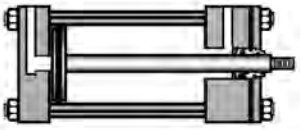
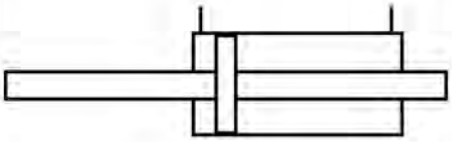
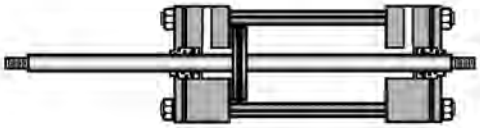
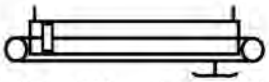
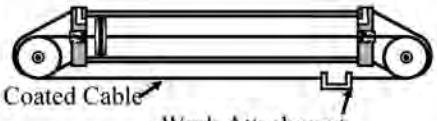
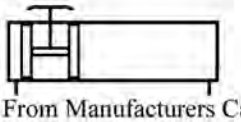
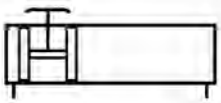
ISO Designation	Symbol	Picture Representation
Infinitely Variable Directional Valves (Cont'd.)		
Complete Symbol for a Solenoid Pilot Operated Valve With Spool Position Feedback		
Servo Valves	 ANSI Symbol  ISO Symbol	
Flow Control Valves Chapter 13		
Fixed Orifice	 Affected by Viscosity Unaffected by	 Standard Orifice Knife Edge Orifice
Needle Valve		
Non-Compensated Flow Control Valve		
Pressure and Temperature Compensated Flow Control	 ANSI Symbol ISO Symbol	

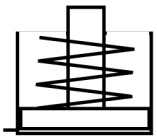
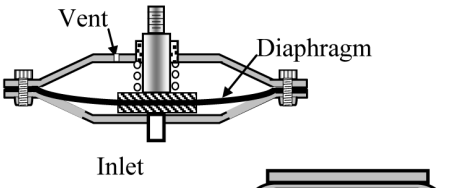
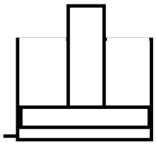
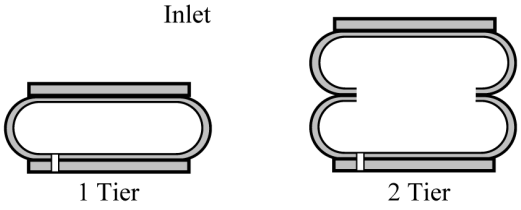
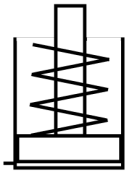
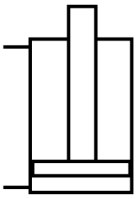
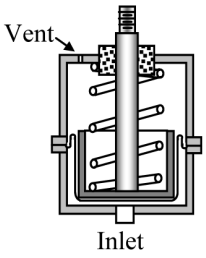
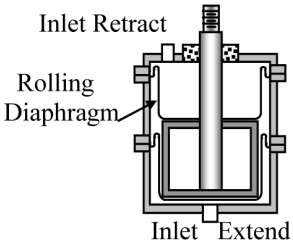
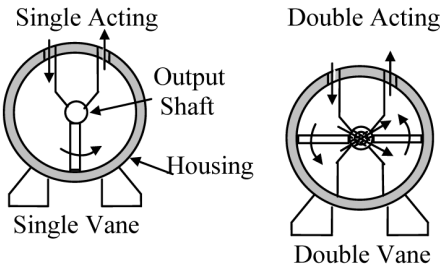
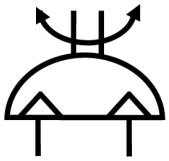
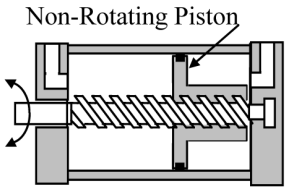
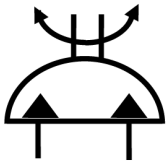
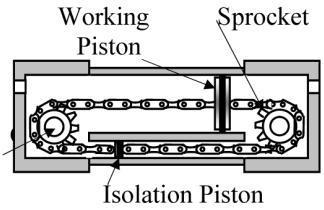
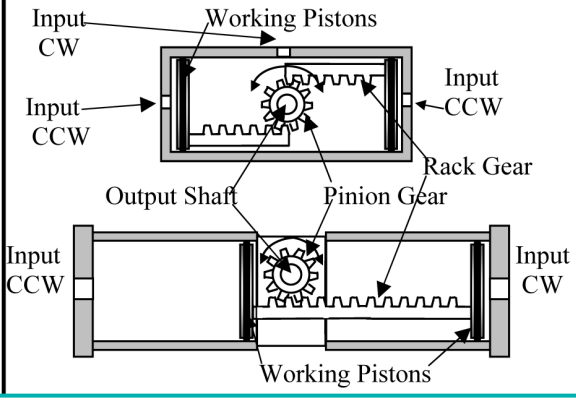
ISO Designation	Symbol	Picture Representation
Flow Control Valves (Cont'd.)		
Proportional Flow Control Valves Without Feedback		
With Feedback		
Three Port Flow Control With Built-in Relief Valve. Also Called a Bypass Flow Control. Bypass Flow Cannot be Used to do Work		
Priority Type Three Port Flow Control Bypass Flow can be Used to do Work		
		
Flow Dividers Spool Type One Direction Only		

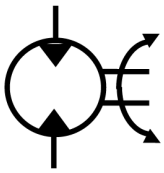
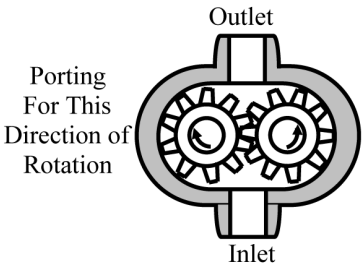
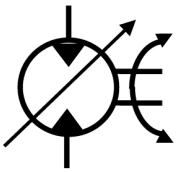
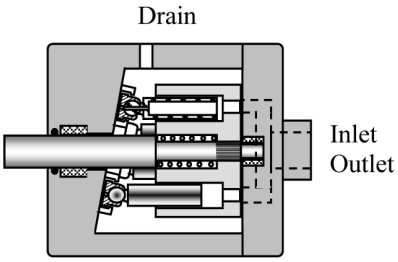
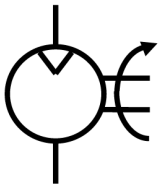
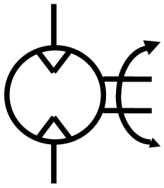
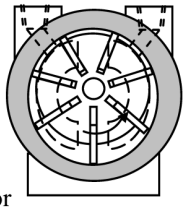
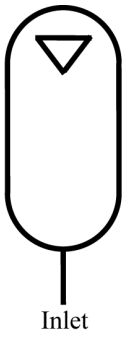
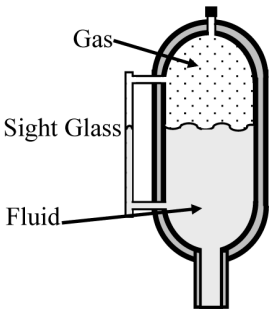
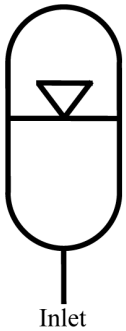
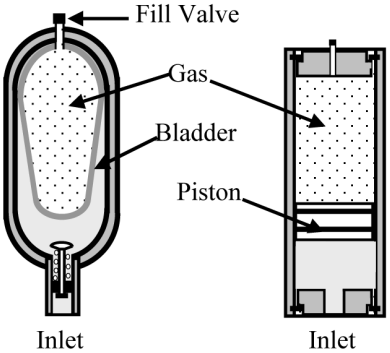
ISO Designation	Symbol	Picture Representation
Flow Control Valves (Cont'd.)		
Flow Dividers Spool Type Bi-directional (Functional Symbol)		
Flow Dividers Motor Type		
		
Pressure Control Valves Chapter 14 (Except Relief and Unloading Valves)		
Sequence Valve Internal Pilot		
External Pilot		
Internal Pilot Without Check Valve		

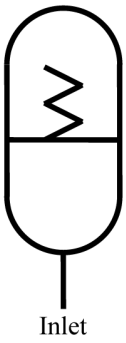
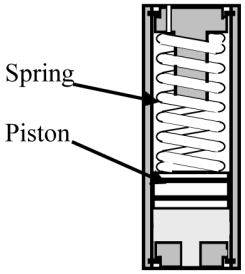
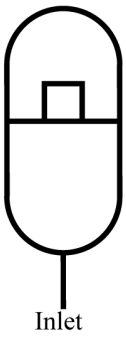
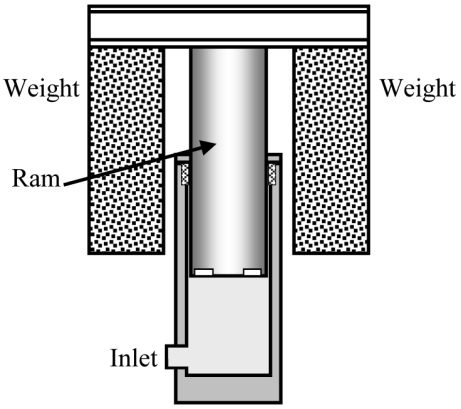
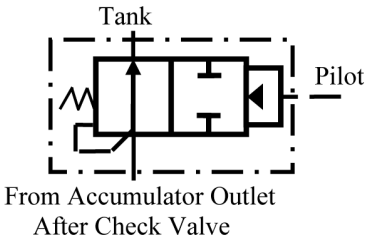
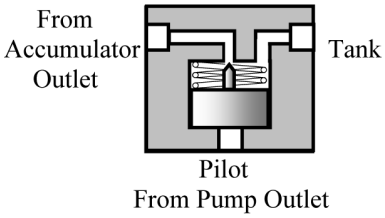
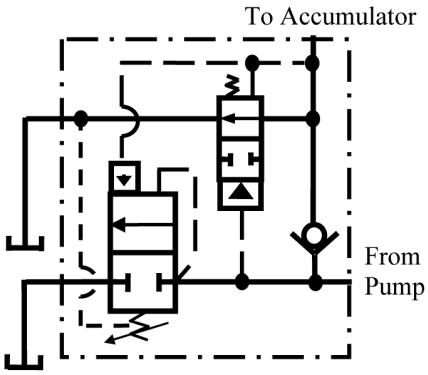
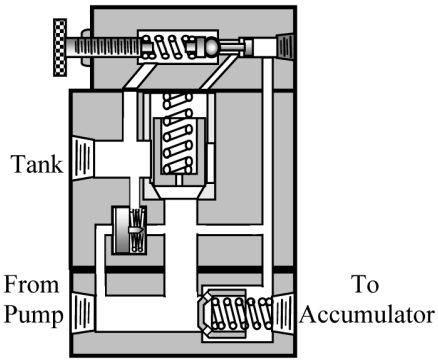
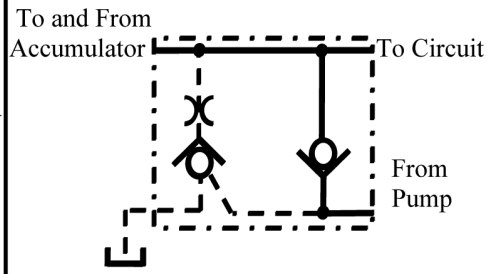
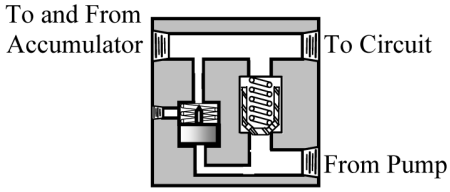
ISO Designation	Symbol	Picture Representation
Pressure Control Valves (Except Relief and Unloading Valves) (Cont'd.)		
Kick Down Sequence Valve		
Counter Balance Valve		
Internal Pilot Internal Drain		
External Pilot Internal Drain		
Internal and External Pilot, Internal Drain		
Reducing Valves		
Reducing Only With and Without Bypass Check		
Reducing, Relieving		
Without Bypass Check		
With Bypass Check		
With External Drain		

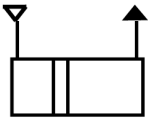
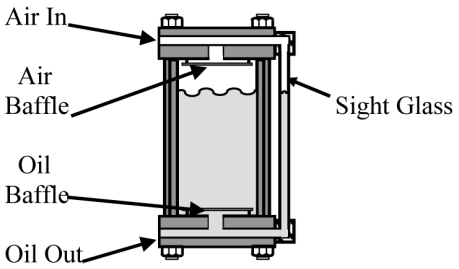
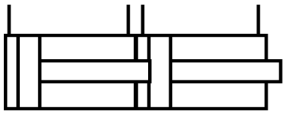
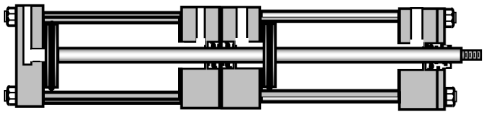
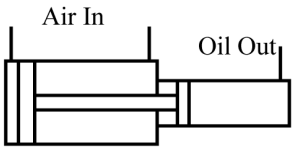
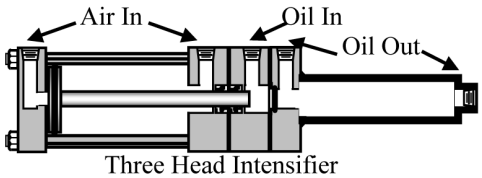
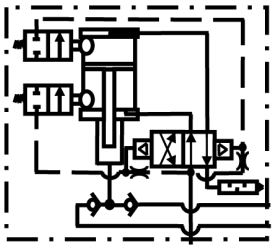
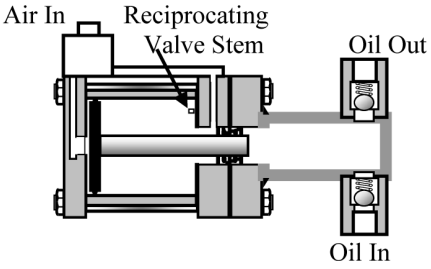
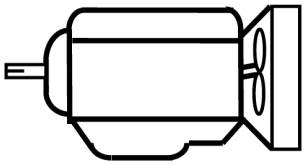
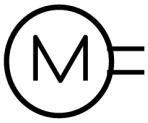
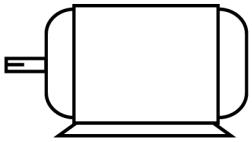
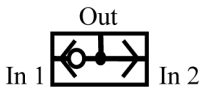
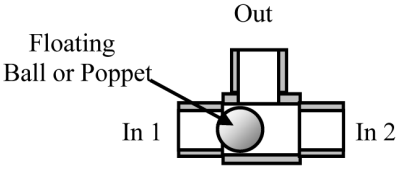
ISO Designation	Symbol	Picture Representation
Pressure Control Valves (Except Relief and Unloading Valves) (Cont'd.) Air Line Regulator (Reducing Valve)	 Non-Relieving  Self Relieving	 
Actuators Chapter 15 Single Acting Rams Push Type The Inlet Port Can be Anyplace on the Body The Only Seals are Where the Rams Exit The Body. Pull Type May Have a Cover Over the Top Ram to Protect Personnel	 	 
Single Acting Telescoping Double Acting Telescoping	 	
Single Acting Spring Return		

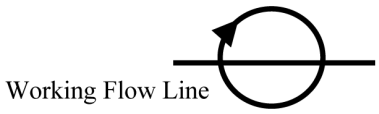
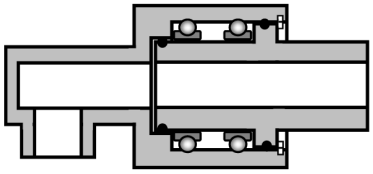
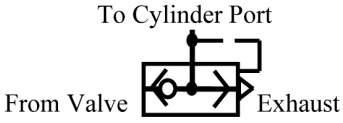
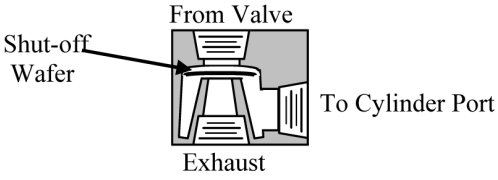
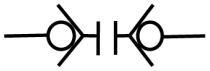
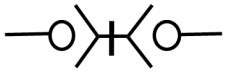
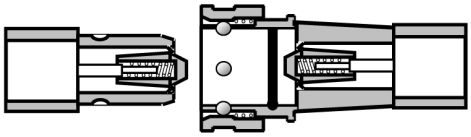
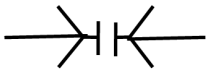
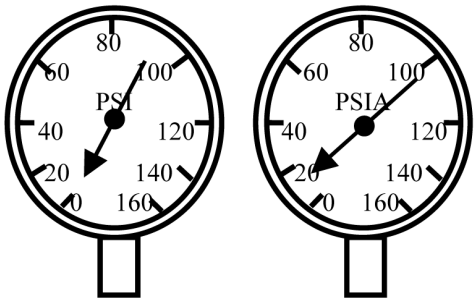
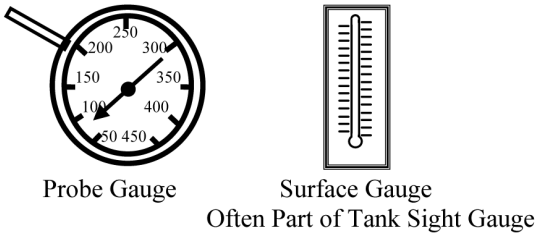
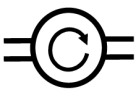
ISO Designation	Symbol	Picture Representation
Actuators (Cont'd.)		
Single Acting Spring Extend		
Non-rotating Rod Cylinder (Twin Rod)	 Symbol From Manufacturers Catalog	
Double Acting Cylinder With Cushions Both Ends. Arrow through Cushion Blocks Indicate They Are Adjustable. May Have Cushions Either End, Neither End or Both Ends..		
Double Acting Cylinder Non-Cushioned		
Double Acting, Double Rod End Cylinder Non-Cushioned		
Cable Cylinder	 Symbol From Manufacturers Catalog	
Rodless Cylinder (Slot Type)	 Symbol From Manufacturers Catalog	
Rodless Cylinder (Band Type)		

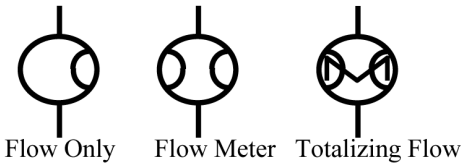
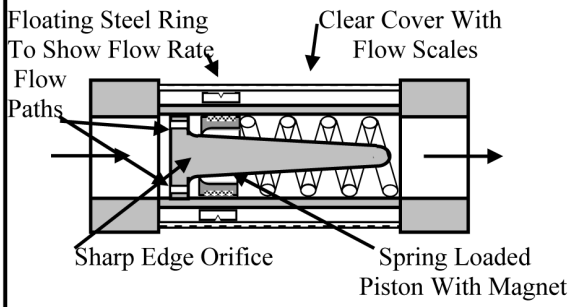
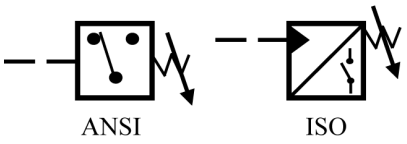
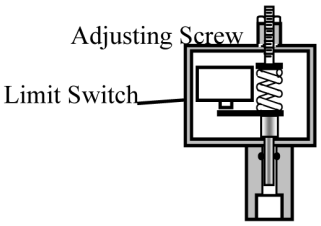
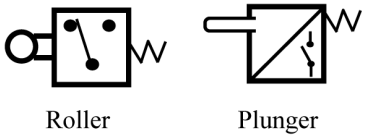
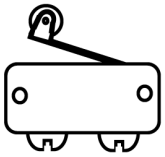
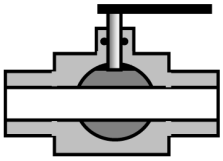
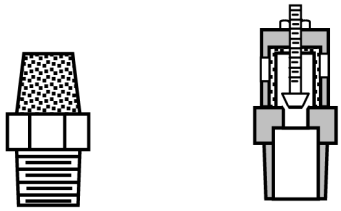
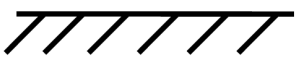
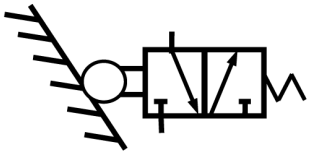
ISO Designation	Symbol	Picture Representation
Actuators (Cont'd.)		
Diaphragm Type		
		
Bellows Type	 	 
Rotary Actuators		
Vane Type		
		
Helical Screw		
Chain & Sprocket		
Rack & Pinion		

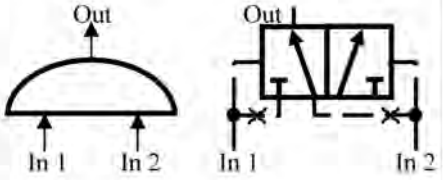
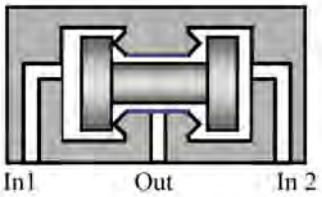
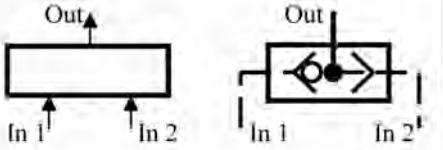
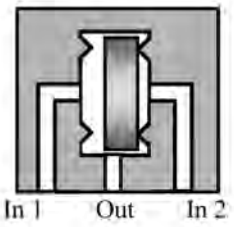
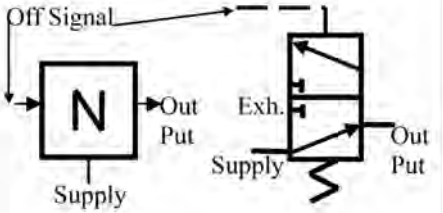
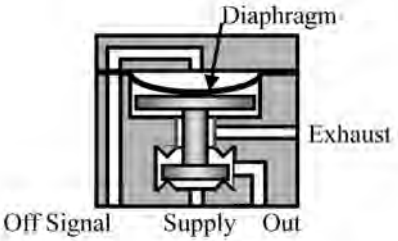
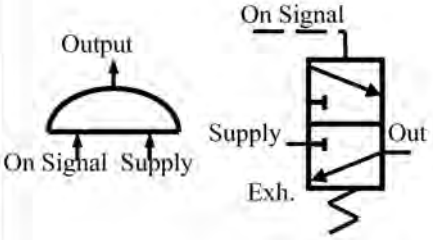
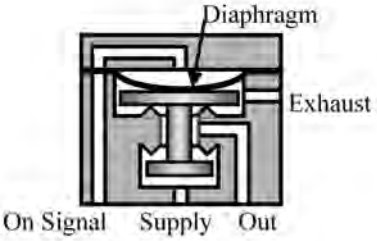
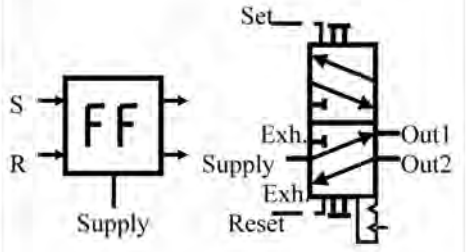
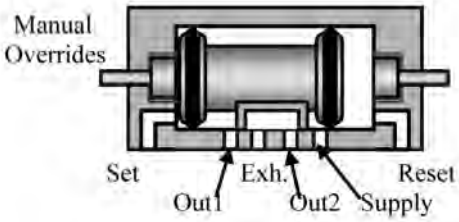
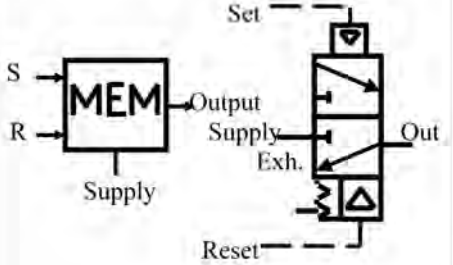
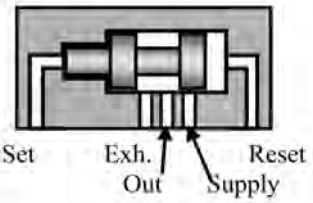
ISO Designation	Symbol	Picture Representation
Actuators (Cont'd.)		
Hydraulic Motors Fixed Displacement Bi-directional		 Gear Motor
Variable Volume Bi-directional		 Piston Motor
Air Motors Uni-directional		
Bi-directional		 Vane Motor
Accumulators Chapter 16		
Gas Charged Non -separated		
Gas Charged Bladder or Piston Type		

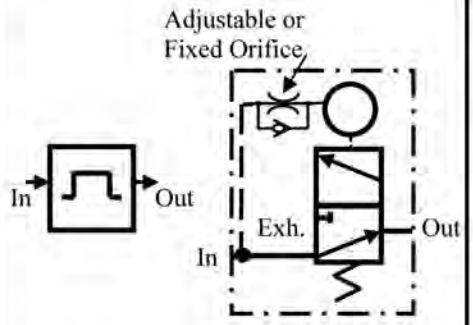
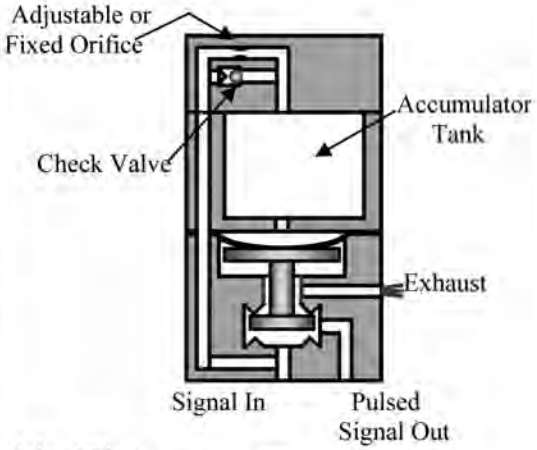
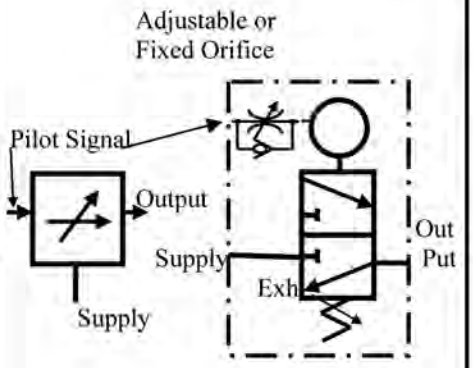
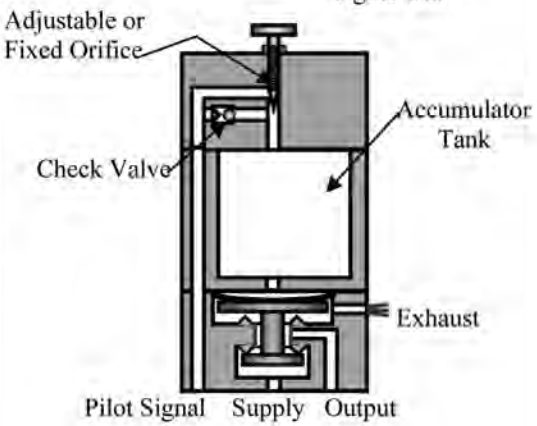
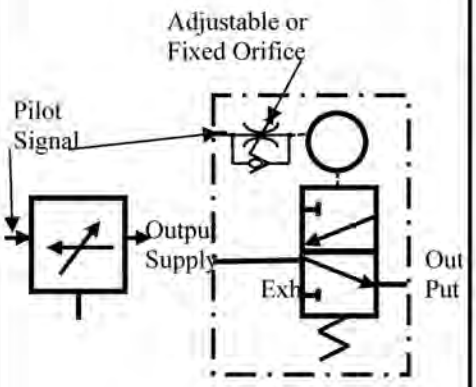
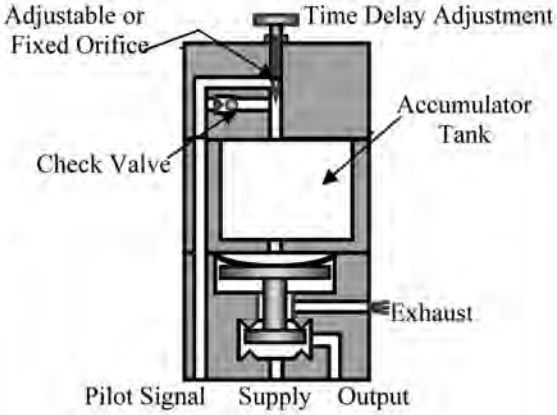
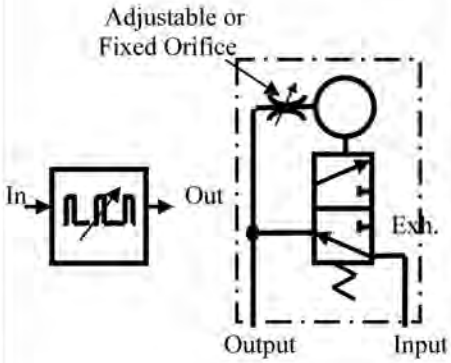
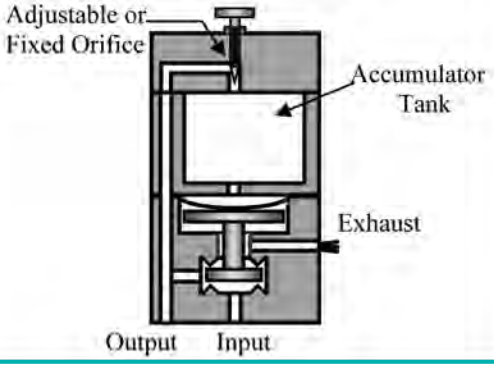
ISO Designation	Symbol	Picture Representation
Accumulators (Cont'd.)		
Spring Loaded		
Weight Loaded		
Accumulator Dump Valve (High Ratio)		
Accumulator Unloading Relief Valve and Dump Valve For Fixed Volume Pumps		
Accumulator Dump and Isolation Check Valve For Pressure Compensated Pumps		

ISO Designation	Symbol	Picture Representation
Air-Oil & Intensifiers Chapter 17		
Air-Oil Tank Changing a Pneumatic Pressure into an Equal Liquid Pressure		
Tandem Cylinder		
Intensifier or Booster	 Single Shot Intensifier	 Three Head Intensifier
Air-Oil Intensifier Reciprocating Type Self Contained		
Miscellaneous Fluid Power Items Chapter 18		
Heat Motor		
Electric Motor		
Shuttle Valve		

ISO Designation	Symbol	Picture Representation
Miscellaneous Fluid Power Items (Cont'd.)		
Rotating Union		
Quick Exhaust		
Quick Disconnects Without Shut-offs Shown Dis-connected		
Shown Connected		
Shown Without Shut-offs		
Pressure Gauges		
Temperature Gauges		
Tachometer Measurement of Rotation in RPM		
Torque Meter Measurement of Torque		

ISO Designation	Symbol	Picture Representation
Miscellaneous Fluid Power Items (Cont'd.)		
Liquid or Gas Flow Meter		
Pressure Switch		
Limit Switch		
Shut-off Valve		
Muffler (Silencer)		
Machine Member To Show Contact by a Valve or Other Device With a Machine Part		

ISO Designation	Symbol	Picture Representation
Air Logic Chapter 19	**ANSI B93-38 1976** **ISO Graphic Symbol**	
"AND" Element A Signal at "In 1" AND "In 2" Gives an Output. Output Always Lowest Of Input Signals		
"OR" Element A Signal at "In 1" OR "In 2" Gives an Output. Output Always Highest Input Signals		
"NOT" Element 3-Way Normally Open "Supply" Goes to Output Until an "OFF Signal" is Received.		
"YES" Element 3-Way Normally Closed Similar to "AND" But Can Choose "Output" Signal		
"FLIP-FLOP" Element 4-Way Double Pilot Operated Supplies and Exhaust's Either of Two Outputs		
"MEMORY" Element 3-Way Normally Closed Supplies or Exhaust's a Single Port. Can be "Reset" While "Set" Signal is on.		

ISO Designation	Symbol	Picture Representation
Air Logic (Cont'd.)	**ANSI B93-38 1976 ISO Graphic Symbol**	
"DIFFERENTIATOR" Element 3-Way Normally Open (Also Called a "ONE SHOT" Element or an ("IMPULSE GENERATOR") A Fixed Signal in Gives a Fixed or Adjustable Timed Signal Out		
"ON DELAY TIMER" Element 3-Way Normally Closed An. "Output" is Given After a "Pilot Signal" Has Been on for a Fixed or Adjustable Time Delay.		
"OFF DELAY TIMER" Element 3-Way Normally Open The "Output" is Lost After a "Pilot Signal" Has Been on for a Fixed or Adjustable Time Delay		
"FREQUENCY GENERATOR" Element 3-Way Normally Open An "Input" Signal Gives a Fixed or Adjustable Output for One Time Period, Then it Goes Off for Two Time Periods.		

ISO Designation	Symbol	Picture Representation
Air Logic (Cont'd.)		
"PRESSURE DECAY SENSOR" Element 3-Way Normally Open Same as a "NOT" except it Takes a lower "Pilot Signal" to Shift it. Used for Indicating end of Stroke of Cylinders.	Manufacturers Symbol / ISO Equivalent Pilot Signal, &, Output, Supply, Supply Exh, Out Put	Diaphragm, Exhaust Pilot Signal, Supply, Output
"SEQUENCER" Element A Combination of an "AND", "OR" and a "MEMORY" Element in a Common Housing to Simplify Circuit Design. Used in Series To Make a Complete Operating System.	1, 3 1, 3, MEM Out, In3, Start, Supply, Reset, Set, Supply, Reset	Output 1, Start, Supply, Reset, Set, Supply, Reset In3 Feedback
"SEQUENCE" Element Gives or Drops a Signal at a pre-determined Adjustable Pressure.	Normally Closed, Normally Open Pressure Sequence	Normally Closed, Normally Open Vacuum Sequence
"ANTI TIEDOWN" Element Requires Two Inputs to Get an Output	From PB1, From PB2, Output	Guarded Palm Buttons Box With Logic Parts
Counters: Counts Number of Cycles	Input Signal, Reset	0000432, Reset
Counts Down From Pre-selected Number Then Gives an Output	Output Signal, Reset, Input Signal, Output Signal, Supply	Set Buttons 0000000, 0000021, Reset
Pressure Indicator		

Absolute viscosity - the ratio of shear stress to shear rate. It is a fluid's internal resistance to flow. The common unit of absolute viscosity is the poise. Absolute viscosity divided by fluid density equals kinematic viscosity.
Absorption - The physical mechanism by which one substance attracts and takes up another substance (liquid, gas, or vapor) into its interior.
Accelerator - A substance that hastens the vulcanization of an elastomer, causing it to take place in a shorter time period or at a lower temperature.
Accumulator -A container in which fluid is stored under pressure as a source of fluid power.
Accumulator, hydro-pneumatic bladder - A hydro-pneumatic accumulator in which the liquid and gas are separated by an elastic bag or bladder.
Actuator, pneumatic/hydraulic - A device in which power is transferred from one pressurized medium (pneumatic) to another (hydraulic) without intensification.
Additive - A chemical added to a fluid to impart new properties or to enhance those that already exist.
Adiabatic compression occurs when no heat is transferred to or from the air during compression.
Adsorption - The physical mechanism by which one substance attracts another substance (either solid, liquid, gas, or vapor) to its surface and through molecular forces causes the incident substance to adhere to that surface.
Aftercooler - A device that cools a gas after it has been compressed.
Aftercooling - The cooling of air after it has been compressed to lower its temperature and precipitate condensed vapors.
Afterfilter - A filter that follows the compressed air dryer, usually for the protection of downstream equipment from desiccant dust.
Air bleeder - A device for removal of air.
Air breather - A device permitting air movement between atmosphere and the component in which it is installed.
Air motor - A device that converts pneumatic fluid power into mechanical torque and motion. It usually provides rotary mechanical motion.
Air, compressed (pressure) - Air at any pressure greater than atmospheric pressure.
Air, dried - Air with moisture content lower than the maximum allowable for a given application.
Air, free - Air at ambient temperature, pressure, relative humidity, and density.
Air, saturated - Air at 100% relative humidity, with a dew point equal to temperature.
Air, standard - Air at a temperature of 68.8 F, a pressure of 14.70 pounds per square inch absolute, and a relative humidity of 36% (0.0750 pounds per cubic foot). In gas industries the temperature of "standard air" is usually given as 60.8 F. producing tiny bubbles that expand explosively at the pump outlet, causing metal erosion and eventual pump destruction.
Air - A gas mixture consisting of nitrogen, oxygen, argon, carbon dioxide, hydrogen, small quantities of neon, helium and other gases.
Amplification, power - The ratio between the output signal variations and the corresponding input (control) power variation (for analog devices only).
Amplification, pressure - The ratio between outlet pressure and inlet (control) pressure.
Amplification - The ratio between the output signal variations and the control signal variations (for analog devices only)
Analog - Of or pertaining to the general class of fluidic devices or circuits whose output varies as a continuous function of its input.
AND device - A control device that has its output in the logical 1 state if and only if all the control signals assume the logical 1 state.
Aniline point - The lowest temperature at which a liquid is completely miscible with an equal volume of freshly distilled aniline (ASTM Designation D611-64).
Anti-foam agent - One of two types of additives used to reduce foaming in petroleum products: silicone oil to break up large surface bubbles, and various kinds of polymers that decrease the number of small bubbles entrained in the oils.
Asperities - Microscopic projections on metal surfaces resulting from normal surface-finishing processes. Interference between opposing asperities in sliding or rolling applications is a source of friction, and can lead to metal welding and scoring. Ideally, the lubricating film between two moving surfaces should be thicker than the combined height of the opposing asperities.
Bactericide - An additive included in the formulations of water-mixed fluids to inhibit the growth of bacteria.
Bernoulli's Law - If no work is done on or by a flowing frictionless liquid, its energy due to pressure and velocity remains constant at all points along the streamline.
Beta Ratio - A rating applied to filters (it is also known as the "filtration ratio." It is a measure of the particle-capture efficiency of a filter element.
The ISO 16889 Multipass Test Procedure passes fluid through a test circuit to check for contaminant retention. A measured amount of contaminant is injected upstream of the filter and laser particle counters record particles in to particles out across the filter. When 100,000 particles are measured upstream of a 10 (filter and 10,000 downstream it would have a Beta Ratio of 10. (B_{10}(=100,000/10,000=10)

A Beta Ratio is of no use alone but is required to find the filter's efficiency rating. Efficiency of a filter element is what counts when comparing one filter to another. The higher the efficiency, the fewer contaminants get by. Efficiency coupled with the volume of contaminant retention can make a more expensive filter cost less due to its longer useful life.
Bleeding - Migration to the surface of plasticizers, waxes, or similar materials to form a film or beads.
Boundary lubrication - A form of lubrication between two rubbing surfaces without development of a full-fluid lubricating film. Boundary lubrication can be made more effective by including additives in the lubricating oil that provide a stronger oil film, thus preventing excessive friction and possible scoring. There are varying degrees of boundary lubrication, depending on the severity of service.
Boyle's Law - The absolute pressure of a fixed mass of gas varies inversely as the volume, provided the temperature remains constant.
Breakout - The force necessary to inaugurate sliding. Expressed in the same terms as friction. An excessive breakout value indicates the development of adhesion.
Breathing capacity - A measure of flow

rate through an air breather.Bulk modulus - The measure of a fluid's resistance to compressibility. It is the reciprocal of compressibility.
Cavitation - A localized gaseous condition within a liquid stream that occurs where the pressure is reduced to the liquid's vapor pressure, often as a result of a solid body, such as a propeller or piston, moving through the liquid. Also, the pitting or wearing away of a solid surface as a result of low fluid levels that draw air into the system.
Charles' Law - The volume of a fixed mass of gas varies directly with absolute temperature, provided the pressure remains constant.
Circuit, meter-in - A speed-control circuit in which the control is achieved by regulating the supply flow to the actuator.
Circuit, meter-out - A speed-control circuit in which the control is achieved by regulating the exhaust flow from the actuator.
Circuit, open - A circuit in which return fluid is directed to the reservoir before reciprocation.
Circuit, regenerative - A circuit in which pressurized fluid discharged from a component is returned to the system to reduce input power requirements.
Circuit, sequence - A circuit that establishes the order in which two or more phases of a circuit occur.
Circuit - An arrangement of interconnected components and parts.
Cold flexibility - Flexibility following exposure to a predetermined time.
Cold flow - Continued deformation under stress.
Compatibility, seal - Ability of an elastomer to resist the action of a
fluid on its dimensional and mechanical properties.
Compressibility - The change in volume of a unit volume of a fluid when subjected to a unit change in pressure.
Compression efficiency - The ratio of the theoretical work required (in a given process) to the actual work required to be done to compress and de-liver the air. Expressed as a percentage. Compression efficiency accounts for fluid-friction losses, leakage, and thermodynamic variations from the theoretical process.
Compression modulus - The ratio of the compressive stress to the resulting compressive strain (the latter expressed as a fraction of the original height or thickness in the direction of the force). Compression modulus may be either static or dynamic.
Compression ratio - The ratio of the absolute discharge pressure to the absolute intake pressure.
Compression set - The amount by which a rubber specimen fails to return to original shape after release of the compressive load.
Compressor - A device that converts mechanical force and motion into pneumatic fluid power.
Condensation - The process of changing a vapor into a liquid condensate by the extraction of heat.
Conditioner, air - An assembly comprising a filter, a pressure-reducing valve with gauge, and a lubricator, intended to deliver compressed air in suitable condition for its application.
Conductor - A component whose primary function is to contain and direct fluid.
Contaminant - Any material or substance that is unwanted or adversely affects the fluid power system or components, or both.
Control - A device used to regulate the function of a component or system.
Controller - A device that senses a change of fluid state and automatically makes adjustments to maintain the state of the fluid between predetermined limits, e.g., pressures, temperatures, etc.
Copolymer - A polymer consisting of two different monomers that are chemically combined.
Corrosion inhibitor - An additive that protects wetted metal surfaces from chemical attack by water or other contaminants. Polar compounds wet the metal surface preferentially, protecting it with a film of oil. Other compounds may absorb water by incorporating it in a water-in-oil emulsion so that only the oil touches the metal surface. Another type of corrosion inhibitor combines chemically with the metal to present a non-reactive surface.
Creep - The progressive relaxation of a given rubber material while it is under stress. This relaxation eventually results in permanent deformation or "set."
Cushion - A device that provides controlled resistance to motion.
Cylinder - A device that converts fluid power into linear mechanical force and motion. It usually consists of a movable element such as a piston and piston rod, plunger or ram, operating within a cylindrical bore.
Cylinder, adjustable stroke - A cylinder equipped with adjustable stops at one or both ends to limit piston travel.
Cylinder, area, piston, effective - The area upon which fluid pressure acts to provide a mechanical force.
Cylinder, area, piston rod - The cross-sectional area of the piston rod.
Cylinder, bore - The internal diameter of the cylinder body.
Cylinder cap - An end closure for a cylinder that completely covers the bore area.
Cylinder capacity - The volume of a theoretically incompressible fluid that would be displaced by the piston during a complete stroke. (For double-acting cylinders it must be given for both directions of stroke.)
Cylinder capacity, extending - The volume required for one full extension of a cylinder.
Cylinder capacity, retracting-Volume (annular) absorbed by one full retraction of the cylinder oscillation of an output shaft.
Cylinder, cushioned - A cylinder with a piston-assembly deceleration device at one or both ends of the stroke.
Cylinder, differential - A double-acting cylinder in which the ratio of the area of the bore to the annular area between the bore and the piston rod is significant in circuit function.
Cylinder, double-acting - A cylinder in which fluid force can be applied to the moveable element (piston) in either direction.
Cylinder, double rod - A cylinder with a single piston and a piston rod extending from each end.
Cylinder, dual stroke - A cylinder combination that provides two working strokes.
Cylinder, duplex - A unit comprised of two cylinders with independent control, mechanically connected on a common axis to provide three or four positions depending on the method of application.
Cylinder force, theoretical - The pressure multiplied by the effective piston area (ignoring friction). For double-acting cylinders, the value must be given for both directions of stroke.
Cylinder, piston-type - A cylinder in

which the piston has a greater cross-sectional area than the piston rod.
Cylinder, plunger (ram) - A cylinder in which the piston has the same cross-sectional area as the piston rod.
Cylinder, rotating - A cylinder in which the piston and piston rod, plunger or ram, is permitted to rotate with reference to the cylinder housing.
Cylinder, single-acting - A cylinder in which the fluid force can be applied to the movable element in only one direction.
Cylinder, tandem - An arrangement of at least two pistons on the same rod moving in separate chambers on the same cylinder body allowing the compounding of force on the piston rod.
Cylinder, telescoping - A cylinder with two or more stages or extensions, achieved by hollow piston rods sliding one within the other (may be single- or double-acting).
Cylinder, tie rod - A cylinder with head and cap end closures that are secured by tie rods.
Cylinder, rotary actuator - A cylinder that translates piston reciprocation into oscillation of an output shaft.
Darcy's Formula - A formula used to determine the pressure drop due to flow friction through a conduit.
Deliquescent -A special hygroscopic compound with absorptive properties that can separate moisture.
Demulsibility - The ability of an oil to separate from water.
Density - The weight of a given volume of air, usually expressed in lb/ft3 at standard temperature and pressure.
Desiccant - A material that tends to remove moisture from compressed air.
Dew point - The temperature at which vapors in a gas condense. For practical purposes, it must be referred to a stated pressure.
Dewaxing - Removal of paraffin wax from lubricating oils to improve low-temperature properties, especially to lower the cloud point and pour point.
Digital - Of or pertaining to the general class of fluidic devices or circuits whose output varies in discrete steps (i.e., pulses or "on-off" characteristics).
Displacement - The net volume swept by the moving parts of a compressor per unit of time. This term applies only to positive-displacement compressors.
Displacement, volumetric - The volume absorbed or displaced per stroke of a cylinder or per cycle of a pump or motor.
Dissolved air - Air that is dispersed at a molecular level in hydraulic fluid to form a single phase.
Dissolved water - Water that is dispersed at a molecular level in hydraulic fluid to form a single phase.
Dither - A low-amplitude, relatively high-frequency periodic electrical signal, sometimes superimposed on a servovalve input to improve system resolution. Dither is expressed by the dither frequency (Hz) and the peak-to-peak dither current amplitude.
Droop - The deviation between no flow secondary pressure and secondary pressure at a given flow.
Dryer, compressed air - A device for reducing the moisture content of the working compressed air.
Durometer - 1. An instrument for measuring the hardness of rubber. It measures the resistance to the penetration of an indenter point into the surface of rubber. 2. Numerical scale of rubber hardness.
EP additive - A lubricant additive that prevents sliding metal surfaces from seizing under conditions of extreme pressure (EP). At the high local temperatures associated with metal-to-metal contact, an EP additive com-bines chemically with the metal to form a surface film that prevents the welding of opposing asperities, and the consequent scoring that is destructive to sliding surfaces under high loads. Reactive compounds of sulfur, chlorine, or phosphorus are used to form these inorganic films.
Efficiency - The ratio of output to the corresponding input.
Elasticity - The property of a material that tends to return to its original shape after deformation.
Elastomer - Any synthetic or natural material with resilience or memory sufficient to return to its original shape after distortion.
Elongation - Generally means "ultimate elongation" or percent increase in original length of a specimen when it breaks.
Emulsion, oil-in-water - A dispersion of oil in a continuous phase of water.
Emulsion, water-in-oil - A dispersion of water in a continuous phase of oil.
Emulsifier- An additive the promotes formation of a stable mixture...or emulsion ..of oil and water.
Emulsion - A homogeneous dispersion of two immiscible liquids, generally of a milky or cloudy appearance.
Entrained air - A mechanical mixture of air bubbles having a tendency to separate from the liquid phase.
Expectancy, life - The predicted working period during which a component or system will maintain a specified level of performance under specified conditions. Sometimes expressed in statistical terms as a probability.
Filter - 1. A device whose primary function is the removal by porous media of insoluble contaminants from a liquid or a gas. 2. Chemically inert, finely divided material added to the elastomer to aid in processing and improve physical properties.
Filter, strainer - A coarse hydraulic filter, usually of woven wire construction. This may be in the form of a complete filter or just an element.
Filter, bypass (reserve) - A filter which provides an alternate unfiltered flow path around the filter element when a preset differential pressure is reached.
Filter, spin-on - A filter with spin-on element sealed in its own pressure housing for independent mounting to the filter.
Filtration ratio (bm) - The ratio of the number of particles greater than a given size (b) in the influent fluid to the number of particles greater than the same size (m) in the effluent fluid.
Filter, strainer - A coarse hydraulic filter usually of woven wire construction. This may be in the form of a complete filter or just an element.
Fire-resistant fluid - Hydraulic oil used especially in high-temperature or hazardous applications. Three common types of fire-resistant fluids are: (1) water-petroleum oil emulsions, in which the water prevents burning of the petroleum constituent; (2) water-glycol fluids; and (3) non-aqueous fluids of low volatility, such as phosphate esters, silicones, polyolesters, and halogenated hydrocarbon-type fluids.
Fitting - A connector or closure for fluid power lines and passages.
Fitting, compression - A fitting that seals and grips by manual adjustable deformation.
Fitting, flange - A fitting that utilizes a

radially extending collar for sealing and connection.
Fitting, flared - A fitting that seals and grips by a pre-formed flare at the end of the tube.
Fitting, flareless - A fitting that seals and grips by means other than a flare.
Flash point - The temperature to which a liquid must be heated under specified conditions of the test method to give off sufficient vapor to form a mixture with air that can be ignited momentarily by a flame.
Flip flop - A digital component or circuit with two stable states and sufficient hysteresis so that it has "memory." Its state is changed with a control pulse; a continuous control signal is not necessary for it to remain in a given state.
Flow characteristic curve - The change in regulated (secondary) pressure occurring as a result of a change in the rate of air flow over the operating range of the regulator.
Flow rate - The volume, mass, or weight of a fluid passing through any conductor per unit of time.
Flow, laminar (streamline) - A flow situation in which fluid moves in parallel lamination or layers.
Flow, output - The flow rate discharged at the outlet port.
Flow, turbulent - A flow situation in which the fluid particles move in a random fluctuating manner.
Flow - Movement of fluid generated by pressure differences.
Fluid capacity - The liquid volumecoincident with the "high" mark of the level indicator.
Fluid friction - Friction due to the viscosity of fluids.
Fluid logic - A branch of fluid power associated with digital signal sensing and information processing, using components with or without moving parts.
Fluid miscibility - Capacity of fluids to be mixed in any ratio without separation into phases.
Fluid power system - A system that transmits and controls power through use of a pressurized fluid within an enclosed circuit.
Fluid power - Energy transmitted and controlled through use of a pressurized fluid.
Fluid stability - The resistance of a fluid to permanent changes in properties.
Fluid stability, oxidation - The resistance of a fluid to permanent changes caused by chemical reaction with oxygen.
Fluid, anti-corrosive - A fluid containing metal corrosion inhibitors.
Fluid, aqueous - A fluid that contains water as a major constituent besides the organic material. The fire-resistance properties are derived from the water content.
Fluid, fire-resistant - A fluid that is difficult to ignite and shows little tendency to propagate flame.
Fluid, hydraulic - A fluid suitable for use in a hydraulic system.
Fluid, Newtonian - Fluid having a viscosity that is always independent of the rate of shear.
Fluid, pneumatic - A fluid suitable for use in pneumatic systems . . . usually air.
Fluid, rust protection - Capacity of a fluid to prevent the formation of rust under specified conditions.
Fluid - A liquid, gas, or combination thereof.
Force motor - A type of electromechanical transducer having linear motion used in the input stages of servovalves.
Free air - Any compressible gas, air, or vapor trapped within a hydraulic system that does not condense or dissolve to form a part of the system fluid.
Free water - Water droplets or globules in the system fluid that tend to accumulate at the bottom or top of the system fluid depending on the fluid's specific gravity.
Frequency response - The changes, under steady-state conditions, in the output variable that are caused by a sinusoidal input variable.
Full-fluid-film lubrication - The presence of a continuous lubricating film sufficient to completely separate two surfaces (as distinct from boundary-layer lubrication). Full-fluid-film lubrication is normally hydrodynamic lubrication, whereby the oil adheres to the moving part and is drawn into the area between the sliding surfaces, where it forms a pressure or hydrodynamic wedge.
Gauge damper (snubber) - A device employing a fixed or variable restrictor inserted in the pipeline leading to a pressure gauge. It prevents damage to the gauge mechanism caused by rapid fluctuations of fluid pressure.
Gauge protector - A device inserted in the pipeline to a pressure gauge and arranged to isolate the pressure gauge from the fluid pressure if this pressure exceeds a predetermined limit. The device can usually be adjusted to suit the range of the pressure gauge.
Gauge, bourdon tube - A pressure gauge in which the sensing element is a curved tube that tends to straighten out when subjected to internal fluid pressure.
Gauge, diaphragm - A gauge in which the sensing element is relatively thin and its inner portion is free to deflect with respect to its fixed periphery.
Gauge, instrument - An instrument or device for measuring, indicating, or comparing a physical characteristic.
Gauge, pressure - A gauge that indicates the pressure in the system to which it is connected.
Head - The height of a column or body of fluid above a given point expressed in linear units. Head is often used to indicate gauge pressure. Pressure is equal to the height times the density of the fluid.
Head, cylinder - The cylinder end closure that covers the differential area between the bore area and the piston rod area.
Head, friction - The pressure required to overcome the friction at the interior surface of a conductor and between fluid particles in motion. It varies with flow, size, type and condition of conductors and fittings, and the fluid characteristics.
Head, pressure - The pressure due to the height of a column or body of fluid.
Head, static - The height of a column or body of fluid above a given point.
Heat exchanger - A device that transfers heat through a conducting wall from one fluid to another. (Typically to cool a system.)
Heater - A device that transfers heat through a conducting wall from one fluid to another. (Typically to warm up a system.)
Hose, wire-braided - Hose consisting of a flexible material reinforced with woven wire braid.
Hose - A flexible line or conductor whose nominal size is its inside diameter.
Hydraulic amplifier - A fluid device that enables one or more inputs to control a source of fluid power and thus is capable of delivering at its output an enlarged reproduction of the essential characteristics of the input. Hydraulic amplifiers may utilize

sliding spools, nozzle-flappers, jet pipes, etc.

Hydraulic fluid - The fluid that serves as the power transmission medium in a hydraulic system. The most commonly used fluids are petroleum and synthetic oils, oil-water emulsions, and water-glycol mixtures. The principal requirements of a premium hydraulic fluid are: proper viscosity, high viscosity index, anti-wear protection (if needed), good oxidation stability, adequate pour point, good demulsibility, rust inhibition, resistance to foaming, and compatibility with seal materials. Anti-wear oils are frequently used in compact, high-pressure, and high-capacity pumps that require extra lubrication protection.

Hydraulic motor - A device that converts hydraulic fluid power into mechanical force and motion. It usually provides rotary mechanical motion.

Hydraulic motor efficiency, hydro-mechanical - The ratio of the effective torque to the derived torque.

Hydraulic motor efficiency, overall - The ratio of the output power to the effective power.

Hydraulic motor efficiency, volumetric - The ratio of the derived output flow to the effective input flow.

Hydraulic motor, fixed displacement - A hydraulic motor in which the displacement per unit of output motion cannot be varied.

Hydraulic motor, flow, input - The flow rate crossing the transverse plane of the inlet port.

Hydraulic motor, gear, external - A motor having two or more external gears.

Hydraulic motor, gear, internal - A motor with an internal gear in engagement with one or more external gears.

Hydraulic motor, gear - A motor in which two or more gears act in arrangement as working members.

Hydraulic motor, vane - A motor in which the fluid under pressure acting on a set of radial vanes causes rotation of an internal member.

Hydraulic stepping motor - A hydraulic motor that follows the commands of a stepped input signal to achieve positional accuracy.

Hydraulics - The engineering science that pertains to liquid pressure and flow.

Hydrodynamics - The engineering science that governs the movement of liquids and the forces opposing that movement.

Hydrokinetics - The engineering science that pertains to the energy of liquid flow and pressure.

Hydro-pneumatics - The combination of hydraulic and pneumatic fluid power.

Hydrostatic transmission - The combination of one or more hydraulic pumps and motors to form a unit.

Hydrostatics - The engineering science that pertains to the energy of liquids at rest.

Ideal gases - Gasses that follow the perfect gas laws without deviation. There are no real ideal gases, but they provide a common starting point for calculations and corrections.

Immiscible - Incapable of being mixed without separation of phases. Water and petroleum oil are immiscible under most conditions, although they can be made miscible with the addition of a proper emulsifier.

Indicator, differential pressure - An indicator that signals a difference in pressure between two points in a fluid power system.

Inhibitor - Any substance that, when present in very small proportions, slows, prevents or modifies chemical reactions such as corrosion or oxidation.

Intensification, ratio of - The ratio of the secondary pressure to the primary pressure or of the primary flow rate to the secondary flow rate.

Intensifier, double-acting - A unit that magnifies the secondary fluid pressure regardless of the direction of flow of the primary fluid.

Intensifier, single-acting - A unit that only magnifies the fluid pressure in one direction of flow of the primary fluid.

Intensifier, single shot - An intensifier in which the continuous application of primary fluid at the inlet port can only give a limited volume of secondary fluid.

Intensifier - A device that converts low-pressure fluid power into higher-pressure fluid power.

Inter-cooling - The process of cooling air between stages of compression to liquefy condensed vapors and save power by reducing the temperature of air entering the next stage.

Isothermal compression - A compression arrangement in which the temperature of the air remains constant during compression.

Joint - A line-positioning connector.

Joint, rotary - A joint connecting lines that have relative operational rotation.

Kinematic viscosity - The absolute viscosity of a fluid divided by its density at the same temperature of measurement. It is the measure of a fluid's resistance to flow under gravity.

Leakage rate - The rate at which a gas or liquid passes through a barrier. Total leakage rate includes the amounts that diffuse or permeate through the material of the barrier as well as the amount that escapes around it.

Line, return - A conductor (pipe) to return the working fluid to the reservoir.

Line, working - A line that conducts fluid power.

Line - A tube, pipe, or hose for conducting fluid.

Lubricator - A device that adds controlled or metered amounts of lubricants into a pneumatic system.

Lubricity - The ability of an oil or grease to lubricate (also called film strength).

Magnetic plug - A plug that attracts and holds ferromagnetic particles.

Manifold - A conductor that provides multiple connection ports.

Maximum inlet pressure - The maximum rated gauge pressure applied to the inlet port of the regulator.

Mechanical efficiency - The ratio of the indicated horsepower to the actual shaft horsepower.

Memory - The tendency of a material to return to its original shape after deformation.

Miscible - Capable of being mixed in any concentration without separation of phases; e.g., water and ethyl alcohol are miscible.

Modulus of elasticity - One of the several measurements of stiffness or resistance to deformation. Often incorrectly used to indicate specifically static tension modulus.

Modulus - Tensile stress at a specified elongation. (Usually 100% elongation for elastomers.)

Moving parts logic - The technology of achieving logic control by means of fluid devices having moving parts.

Muffler - A device for reducing gas flow noise. Noise is decreased by backpressure control of gas expansion.

Newt - A unit of kinematic viscosity in the English system. It is expressed in square inches per second (see Stokes).
Newtonian fluid - A fluid, such as a straight mineral oil, whose viscosity does not change with rate of flow.
Non-Newtonian fluid - A fluid, such as a grease or a polymer containing oil (e.g. multi-grade oil), in which shear stress is not proportional to shear rate.
NOR device - A control device that has its output in the logical 1 state if and only if all the control signals assume the logical 0 state.
NOT device - A control device that has its output in the logical 1 state if and only if the control signal assumes the logical 0 state. (The NOT device is a single-input NOR device.)
Oil swell - The change in volume of a rubber article due to absorption of oil or other fluid.
OR device - A control device that has its output in the logical 0 state if and only if all the control signals assume the logical 0 state.
Out-gassing - A vacuum phenomenon wherein a substance spontaneously releases volatile constituents in the form of vapors or gases. In rubber compounds, these constituents may include water vapor, plasticizers, air, inhibitors, etc.
Output stage - The final stage of hydraulic amplifications used in a servovalve.
Oxidation inhibitor - A substance added in small quantities to a petroleum product to increase its oxidation resistance, thereby lengthening its service or storage life. Also called an anti-oxidant.
Ozone resistance - A material's ability to withstand the deteriorating effect of ozone (which generally causes cracking).
Packing - A sealing device consisting of bulk deformable material of one or more mating deformable elements, reshaped by manually adjustable compression to obtain and maintain effectiveness. Packing usually uses axial compression to obtain radial sealing.
Pascal's Law - A pressure applied to a confined fluid at rest is transmitted with equal intensity throughout the fluid.
Permanent set - The deformation remaining after a specimen has been stressed in tension for a definite period and released for a definite period.
Permeability - The rate at which a liquid or gas under pressure passes through a solid material by diffusion and solution. In rubber terminology, it is the rate of gas flow expressed in atmospheric cubic centimeters per second through an elastomeric material one centimeter square and one centimeter thick (atm cm3/cm2 ∞cm∞sec)
Petroleum fluid - A fluid composed of petroleum oil that may contain additives and/or inhibitors.
Pipe - A conductor whose outside diameter is standardized for threading. Pipe is available in standard, extra-strong, or double extra-strong wall thickness.
Piston rod - The element that transmits mechanical force and motion from the piston.
Plasticizer - A substance -- usually a heavy liquid -- added to an elastomer to decrease stiffness, improve low-temperature properties, and improve processing.
Pneumatics - The engineering science pertaining to gaseous pressure and flow.
Poise - The standard unit of dynamic viscosity in the centimeter-gram-second (CGS) system. It is the ratio of the shearing stress to the shear rate of fluid and is expressed in millipascal sec. (equals 1 centipoise).
Polar compound - A chemical compound whose molecules exhibit electrically positive characteristics at one extremity and negative characteristics at the other. Polar compounds are used as additives in many petroleum products.
Polymer - A material formed by the joining together of many (poly) individual units (mer) of one or more monomers. Synonymous with elastomers.
Polytropic compression - A process that occurs when heat is transferred to or from air at a precise rate during compression so that PVn is constant.
Port - A terminus of a passage in a component to which conductors can be connected.
Port, differential pressure - A port that provides a passage to the upstream and downstream sides of a component.
Post cure - The second step in the vulcanization process for the more exotic elastomers. It provides stabilization of parts and drives off decomposition products resulting from the vulcanization process.
Pour point - The lowest temperature at which an oil or distillate fuel is observed to flow when cooled under conditions prescribed by test method ATM D 97. The pour point is 3C (5 F) above the temperature at which the oil in a test vessel shows no movement.
Power unit - A combination of pump, pump drive, reservoir, controls and conditioning components to supply hydraulic power to a system.
Pressure, absolute - The pressure above zero absolute, i.e., the sum of atmospheric and gauge pressure. In vacuum related work it is usually expressed in millimeters of mercury (mm-Hg).
Pressure, atmospheric - Pressure exerted by the atmosphere at any specific location. (Sea level pressure is approximately 14.7 pounds per square inch absolute. 1 bar = 14.5 PSI).
Pressure, back - The pressure encountered on the return side of a system.
Pressure, break loose (breakout) - The minimum pressure that initiates movement.
Pressure, burst - The pressure that causes failure of and consequential loss of fluid through the product envelope.
Pressure, charge - The pressure at which replenishing fluid is formed into a fluid power system.
Pressure, control range - The permissible limits between which system pressure may be set.
Pressure, cracking - The pressure at which a pressure-operated valve begins to pass fluid.
Pressure, differential (pressure drop) - The difference in pressure between any two points of a system or a component.
Pressure, gauge - The pressure differential above or below ambient atmospheric pressure.
Pressure, induced - The pressure generated by an externally applied force.
Pressure, inlet - The pressure at the apparatus inlet port.
Pressure, intensified - The outlet pressure in a fluid power cylinder required to slow the piston rod extending under regulated pressure introduced at the cap end.
Pressure, maximum inlet - The maximum rated gauge pressure applied to the inlet.
Pressure, nominal - A pressure value assigned to a component or system for the purpose of convenient designation. ■

Pressure, outlet - Pressure at the apparatus outlet port.
Pressure, override - The difference between the cracking pressure of a valve and the pressure reached when the valve is passing its rated flow.
Pressure, peak - The maximum pressure encountered in the operation of a component.
Pressure, pilot - The pressure in the pilot circuit.
Pressure, precharge - The pressure of compressed gas in an accumulator prior to the admission of a liquid.
Pressure, proof - The non-destructive test pressure -- in excess of the maximum rated operating pressure -- that causes no permanent deformation, excessive external leakage, or other resulting malfunction.
Pressure, rated - The qualified operating pressure that is recommended for a component or system by the manufacturer.
Pressure, shock - The pressure existing in a wave moving at sonic velocity.
Pressure, static - The pressure in a fluid at rest.
Pressure, surge - The pressure resulting from surge conditions.
Pressure, system - The pressure that overcomes the total resistance in a system. It includes all losses as well as useful work.
Pressure - Force per unit area, usually expressed in pounds per square inch or bar.
Pump, fixed-displacement - A hydraulic pump in which the volume displaced per cycle cannot be varied.
Pump, gear, external - A pump with two or more external gears.
Pump, gear, internal - A pump with an internal gear in engagement with one or more external gears.
Pump, gear - A pump in which two or more gears act in engagement as pumping members.
Pump, hydraulic - A device that converts mechanical force and motion into hydraulic fluid power.
Pump, multiple-stage - Two or more hydraulic pumps connected in series.
Pump, piston, axial - A pump having several pistons with mutually parallel axes that are arranged around and parallel to a common axis.
Pump, piston, inline- a pump having several pistons with mutually parallel axes arranged on a common plane.
Pump, piston, radial - A pump having several pistons arranged to operate radially.
Pump, piston - A pump in which the fluid volume is displaced by one or more reciprocating pistons.
Pump, screw - A hydraulic pump having one or more screws rotating in a housing.
Pump, vane, balanced - A pump in which the transverse forces on the rotor are balanced.
Pump, vane, unbalanced - A pump in which the transverse forces on the rotor are not balanced.
Pump, vane - A hydraulic pump having multiple radial vanes within a supporting rotor.
Pump, variable-displacement - A hydraulic pump in which the volume displaced per cycle can be varied.
Pump-motor - A unit that functions either as a pump or as a rotary motor.
Pump - A device that converts mechanical torque and motion into hydraulic fluid power.
Pumping or surge - The reversal of flow within a dynamic compressor. It takes place when insufficient pressure is generated to maintain flow.
Quick-disconnect coupling - A component that can quickly join or separate a fluid line without the use of tools or special devices.
Refrigerated dryer - A device that separates moisture by lowering the air temperature by means of refrigeration compressor and heat exchanger.
Regenerative dryer - A dryer that restores its own capacity to separate moisture -- without replacing the drying compound.
Regenerative circuit - see Circuit, regenerative.
Regulator, air line pressure - A regulator that transforms a fluctuating air pressure supply to provide a constant lower pressure output.
Reinforcing agent- a material dispersed in an elastomer to improve compression, shear or other stress properties.
Reservoir (tank) - A container for storage of liquid in a fluid power system.
Reservoir, hydraulic - A reservoir for storing and conditioning a liquid in a hydraulic system.
Reservoir, pressure-sealed - A sealed reservoir for storage of fluids under pressure.
Resilient - Capable of returning to original size and shape after deformation.
Reyn - The standard unit of absolute viscosity in the English system. It is expressed in pound-seconds per square inch.
Reynolds Number - A numerical ratio of the dynamic forces of mass flow to the shear stress due to viscosity. Flow usually changes from laminar to turbulent between Reynolds Numbers 2000 and 4000.
Ring, O- A ring that has a round cross-section.
Ring, piston - A piston sealing ring. It is usually one of a series and often is split to facilitate expansion or contraction.
Ring, scraper - A ring that removes material by a scraping action.
Rotation - The direction of rotation is always quoted as viewed looking at the shaft end. In dubious cases, provide a sketch.
Seal, cup - A sealing device with a radial base integral with an axial cylindrical projection at its outer diameter.
Seal, dynamic - A sealing device used between parts that have relative motion.
Seal, elastomer - A material having rubber-like properties; i.e., having the capacity for large deformation, with rapid and substantially complete recovery on release from the deforming force.
Seal, rod (shaft) - A sealing device that seals the periphery of a piston rod.
Seal, static (gasket) - A sealing device used between parts that have no relative motion.
Sensor - A device that detects and transmits changes in external conditions.
Separator - A device whose primary function is to isolate contaminants by physical properties other than size. (Separators remove gas from liquid media or remove liquid from gaseous media).
Servovalve - A valve that modulates output as a function of an input command.
Servovalve, electrohydraulic - A servovalve that is capable of continuously controlling hydraulic output as a function of an electrical input.
Servovalve, electrohydraulic, flow-control - An electrohydraulic servovalve whose primary function is control of output flow.
Servovalve hysteresis - The difference in the servovalve input currents required to produce the same output during a single cycle of valve input current when cycled at a rate below that at which dynamic effects are important.

Servovalve null leakage - The total internal leakage from the valve in the null position.
Servovalve, pressure-control - A hydraulic servo valve whose primary function is the control of output pressure.
Shear rate - The rate at which adjacent layers of fluid move with respect to each other, usually expressed as reciprocal seconds.
Shear stress - The frictional force overcome in sliding one layer of fluid along another, as in any fluid flow. The shear stress of petroleum oil -- or other Newtonian fluid -- at a given temperature varies directly with shear rate (velocity). The ratio between shear stress and shear rate is constant; this ratio is termed viscosity.
Shrinkage - The decreased volume of a seal, usually caused by extraction of soluble constituents from fluids, followed by air drying.
Silencer - A device for reducing gas flow noise. Noise is decreased by tuned resonant control of gas expansion.
Snubber-See gauge damper.
Solenoid, digital - An electronically energized device that generates on and off signals.
Solenoid, proportional - An electrical device that reacts proportionally to the strength of electrical signals.
Sorption - The term used to denote the combination of absorption and adsorption processes in the same substance.
Specific gravity, liquid - The ratio of the weight of a given volume of liquid to the weight of an equal volume of water.
Squeeze - The cross-sectional diametrical compression of an O-ring between the surface of the groove bottom and the surface of the other mating metal part in the gland assembly.
Stage - A hydraulic amplifier used in a servovalve. Servovalves may be single-stage, two-stage, three-stage, etc.
Standard - A document, or an object for physical comparison, for defining product characteristics, products, or processes; prepared by the consensus of a properly constituted group of those substantially affected and having the qualifications to prepare the standard for voluntary use.
Stokes - The standard unit of kinematic viscosity in the CGS system. It is expressed in square centimeters per second; 1 centistokes equals 0.01 stokes.
Strainer - See filter, strainer.
Surface tension - The surface force of a liquid in contact with a fluid by which it tends to assume a spherical form and to present the least possible surface. It is expressed in pounds per foot or dynes per centimeter.
Surfactant - A surface-active agent that reduces interfacial tension of a liquid. A surfactant used in petroleum oil may increase the oil's affinity for metals and other material.
Surge - A transient rise of pressure or flow.
Swell - The increased volume of specimen caused by immersion in a fluid (usually a liquid).
Switch, float - An electric switch that is responsive to liquid level.
Switch, flow - An electric switch operated by fluid flow.
Switch, pressure-differential - An electric switch operated by a difference in pressure.
Switch, pressure - An electric switch operated by fluid pressure.
Standard pressure and temperature - Generally defined as 68… F and 14.70 psia with a relative humidity of 36%.
Synthetic fluid, silicate ester - A fluid compound of organic silicates. It may contain additives.
Synthetic fluid - Fluid other than mineral that has been artificially compounded for use in a fluid power system.
Temperature, ambient - The temperature of the environment in which an apparatus is working.
Tensile strength - The force in pounds per square inch required to cause the rupture of a specimen of a rubber material.
Terpolymer - A polymer consisting of three different monomers chemically combined.
Tie rod - An axial external element that traverses the length of the cylinder. It is pre-stressed at assembly to hold the ends of the cylinder against the tubing. (Tie rod extensions can serve as mounting devices.)
Torque motor - A type of electro-mechanical transducer having rotary motion used in the input stages of servovalves.
Torque - Rotary force transmitted by the driving shaft of a motor or pump.
Torr - A unit of pressure equal to 1/760 of an atmosphere.
Torricelli's Theorem - The liquid velocity at an outlet discharging into the free atmosphere is proportional to the square root of the head.
Transducer, flow - A device that converts fluid flow to an electrical signal.
Transducer, pressure - A device that converts fluid pressure to an electrical signal.
Trunnion - A cylinder-mounting device consisting of a pair of opposite projecting cylindrical pivots. The cylindrical pivot pins are at right angle or normal to the piston rod centerline to permit the cylinder to swing in a plane.
Tube - A conductor whose size is its outside diameter. Tube is available in varied wall thickness and material.
Vacuum - Pressure less than ambient atmospheric pressure.
Vacuum pump - A device that uses mechanical force and motion to evacuate gas from a connected chamber to create subatmospheric pressure.
Valve - A device that controls fluid flow direction, pressure, or flow rate.
Valve actuator - The valve part(s) through which force is applied to move or position flow-directing elements.
Valve, air - A valve for controlling air flow direction, pressure, or flow rate.
Valve, cartridge - A valve with working parts contained in a cylindrical body. The cylindrical body must be inserted into a housing for use. Ports through the body cooperate with ports in the containing housing.
Valve, directional-control - A valve whose primary function is to direct or prevent flow through selected passages.
Valve, directional-control, 3-way - A directional control valve whose primary function is to pressurize and exhaust a port.
Valve, directional-control, 4-way - A directional control valve whose primary function is to pressurize and exhaust two ports.
Valve, directional-control, check - A directional control valve that permits flow of fluid in only one direction.
Valve, directly operated - A valve in which the controlling forces acting on the element directly influence the movement of the control elements.
Valve, electrohydraulic, proportional - A valve that responds proportionally to input signals.
Valve, flow-control (flow-metering)-a valve whose primary function is to control

flow rate.

Valve, flow-control, bypass - A pressure-compensated flow-control valve that regulates the working flow, diverting surplus fluid to the reservoir or to a second service.

Valve, flow-control, deceleration - A flow control valve that gradually reduces flow rate to provide deceleration.

Valve, flow-control, pressure-compensated - A flow control valve that controls the rate of flow independent of pressure fluctuations.

Valve, flow-dividing, pressure-compensated - A flow-dividing valve that divides flow at a constant ratio regardless of difference in resistance of the branches.

Valve, flow-dividing - A valve that divides the flow from a single source into two or more branches.

Valve, hydraulic - A valve for controlling liquid flow direction, pressure, or flow rate.

Valve, needle - A flow-control valve in which the adjustable control element is a tapered needle. Its usual purpose is to accurately control the rate of flow.

Valve, pilot-operated (indirect) - A valve in which a relatively small flow through an integral vent line relief (pilot) controls the movement of the main element.

Valve, pilot - A valve applied to operate another valve or control.

Vapor pressure - The pressure of a confined vapor in equilibrium with its liquid at a specified temperature; thus, a measure of a liquid's volatility.

Viscosity - The measurement of a fluid's resistance to flow. The common metric unit of absolute viscosity is the poise, which is defined as the force in dynes required to move a surface one square centimeter in area past a parallel surface at a speed of one centimeter per second, with the surfaces separated by a fluid film one centimeter thick. In addition to kinematic viscosity, there are other methods for determining viscosity, including, Saybolt Universal viscosity, Saybolt Furol viscosity, Engier viscosity, and Redwood viscosity. Because viscosity varies inversely with temperature, its value is meaningless until the temperature at which it is determined is reported.

Viscosity index (V.I.) - An empirical, unit-less number indicating the effect of temperature changes on the kinematic viscosity of an oil. Liquids change viscosity with temperature, becoming less viscous when heated. The higher the V.I. of an oil, the lower its tendency to change viscosity with temperature.

Volume change - A change in the volume of a seal as a result of immersion in a fluid -- expressed as a percentage of the original volume.

Volumetric efficiency - The ratio of the actual volume of air admitted (at a specified temperature and pressure), to the full piston-displacement volume- obviously for reciprocating compressors only. Volume (annular) absorbed by one full retraction of the cylinder.

Vulcanization - A thermo-setting reaction involving the use of heat and pressure, resulting in greatly increased strength and elasticity of rubber-like materials.

Efficiency is figured by the formula:

Efficiency = (1 1/Beta) 100

Efficiency10= (1 1/10) 100

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